

Czech Technical University in Prague
Faculty of Nuclear Sciences and Physical Engineering
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Diploma Thesis

Study of soliton and black hole dynamics within the
ultra-light dark matter model

Studium dynamiky solitonů a černých děr v rámci modelu
ultra-lehké temné hmoty

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- [1] J. N. Lopez-Sanchez, E. Munive-Villa, C. Skordis and F. R. Urban, "Scaling relations and tidal disruption in spin-s ultralight dark matter models," [arXiv:2502.03561 [astro-ph.CO]].
- [2] A. Eberhardt and E. G. M. Ferreira, "Ultralight fuzzy dark matter review," [arXiv:2507.00705 [astro-ph.CO]].
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DECLARATION

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***Title:* Study of soliton and black hole dynamics within the ultra-light dark matter model**

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Abstract: This thesis presents the development and validation of a numerical simulation framework for studying ultra-light dark matter in the presence of baryonic matter. The framework is based on a particle-mesh approach, combining a Fourier split-step solver for the Schrödinger–Poisson system with a grid-based gas component, collisionless N-body particles representing stars, and sink particles used as proxies for central black holes.

The implemented framework is validated through a series of numerical tests (Chapter 4), confirming the correct behaviour of all components. It is then applied to study solitonic core evolution and spin dependence, the suppression of soliton mass by a central black hole, enhanced gas accretion in the presence of a solitonic core, and exploratory mergers of composite multi-component systems (Chapter 5). The results are discussed in terms of their physical interpretation and numerical limitations in Chapter 6.

Key words: ultralight dark matter, Schrödinger-Poisson system, baryonic matter, particle-mesh method, sink particles, galaxy mergers

Název práce: **Studium dynamiky solitonů a černých děr v rámci modelu ultra-lehké temné hmoty**

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Abstrakt: V této diplomové práci představuji vývoj a validaci numerického simulačního kódu pro studium ultralehké temné hmoty v přítomnosti baryonické hmoty. Framework je založen na particle-mesh přístupu, který kombinuje Fourierovu split-step spektrální metodu pro řešení soustavy Schrödingerových–Poissonových rovnic s dalšími komponentami. Těmito komponentami jsou: plyn modelovaný na mřížce, bezsrážkové N-body částice reprezentující hvězdy a tzv. sink částice, které slouží jako aproximace centrálních černých děr.

Simulační kód je validován pomocí série testů (Kapitola 4), které ověřují správné chování všech jeho komponent. Následně je použit ke studiu vývoje solitonových jader a jejich závislosti na spinu, potlačení hmotnosti solitonu centrální černou dírou, zesílené akrece plynu v přítomnosti solitonového jádra a k provedení simulací srážek systémů využívajících všechny dostupné komponenty kódu (Kapitola 5). Interpretace výsledků je poté diskutována v Kapitole 6.

Klíčová slova: ultralehká temná hmota, Schrödingerova–Poissonova soustava rovnic, baryonická hmota, particle-mesh metoda, sink částice, srážky galaxií

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Contents

Introduction	11
1 Theoretical Background	12
1.1 Introduction to Dark Matter	12
1.2 Alternative Dark Matter Models	14
1.3 Cold Dark Matter Models	15
1.3.1 Weakly Interacting Massive Particles	15
1.3.2 Ultra-Light Scalar Dark Matter	16
2 Theoretical and Numerical Framework for Gravitationally Bound Systems in the ULDM Paradigm	22
2.1 Axion-Like Ultra-Light Dark Matter Field	23
2.1.1 Split-Step Fourier Method for the Schrödinger–Poisson System	23
2.1.2 Extension to the Schrödinger–Poisson System	24
2.1.3 Construction of a Stationary Soliton Profile	25
2.1.4 Formation and Growth of Solitonic Cores	27
2.1.5 Spin Implementation in the Simulation	28
2.2 Simulating Gas	30
2.2.1 Modelling gas	30
2.2.2 Physical Description of the Gas	30
2.2.3 How is the Numerical Scheme Implemented?	31
2.2.4 Coupling to Gravity: Strang Splitting	32
2.2.5 Optional Cooling	33
2.2.6 Time Step Constraint and CFL Condition	33
2.3 N-body Stellar Component	33
2.3.1 Particle Representation	34
2.3.2 Mass Deposition onto the Grid	34
2.3.3 Force Interpolation	35
2.3.4 Time Integration	35
2.4 Sink Particles as Black Hole Proxies	36
2.4.1 Accretion Criteria	36
2.4.2 Accretion Model	37
2.5 Astrophysical Scaling Estimates for Composite Systems	38

2.5.1	Initial Conditions	38
3	Code Implementation	40
3.1	Discretization and computational representation	41
3.2	Component representation	43
3.2.1	The ULDM Sector	43
3.2.2	The Baryonic Sector	46
3.3	Time Integration and Time-Step Control	47
3.3.1	Split-Step Evolution of the ULDM Field.	47
3.3.2	Time-Step Constraint for the Schrödinger Solver	47
3.3.3	Hydrodynamic and Particle Time Stepping	48
3.4	One Global Simulation Step	49
3.4.1	Density Assembly	49
3.4.2	Poisson Solver	50
3.4.3	ULDM Update	50
3.4.4	Particle Update	50
3.4.5	Gas Update	51
3.4.6	Sink Formation and Accretion	51
3.4.7	Diagnostics and Output	52
3.5	Manipulation with Small Numbers	53
3.6	Higher Order Methods	54
4	Code Validation	55
4.1	Quantum Harmonic Oscillator	55
4.1.1	Analytical Solution of Quantum Harmonic Oscillator	55
4.1.2	Test Results	56
4.2	Stability Testing	56
4.3	Movement Testing	58
4.3.1	Linear Motion	58
4.3.2	Rotational Motion Around a Point like Source	59
4.4	Sod Shock Tube Test	64
4.5	Energy Conservation	65
4.6	Performance Overview	66
5	Results	70
5.1	Dark Matter Density Profiles	70
5.2	Core Radius Growth	72
5.3	Core Mass Dependence on Black Hole Mass	75
5.3.1	The ULDM and BH System	75
5.3.2	Difference from ULDM Only System	76
5.3.3	Dependence on Mass of the Scalar Field	77

5.3.4	Scaling Relation	78
5.3.5	Exploratory scaling estimate	79
5.4	Gas–Soliton Dynamics and Sink Accretion	81
5.5	Composite System Mergers	82
5.5.1	Initial Conditions for Composite Systems	83
5.5.2	Stability Testing	85
5.5.3	System Mergers	87
5.5.4	ULDM + Sink + N-body Mergers	87
5.5.5	Composite Mergers with Gas	89
6	Discussion	96
6.1	Spin Dependence of Density Profiles	96
6.2	Spin Dependence of Solitonic Core Growth	97
6.3	Study of ULDM Black Hole Systems	98
6.3.1	Scalar Field Mass Dependence	98
6.3.2	Mass Scaling Relations	98
6.3.3	Scaling Model Estimate	99
6.4	Gas–Soliton Dynamics and Sink Accretion	100
6.5	Composite System Mergers	100
6.5.1	Stable Systems	100
6.5.2	System Mergers	101
	Conclusion	104
	Bibliography	109
	A Table of Initial Conditions	110
	B A Note on Self-Interaction	111

Introduction

The nature of dark matter remains one of the major open questions in modern astrophysics and cosmology. While the standard cold dark matter picture successfully describes the large-scale structure of the Universe, several open questions on galactic scales motivate the study of alternative models. One such possibility is ultra-light dark matter, whose wave-like behaviour can lead to the formation of gravitationally bound solitonic cores, making it interesting to study how these structures interact with baryonic matter and compact central objects.

The aim of this thesis is to develop and test a numerical model for studying gravitationally bound systems in the ULDM paradigm. The core of the model is based on the Fourier split-step method for solving the Schrödinger–Poisson equations. This is extended by additional components representing baryonic matter: a grid-based gas component, collisionless N-body particles representing stars, and sink particles used as effective proxies for unresolved compact objects such as black holes. The purpose of the implementation is not to construct a fully realistic galaxy-formation simulation, but rather to build a controlled and modular framework in which the basic dynamical effects of the individual components can be studied.

Before applying the model to more complex systems, its numerical behaviour must first be tested. For this reason, several validation tests are performed, including comparisons with analytical solutions, stability tests of isolated solitons, tests of particle motion in a gravitational potential, hydrodynamic benchmark tests, and energy-conservation checks. These tests are used to verify that the implemented components behave as expected and to identify the numerical limitations of the framework.

After validation, the model is used to study selected systems relevant for ULDM phenomenology. These include the stability and growth of solitonic cores, the influence of a central black-hole-like sink on the soliton structure, gas accretion in the presence of a solitonic core, and exploratory merger simulations of composite systems containing ULDM, sinks, collisionless particles, and gas. The simulations are interpreted as controlled toy models rather than fully self-consistent galaxy mergers. Nevertheless, they provide useful insight into how baryonic components can affect the evolution of ULDM-dominated systems and what kinds of physical behaviour may be worth studying in more detail in future work.

The structure of this thesis follows the development of the simulation framework. First, Chapter 1 introduces the theoretical motivation for dark matter and the ultra-light dark matter model. The physical and numerical description of the simulated components is then presented in Chapter 2, followed by the practical implementation of the code in Chapter 3. The implemented methods are tested and validated in Chapter 4. Finally, the results of selected simulation runs are presented in Chapter 5 and their physical interpretation and limitations are discussed in Chapter 6.

Chapter 1

Theoretical Background

1.1 Introduction to Dark Matter

The existence of DM was first proposed by Fritz Zwicky in his work on galaxy clusters in 1933. While studying the Coma cluster, Zwicky used spectroscopic methods to measure the velocities within the individual galaxies in this dense system of approximately 800 nebulae. He observed large differences in velocities, with individual galaxies showing recession speeds ranging from about 5100 to 8500 km/s, differences of at least 1500 to 2000 km/s from the cluster's mean velocity. To understand whether this system could be gravitationally bound, Zwicky used the virial theorem to compare the observed kinetic energies to the gravitational potential energy that would arise from visible matter alone. His calculations revealed a discrepancy showing that the cluster would need to be at least 400 times more massive than the luminous matter suggested it would be to maintain gravitational equilibrium and prevent the system from dispersing. This led Zwicky to conclude the following: dark matter is present in much greater amounts than luminous matter. His analysis provided the first quantitative evidence for the existence of a non-luminous mass dominating the dynamics of cosmic structures decades before galactic rotation curves would reinforce this mystery [1].

As observational techniques and instruments have improved, the need for dark matter has appeared across a wide range of cosmic scales. Discrepancies between the amount of visible matter and the gravitational effects we observe show up not just within galaxies and smaller structures, but also in galaxy clusters and even across the large-scale structure of the universe. Measurements of the cosmic microwave background [2], along with luminosity distances from Type Ia supernovae, provide precise estimates of the total matter energy content of the universe.

When these observations are combined with predictions from Big Bang Nucleosynthesis for the primordial abundances of light elements, a clear picture emerges: most of the matter in the universe must be non-baryonic, and not interacting with light. Because dark matter does not emit, absorb, or reflect electromagnetic radiation, it must interact with regular matter mainly through gravity. Although interactions through other forces are possible, any electric charge would have to be extremely small to remain consistent with the observed absence of electromagnetic signatures and with existing constraints [3, 4, 5, 6]. Although dark matter has not yet been detected directly in the laboratory, its presence is strongly supported by several independent gravitational observations.

Observational Evidence

Rotation curves One of the first indications for a massive, non-luminous component in galaxies comes from their rotation curves. A rotation curve describes the orbital velocity of tracers (e.g., stars or gas) as a function of their galactocentric radius. For a test particle on a circular orbit at radius r , the gravitational force provides the required centripetal acceleration,

$$\frac{mv(r)^2}{r} = \frac{GmM(r)}{r^2}, \quad (1.1)$$

where $M(r)$ is the mass enclosed within radius r and m is the mass of the observed tracer. In the inner region of a spiral galaxy, the mass distribution can be approximated by roughly uniform, so that

$$M(r) \simeq \frac{4}{3}\pi r^3 \rho, \quad (1.2)$$

with ρ the mean mass density in the central part of the galaxy. Substituting this scaling into Eq. (1.1) gives $v(r) \propto r$, meaning, the rotation curve should rise approximately as a straight line for small radii.

At sufficiently large radii, beyond most of the observable matter, one would expect $M(r)$ to approach a constant. In that case, Eq. (1.1) implies a decline, $v(r) \propto r^{-1/2}$. The resulting rotation curve would therefore rise in the inner galaxy and then fall off for larger radii. Observationally, however, rotation curves of many spiral galaxies remain approximately flat far beyond the visible disk. Within the Newtonian framework used above, a flat rotation curve requires $M(r) \propto r$, meaning that the enclosed mass must increase with radius even in the regions where we do not observe enough baryonic mass. This inconsistency points to the presence of a non-luminous massive halo component at large radii and is commonly interpreted as evidence for dark matter. Examples of measured flat rotation curves are shown in Figure 1.1.

Gravitational Lensing Gravitational lensing occurs when light from a distant source is bent by a massive object that warps the geometry of spacetime. As a result, background galaxies appear distorted or displaced. For a compact lens, the typical deflection angle can be approximated as:

$$\alpha_D \approx \frac{4Gm}{c^2 R}, \quad (1.3)$$

where m is the lens mass and R is the characteristic impact parameter. By measuring these distortions statistically, one can estimate the total mass of the lensing system, including any non-luminous contribution [7].

Gravitational lensing provides another indication that a significant fraction of matter in the Universe is not directly visible. In galaxy clusters, the mass inferred from gravitational lensing is often much larger than the mass we can account for from stars and hot gas alone, implying an additional unseen component. A particularly illustrative case is the Bullet Cluster, where the lensing-derived mass distribution is spatially offset from the X-ray emitting gas. This suggests that the dominant gravitational component effectively behaves as collisionless and does not follow baryonic matter during the merger [8].

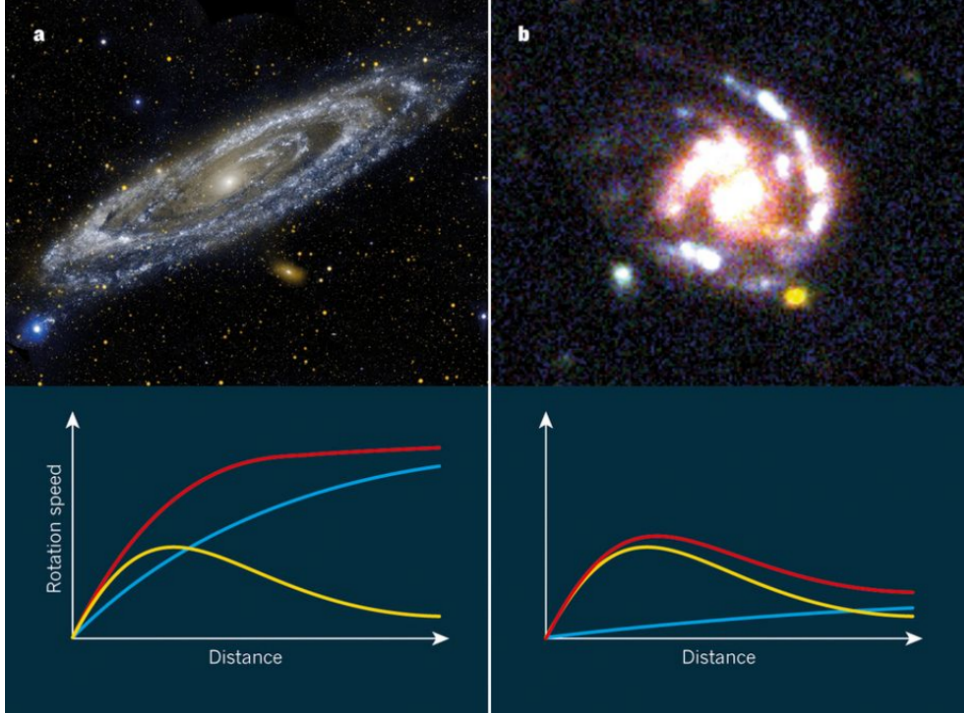


Figure 1.1: Illustration of galaxy rotation curves (bottom panels) for the corresponding galaxies (top panels). The yellow curves show the contribution expected from the observed baryonic component alone, the blue curve represents an additional extended component, and the red curve is the resulting total rotation curve. A nearly flat total curve at large radii implies that the enclosed mass continues to increase with radius beyond the visible disk. The two examples shown correspond to a nearby spiral galaxy and a distant galaxy observed at earlier cosmic time. Adapted from [9].

Cosmic Microwave Background anisotropies The Cosmic Microwave Background (CMB) provides another way to observe the presence of dark matter in the Universe. Its temperature anisotropies provide a snapshot of the state of the photon-baryon plasma at the time of recombination, about 3.8×10^5 years after the Big Bang. These anisotropies are commonly analyzed through the angular power spectrum, which quantifies how strongly temperature fluctuations appear on different angular scales on the sky [10, 11].

A characteristic feature of this spectrum is the presence of acoustic peaks, which arise from oscillations of the photon-baryon fluid in the early Universe. The positions and relative heights of these peaks depend in a predictable way on the baryon density and on the total matter density. In practice, the observed CMB power spectrum cannot be fitted using only baryonic matter and requires an additional non-baryonic matter component [12, 11].

1.2 Alternative Dark Matter Models

One of the key unknowns in identifying the nature of DM is its mass. The particle candidates for DM span an enormous range of mass scales, illustrated in Figure 1.2, covering more than 80 orders of magnitude. The figure presents various broad classes of DM models, each encompassing several specific candidates. These range from hypothetical elementary particles such as WIMPs, axions, and sterile neutrinos to composite states [13, 14], and even extend to objects on an astrophysical scale such as primordial black holes [15].

Dark matter candidates differ not only in their masses and possible interactions, but also

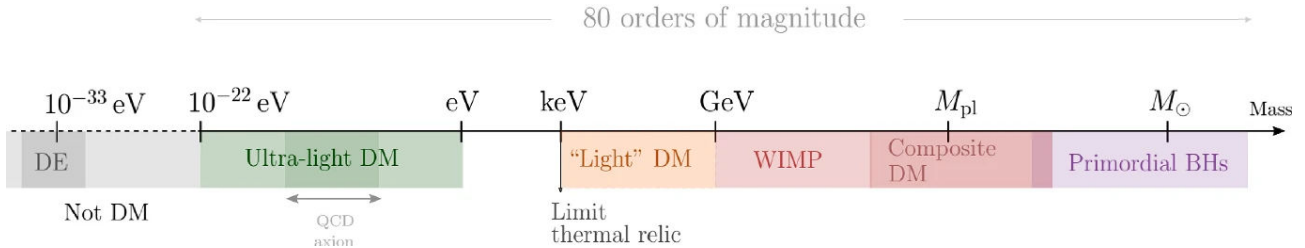


Figure 1.2: Diagram illustrating the wide range of proposed dark matter models, spanning many orders of magnitude in mass. These candidates represent fundamentally different physical scenarios, from new elementary particles to astrophysical-scale objects such as black holes. Figure adopted from [15].

in the way they could have been produced in the early Universe. Some scenarios assume that DM was once in thermal equilibrium with the hot primordial plasma, while others rely on non-thermal production mechanisms. Detailed discussion of these possibilities is beyond the scope of this work. Here, the focus is on ultralight bosonic scalar dark matter, whose phenomenology differs from standard particle-like CDM on small scales.

1.3 Cold Dark Matter Models

The Standard Model of particle physics is the most successful description of the Universe on the smallest scales that we have up to date. It is built on the idea of gauge symmetry, which provides a remarkably elegant way to encode electromagnetic, weak, and strong interactions within a single theoretical structure. It is summarised by the Standard Model Lagrangian:

$$\mathcal{L}_{\text{SM}} = \mathcal{L}_{\text{EM}} + \mathcal{L}_{\text{weak}} + \mathcal{L}_{\text{strong}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yukawa}}, \quad (1.4)$$

where the individual terms describe the electromagnetic, weak, and strong gauge interactions, the dynamics of the Higgs field, and the Yukawa couplings responsible for the fermion masses. The Standard Model has been extraordinarily successful in describing laboratory-scale physics and remains one of the most experimentally tested theories in modern physics. At the same time, it does not contain a viable candidate for DM. However, because of this success, it is natural to treat it as the starting point and extend it by introducing additional fields and interaction terms in the Lagrangian that could explain DM.

1.3.1 Weakly Interacting Massive Particles

Among the possible dark matter candidates arising from extensions of the Standard Model are weakly interacting massive particles (WIMPs). These are hypothetical particles that interact not only gravitationally, but also through the weak interaction or through a new force of similar strength. WIMP models are among the most studied dark matter scenarios, although no direct detection has been confirmed so far.

Two examples of such models are Kaluza-Klein dark matter and models with a scalar singlet. Kaluza-Klein models arise in theories with an extra spatial dimension. Since such an extra dimension is not observed directly, it is usually assumed to be compactified, for example on a circle of radius R . In that case, fields propagating in the extra dimension appear in four dimensions as a tower of massive states, called Kaluza-Klein modes. The motivation for these models is that extra dimensions may help connect gravity with the gauge interactions of the Standard Model. At the same time, the lightest stable Kaluza-Klein particle can act as a dark matter candidate [16, 17].

Another extension of the Standard Model is obtained by introducing a scalar singlet field. This field is neutral under the Standard Model gauge symmetries, but it can still interact with the Higgs field. In this way, the Higgs can serve as a portal between the visible sector and the dark sector. If the scalar field is stable, it can provide a viable dark matter candidate [16, 18].

1.3.2 Ultra-Light Scalar Dark Matter

Ultra-light scalar dark matter (ULDM) represents an alternative class of dark matter models in which the dark matter is composed of extremely light bosonic particles. These models typically involve particle masses in the broad range $10^{-25} \text{ eV} \lesssim m \lesssim 2 \text{ eV}$, although the phenomenology relevant for galactic structure usually appears only in a narrower part of this interval [15]. In contrast to more conventional cold dark matter candidates, these particles have masses so small that their de Broglie wavelengths can become comparable to astrophysical length scales, in some cases reaching the size of dwarf galaxies [19, 15]. Because of this, ULDM preserves the large-scale successes of the cold dark matter paradigm while introducing new wave-like effects on galactic scales.

The de Broglie wavelength of a non-relativistic particle is given by

$$\lambda_{dB} = \frac{\hbar}{mv}, \quad (1.5)$$

where m is the particle mass and v is its velocity. For the mass range studied closely in this thesis, $m \approx 10^{-22} \text{ eV}$ and typical halo velocities $v \approx 100 \text{ km/s}$, Equation 1.5 yields a de Broglie wavelength of approximately 190 pc. This large scale leads to collective quantum behaviour that is absent in standard particle dark matter models [15, 20].

This wave nature is one of the main reasons why ULDM has attracted considerable interest in recent years. On cosmological scales, its behaviour remains similar to that of cold dark matter, but on smaller scales it can lead to distinctive phenomena, such as the suppression of structure formation below a characteristic scale, which alleviates the missing satellite problem [15]. A key prediction of ULDM models is that, inside virialised dark matter halos, the wave-like field can relax into a dense and coherent central configuration. Such a self-supported, approximately stationary core is commonly referred to as a solitonic core, or simply a soliton, which has been proposed as a solution to the core-cusp problem [21, 19, 22]. In the central region of the halo, gravity is balanced by the dispersive effects associated with the wave nature of the field, preventing the formation of a singular density cusp. A schematic illustration of this behaviour is shown in Figure 1.3, where the halo consists of a central solitonic core surrounded by an outer region with more particle-like behaviour [23, 21]. These effects have been discussed as possible ways to alleviate some of the long-standing small-scale tensions of the Λ CDM model, such as the cusp-core problem and the missing-satellites problem [15, 19].

Since particles in the ULDM regime are far lighter than typical thermal dark matter candidates, they are generally expected to be produced through non-thermal mechanisms in the early Universe [19]. Among the most studied realisations of this idea are axions and axion-like particles, which provide a well-motivated particle-physics origin for ultra-light bosonic dark matter. These candidates are introduced in the following subsection.

Axions and Axion-like Particles

Among the best-motivated realizations of ultra-light scalar dark matter are axions and axion-like particles (ALPs). The QCD axion was originally introduced as a solution to the strong CP problem in quantum chromodynamics, where the Lagrangian admits a CP-violating term of

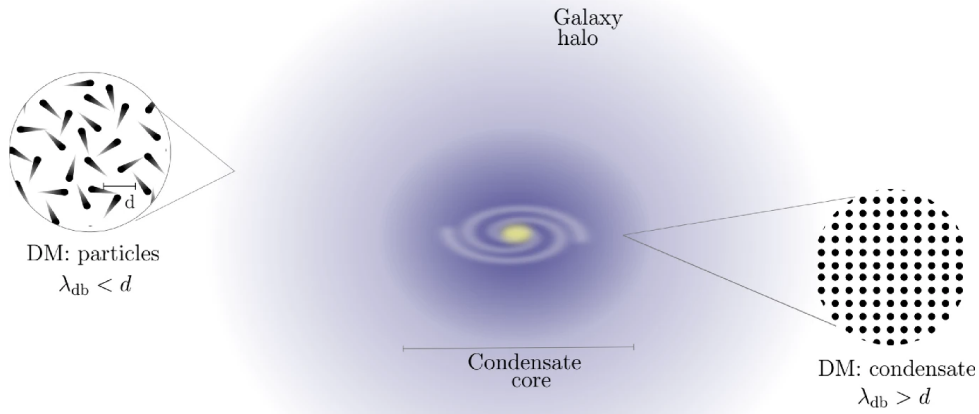


Figure 1.3: Schematic sketch of how ULDM behaves within a galaxy, forming a condensate core in the central part of a galaxy while on the outskirts exhibiting particle-like behaviour. Taken from [15].

the form

$$\mathcal{L}_\theta = -\bar{\Theta} \frac{\alpha_s}{8\pi} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}. \quad (1.6)$$

Experimental bounds imply that the parameter $\bar{\Theta}$ must be extremely small, which is difficult to explain naturally. The Peccei–Quinn mechanism resolves this problem by promoting $\bar{\Theta}$ to a dynamical field. After spontaneous breaking of the corresponding global $U(1)$ symmetry, a pseudo-Goldstone boson appears, identified as the axion [24, 25, 26].

Axion-like particles are generalisations of the QCD axion. They preserve the basic idea of a light scalar field protected by an approximate shift symmetry, but unlike the QCD axion, their masses and couplings are not fixed by the same relation. This makes them particularly interesting in the context of ultra-light dark matter, where they can occupy a broad range of parameter space and exhibit wave-like behaviour on astrophysical scales [19, 15].

If produced non-thermally in the early Universe, both axions and ALPs can behave as cold dark matter. In the ultra-light regime relevant for this thesis, they are commonly treated effectively as classical scalar fields, whose dynamics can later be reduced to the non-relativistic equations used in numerical simulations [19, 22].

Effective Dynamics of Ultra-Light Dark Matter

Although ultra-light scalar dark matter can arise from different microscopic models, many of them share the same effective behaviour on galactic scales. In the non-relativistic regime, their dynamics are commonly described by the Schrödinger-Poisson system or by suitable generalisations of it [19]. This makes the non-relativistic limit particularly important, since it provides the direct connection between the underlying scalar-field description and the equations used in numerical simulations.

A convenient starting point is to consider a real scalar field ϕ with mass m evolving in an expanding Universe. In the weak-field regime, the spacetime metric can be written as a perturbed FLRW metric of the form

$$ds^2 = \left(1 + \frac{2\Phi}{c^2}\right) c^2 dt^2 - a^2 \left(1 - \frac{2\Phi}{c^2}\right) d\mathbf{r}^2, \quad (1.7)$$

where a is the scale factor and Φ is the gravitational potential [19]. The scalar field is then described by the Klein-Gordon equation. If one further assumes that the field oscillates rapidly with frequency set by its rest mass, while its envelope evolves only slowly in time, it can be rewritten in terms of a complex field ψ as

$$\phi(x, t) = \frac{1}{\sqrt{2ma^3}} (\psi(x, t)e^{-imt} + \psi^*(x, t)e^{imt}), \quad (1.8)$$

where the approximation is valid when the time variation of ψ is slow compared to the fast phase oscillation [19].

Under these assumptions, the Klein-Gordon equation reduces to

$$i\hbar \left(\dot{\psi} + \frac{3}{2}H\psi \right) = \left(-\frac{\hbar^2}{2ma^2}\nabla^2 + m\Phi \right) \psi, \quad (1.9)$$

which is the cosmological Schrödinger equation for the scalar field in the non-relativistic and weak-field limit [19]. The gravitational potential is determined by the Poisson equation, so that together they form the Schrödinger-Poisson system. In the limit $a = 1$ and $\dot{a} = 0$, appropriate for small-scale simulations where cosmic expansion can be neglected, one recovers the usual form

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2 \psi + m\Phi\psi, \quad \nabla^2 \Phi = 4\pi Gm|\psi|^2, \quad (1.10)$$

where the mass density is given by

$$\rho = m|\psi|^2, \quad (1.11)$$

interpreted as coarse-grained density. This system forms the basis of the fuzzy dark matter model and also describes axion-like dark matter in the regime where self-interactions are negligible [19].

The wavefunction ψ can also be rewritten in Madelung form as

$$\psi = \sqrt{\rho/m} e^{im\theta/\hbar} = \sqrt{n} e^{im\theta/\hbar}, \quad (1.12)$$

where n is the comoving number density and θ is the phase of the wave function. The velocity field is then defined as the gradient of the phase, $\vec{v} = \nabla\theta$ [27, 19].

Substituting this into the Schrödinger-Poisson system leads to a set of hydrodynamic-like equations:

$$\partial_t \rho + \frac{1}{a^2} \nabla \cdot (\rho \vec{v}) = 0, \quad (1.13)$$

$$\partial_t \vec{v} + \frac{1}{a^2} (\vec{v} \cdot \nabla) \vec{v} = -\nabla V_N + \frac{\hbar^2}{2m^2 a^2} \nabla \left(\frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}} \right), \quad (1.14)$$

where the last term is often referred to as the quantum pressure or quantum potential, although it arises from wave interference rather than thermodynamics pressure. It appears from spatial gradients in the amplitude of the density and captures the dispersive behaviour of the scalar field [19]. This term is responsible for some of the characteristic small-scale effects of ULDM, including the support of solitonic cores against gravitational collapse [19]. This additional pressure-like contribution becomes dynamically significant on scales where density gradients are steep, typically at or below the de Broglie wavelength λ_{dB} , meaning in the central part of the soliton. To estimate the relative importance of this term, one should use a halo of mass M and radius R , and compare the quantum pressure gradient Q/R to the typical inertial term v^2/R . Doing this, one finds the following relation:

$$\frac{Q/R}{v^2/R} \sim \left(\frac{\lambda_{\text{dB}}}{R} \right)^2, \quad (1.15)$$

which implies that deviations from classical cold dark matter (CDM) dynamics are controlled by the ratio of the de Broglie wavelength to the halo size. For ultra-light axions, λ_{dB} can reach kiloparsec scales, making the quantum pressure term relevant for the internal dynamics of dwarf galaxies and solitonic cores [19]. As shown in [28], the inclusion of this pressure term is necessary to accurately model the stability and structure of fuzzy dark matter halos. It suppresses small-scale fluctuations and supports the formation of coherent solitonic cores, thus playing a defining role in the nonlinear evolution of axion dark matter.

In this framework, gravitational clustering and structure formation can be viewed through the lens of wave turbulence. The process of soliton formation is often interpreted as a form of gravitational Bose-Einstein condensation. These effects occur classically in the field description and do not require quantum statistical mechanics. Consequently, this classical wave dynamics offers a powerful alternative to the collisionless particle picture of CDM, with distinct implications for galactic structures [19].

Classification of Ultra-Light Dark Matter Models

Although their microscopic origin may differ, many of these models reduce in the non-relativistic limit to the Schrödinger–Poisson system or to suitable generalisations of it. Following the classification used in [15], I distinguish here between the purely gravitational case, usually referred to as fuzzy dark matter, and models in which additional self-interactions modify the small-scale behaviour of the field.

Fuzzy Dark Matter The first class is fuzzy dark matter (FDM), in which the scalar field is bound only by gravity and its dynamics are described by the Schrödinger–Poisson system. In natural units, this can be written as

$$i\frac{\partial\psi}{\partial t} = -\frac{1}{2m}\nabla^2\psi + \Phi\psi, \quad (1.16)$$

together with

$$\nabla^2\Phi = 4\pi Gm|\psi|^2. \quad (1.17)$$

Here, $m|\psi|^2$ corresponds to the mass density of the scalar field. In this model, gravitational attraction is balanced by quantum pressure, which allows the formation of stable solitonic cores.

FDM is commonly associated with particle masses of order $m \sim 10^{-22}$ eV, as originally proposed by Hu et al. [23]. This mass scale leads to de Broglie wavelengths large enough to affect halo structure on kiloparsec scales, while preserving the successful large-scale behaviour of the Λ CDM model. In its simplest form, FDM is therefore characterised mainly by a single free parameter, namely the particle mass. A benchmark value of order 10^{-22} eV is often used in the literature [15], and will also be used heavily in this work, although current constraints tend to push this value higher.

Microscopic scalar theories such as QCD axions and axion-like particles can fall into this class when self-interactions are negligible, since in that regime they reduce to the same effective Schrödinger–Poisson dynamics on galactic scales [20]. Despite its appeal, FDM is increasingly constrained by observations, including the Lyman- α forest [29, 30], Milky Way satellite galaxies and stellar streams [31], and galaxy formation at high redshift. These constraints restrict the allowed mass range and limit how strongly FDM can depart from the standard CDM picture [32].

Self-Interacting Scalar Field Dark Matter The second class consists of models in which the scalar field experiences, in addition to gravity, a non-negligible self-interaction. These models are commonly referred to as scalar field dark matter (SFDM), although other names also appear in the literature [20]. The inclusion of self-interactions modifies both the condensation process and the internal structure of dark matter halos.

A general form of the corresponding non-linear Schrödinger equation is

$$i\frac{\partial\psi}{\partial t} = -\frac{1}{2m}\nabla^2\psi + \Phi\psi + g|\psi|^2\psi + g_3|\psi|^4\psi + \dots, \quad (1.18)$$

where the coefficients g , g_3 , etc., describe the strength of the self-interaction terms. The sign of g determines whether the interactions are attractive or repulsive and their precise form depends on the underlying microscopic model or effective description. Unlike in the purely gravitational FDM case, the phenomenology of SIFDM depends not only on the particle mass, but also on the sign and magnitude of the self-interaction coupling.

Self-interactions can significantly alter the structure of the condensate and lead to behaviour that is not present in the purely gravitational limit [33]. In particular, repulsive self-interactions can support more extended cores and, under suitable conditions, give rise to superfluid-like behaviour with long-range coherence. For this reason, SIFDM provides a broader phenomenology than FDM and allows a wider range of possible halo structures and density profiles [15, 33].

Phenomenology of ULDM halos

A characteristic prediction of ultra-light dark matter models is that virialised dark matter halos do not develop the same inner structure as in standard cold dark matter. Instead, numerical simulations and analytical arguments show that ULDM halos tend to form a distinct core-halo structure, consisting of a dense central solitonic core surrounded by an extended outer halo [21, 19]. The central soliton is a gravitationally bound, approximately stationary solution of the Schrödinger-Poisson system, supported against collapse by the quantum pressure term discussed in the previous subsection. Outside the core, the halo is not perfectly smooth, but exhibits interference patterns that arise naturally from the wave nature of the field [19].

This behaviour differs from the standard CDM picture, in which halos are usually described by cuspy density profiles such as the Navarro–Frenk–White (NFW) profile:

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right)\left(1 + \frac{r}{r_s}\right)^2}, \quad (1.19)$$

where r is the radial distance from the halo centre, ρ_s is a characteristic density, and r_s is the scale radius at which the profile transitions from the inner cuspy behaviour to the outer power-law decline. In the ULDM case, the presence of quantum pressure prevents the formation of a singular central cusp and instead produces a flatter density core. For this reason, ULDM has often been discussed as a possible way to alleviate the cusp–core problem, one of the long-standing small-scale tensions of the Λ CDM model [23, 15]. At the same time, the suppression of structure formation below a characteristic scale reduces the abundance of very small halos, which may help with the missing-satellites problem [21, 15].

An important phenomenological feature is that ULDM behaves similarly to CDM on large scales, while departing from it on galactic and sub-galactic scales. The outer parts of halos may still resemble collisionless dark matter systems, but the inner region is modified by the existence of the solitonic core and by wave interference in the surrounding halo [19]. This makes ULDM particularly interesting, since it preserves the large-scale successes of Λ CDM while introducing new and testable effects in the regime where standard CDM faces the greatest challenges.

These differences can in principle be constrained observationally. Signatures of ULDM halos may appear in the internal kinematics of dwarf spheroidal galaxies, in galaxy rotation curves, in low-surface-brightness galaxies, and in the suppression of small-scale power in structure formation [15, 34]. In addition, gravitational lensing and cosmological large-scale-structure data may provide further constraints on the allowed particle mass and on the range in which wave effects remain relevant [15].

Chapter 2

Theoretical and Numerical Framework for Gravitationally Bound Systems in the ULDM Paradigm

One of the main objectives of this work is to study the properties of gravitationally bound multi-component systems within the framework of ultra-light dark matter. A particular motivation for extending the simulation framework beyond the scope of just ULDM was the observation that solitonic cores can represent relatively massive and dense central structures. Although the ULDM soliton itself is stabilised by wave effects and does not undergo gravitational collapse in the same way as ordinary matter, it is natural to ask whether baryonic matter accumulated in its central region could eventually collapse and give rise to a compact object such as a supermassive black hole (SMBH).

This question motivated the implementation of additional physical components into the simulation framework, namely gas, collisionless N-body particles representing stars, and sink particles intended to mimic black holes. The broader goal was to move from simulations of isolated ULDM configurations towards a framework capable of describing more realistic astrophysical systems, including the formation of central compact objects, the interplay between dark matter and baryons, and eventually even interactions between galaxy-like structures.

The resulting framework represents a substantial extension of the baseline ULDM simulation model and provides a tool for studying ULDM systems coupled to baryonic matter and central black holes. This chapter summarises the main physical components that were implemented together with the theoretical background on which their numerical treatment is based.

These include the following components:

1. **Axion-like ultra-light dark matter field 2.1**
2. **Grid-based gas component 2.2**
3. **stellar N-body 2.3**
4. **N-body sink particles representing black holes 2.4**

2.1 Axion-Like Ultra-Light Dark Matter Field

The aim of this section is to introduce the field-theoretical and numerical formulation that forms the basis of the simulation framework. In particular, I summarise how the underlying relativistic scalar field is connected to the complex wave function evolved in the code, and how this description is modified when written in comoving coordinates.

At a fundamental level, the model can be formulated in terms of a real quantum scalar field ϕ governed by a relativistic action. In the non-relativistic and weak-field limit, this field can be re-expressed as a complex scalar wave function ψ using Equation 1.8. The resulting dynamics are then described by the Schrödinger–Poisson system (Equation 1.16) [19].

In cosmological applications, the Schrödinger–Poisson system is often written in comoving coordinates, using a rescaled wave function $\psi_c = a^{3/2}\psi$, a comoving gradient $\nabla_c = a\nabla$, and a comoving gravitational potential $\Phi_c = a\Phi$. The evolution equations then take the form

$$i\hbar\partial_t\psi_c(t, \mathbf{x}) = -\frac{\hbar^2}{2ma(t)^2}\nabla_c^2\psi_c(t, \mathbf{x}) + \frac{m}{a(t)}\Phi_c(t, \mathbf{x})\psi_c(t, \mathbf{x}), \quad (2.1)$$

$$\nabla_c^2\Phi_c(t, \mathbf{x}) = 4\pi Gm(|\psi_c(t, \mathbf{x})|^2 - \langle|\psi_c|^2\rangle(t)), \quad (2.2)$$

where the subtraction of the mean density is the standard way to model the evolution of perturbations using the density contrast defined as $\delta = \rho/\rho_{mean} - 1 = (\rho - \rho_{mean})/\rho_{mean}$.

The wave function ψ_c carries both density and velocity information. The mass density is given by $\rho_c = m|\psi_c|^2$, while the velocity field can be obtained from the phase of the wave function, $\psi_c = \sqrt{\rho_c/m}e^{i\alpha}$, as $\mathbf{v}_c = \hbar/m\nabla_c\alpha$. This polar decomposition allows for a fluid-like interpretation of the system via the Madelung transformation discussed in Section 1.3.2 [35].

In this formulation, the wave nature of the field gives rise to the quantum-pressure term, which suppresses structure formation below the Jeans scale and prevents gravitational collapse on sufficiently small scales. The comoving Jeans wavenumber depends on the redshift and particle mass as

$$k_J = \left(\frac{6m}{1+z}\right)^{1/4} \left(\frac{mH_0}{\hbar}\right)^{1/2}, \quad (2.3)$$

which implies that only perturbations larger than the corresponding Jeans length can grow gravitationally [35]. In this work, however, I adopt a redshift of $z = 0$ and therefore a scale factor $a = 1$, since the focus is on the small-scale dynamics of already formed structures rather than on cosmological evolution [35, 23, 21].

2.1.1 Split-Step Fourier Method for the Schrödinger–Poisson System

Several numerical methods have been developed over the years to integrate the SP system. Among them, the split-step Fourier method stands out for its efficiency and accuracy in evolving the wave function over time [36]. The foundation of the numerical approach used in this work lies in the Fourier split-step method [37], which uses the separability of the Hamiltonian into kinetic and potential components. This technique enables stable and efficient evolution of the wave function by alternating between operations in different representations, the Fourier space for kinetic terms and the real space for potential terms.

To illustrate the method, let us first consider the dimensionless vacuum Schrödinger equation in one spatial dimension:

$$i\frac{\partial\psi(x, t)}{\partial t} = -\frac{1}{2}\partial_x^2\psi(x, t). \quad (2.4)$$

The analytical solution over a small time step h can be written as:

$$\psi(x, t + h) = \exp\left(i\frac{h}{2}\partial_x^2\right) \psi(x, t). \quad (2.5)$$

The key insight is that the exponential of the differential operator can be computed using the Fourier transform. In Fourier space, spatial derivatives become multiplied by the associated wave number k . Applying the Fourier transform $\mathcal{F}$ to Equation 2.5, we obtain:

$$\tilde{\psi}(k, t + h) = \exp\left(-i\frac{h}{2}k^2\right) \tilde{\psi}(k, t), \quad (2.6)$$

where $\tilde{\psi}$ denotes the Fourier transformed ψ and k is the wave number. The resulting wave function in real space is then recovered via the inverse transform:

$$\psi(x, t + h) = \mathcal{F}^{-1} \left[\exp\left(-i\frac{h}{2}k^2\right) \tilde{\psi}(k, t) \right]. \quad (2.7)$$

When an external potential $V(x)$ is added to the Schrödinger equation, the complete time-dependent Schrödinger equation becomes:

$$i\frac{\partial\psi(x, t)}{\partial t} = \left(-\frac{1}{2}\partial_x^2 + V(x)\right) \psi(x, t). \quad (2.8)$$

For finite time steps, we can employ symmetric operator splitting (Strang splitting) [38] to maintain second-order accuracy:

$$\psi(x, t + h) \approx e^{-i\frac{h}{2}V(x)} \mathcal{F}^{-1} \left[e^{-i\frac{h}{2}k^2} \mathcal{F} \left(e^{-i\frac{h}{2}V(x)} \psi(x, t) \right) \right]. \quad (2.9)$$

This advances the equation by alternating between real space for potential operations and Fourier space for kinetic operations. In the Schrödinger–Poisson system, the potential is not prescribed externally, but is recomputed self-consistently from Poisson’s equation using the instantaneous density $\rho = m|\psi|^2$. The method is numerically robust and preserves the norm of the wave function to machine precision. Additionally, this method has a computational complexity of the order $\mathcal{O}(n \log n)$. Compared to finite-difference methods with complexity $\mathcal{O}(N^2)$, this approach is computationally more efficient, which makes it suitable for long-term integrations required in cosmological and astrophysical simulations [39, 35].

2.1.2 Extension to the Schrödinger–Poisson System

To model fuzzy dark matter consistently, the coupled Schrödinger–Poisson system must be solved, including the gravitational potential generated by the wave function itself. In comoving coordinates with the scale factor $a(t)$, the system takes the form given in Equation 2.2. The formal solution over a time step Δt can be written as

$$\psi_c(t + \Delta t, x) = \mathcal{T} \exp \left[-i \int_t^{t+\Delta t} \hat{H}(t') dt' \right] \psi_c(t, x), \quad (2.10)$$

where $\mathcal{T}$ denotes time ordering and $\hat{H}(t')$ is the Hamiltonian operator dependent on time of the system [35].

For sufficiently small time steps, the evolution of the scale factor and gravitational potential can be treated as approximately constant within each step, which allows the use of the split-step method. Using the symmetric operator splitting introduced in Equation 2.9, the evolution over one time step can be approximated as [35, 39]

$$\psi_c(t + \Delta t, x) \approx \underbrace{e^{-i\frac{m}{h}\frac{1}{a(t)}\frac{\Delta t}{2}\Phi_c(t+\Delta t, x)}}_{\text{kick}} \underbrace{e^{i\frac{h}{m}\frac{1}{a(t)^2}\frac{\Delta t}{2}\nabla_c^2}}_{\text{drift}} \underbrace{e^{-i\frac{m}{h}\frac{1}{a(t)}\frac{\Delta t}{2}\Phi_c(t, x)}}_{\text{kick}} \psi_c(t, x). \quad (2.11)$$

2.1.3 Construction of a Stationary Soliton Profile

A natural choice of initial wave function for the simulations is a stationary self-gravitating solution of the Schrödinger–Poisson system. These so called solitons represent a gravitationally bound scalar field that provides a stable and physically motivated initial condition. A schematic illustration of this type of state was shown in the previous chapter in Figure 1.3. In the literature, similar objects also appear under the names Bose stars or axion stars, depending on the physical context in which they are discussed [40, 41, 42, 43].

Mathematically, the soliton corresponds to a stationary solution of the Schrödinger–Poisson system. This is obtained by inserting the time separable ansatz

$$\psi(\mathbf{r}, t) = \psi(\mathbf{r})e^{-imEt/\hbar},$$

of the wave function $\psi(r, t)$ into the time-dependent Schrödinger equation, which leads to the following eigenvalue problem:

$$mE\psi = -\frac{\hbar^2}{2m}\nabla^2\psi + mV_N\psi, \quad (2.12)$$

$$\nabla^2V_N = 4\pi Gm|\psi|^2, \quad (2.13)$$

where E is the energy eigenvalue and V_N is the Newtonian gravitational potential of the SP system. For an isolated self-gravitating object, the boundary condition $\psi(r \rightarrow \infty) = 0$ is imposed. Without external sources of gravity, the resulting density profile is localised around the origin, with a central core and a rapid decrease at large radii [19], but this may change as will be shown in later chapters.

The assumption of spherical symmetry reduces the problem to a radial system, which is more convenient for numerical treatment. In this case, the stationary wave function and the gravitational potential depend only on the radial coordinate, and the problem can be rewritten as a coupled system of ordinary differential equations. To solve this system numerically, it is convenient to introduce dimensionless variables, which reduce the number of free parameters and make the structure of the eigenvalue problem more transparent. Using the rescaling relations from [44], discussed in more detail in Section 3.5, we obtain the following result:

$$\hat{\psi} = \frac{m\sqrt{G\rho_0}}{\hbar}\psi, \quad \hat{\Phi} = \frac{m^2}{\hbar^2}\Phi, \quad \hat{\omega} = \frac{m\omega}{\hbar^2}. \quad (2.14)$$

With these rescaled quantities, the equations transform to:

$$\begin{aligned} \frac{1}{2}\frac{d^2}{dr^2}(r\hat{\psi}) &= r\hat{\psi}(\hat{\Phi} - \hat{\omega}), \\ \frac{d^2}{dr^2}(r\hat{\Phi}) &= 4\pi r\hat{\psi}^2, \end{aligned} \quad (2.15)$$

where $\hat{\omega}$ plays the role of the eigenvalue of the system. Solving this system with appropriate boundary conditions yields localised, gravitationally bound states that can be used as stable initial conditions for the subsequent time evolution. The boundary conditions are

$$\begin{aligned} \hat{\psi}(r \rightarrow \infty) &\rightarrow 0, & \hat{\Phi}(r \rightarrow \infty) &= -\frac{GM}{\hbar^2 r}, \\ \hat{\psi}(0) &= 1, & \left.\frac{d\hat{\psi}}{dr}\right|_{r=0} &= 0, \\ \left.\frac{d\hat{\Phi}}{dr}\right|_{r=0} &= 0, & \left.\frac{d\hat{\psi}}{dr}\right|_{r \rightarrow \infty} &\rightarrow 0, \end{aligned} \quad (2.16)$$

where $M(r) = 4\pi \int_0^r \rho(r') r'^2 dr'$ is the enclosed mass within radius r .

The condition $\hat{\psi}(0) = 1$ is a normalisation choice used to fix the central amplitude of the solution. Choosing a different central value would not change the existence of the bound-state family, but would instead generate a rescaled representative of the same solution class, with correspondingly rescaled physical properties such as the soliton mass, rather than a fundamentally different solution [44].

Solving the eigenvalue problem in Equation 2.13, we obtain a family of localised bound states corresponding to different energy levels. The soliton used in this work is the lowest-energy, which represents the ground state and is the most stable member of this family. Once such a solution is obtained, several of its characteristic properties can be approximated by the following relations [19, 45]:

1. Half-mass radius:

$$R_{1/2} \simeq \frac{4\hbar^2}{GMm^2}, \quad (2.17)$$

where M is the total mass of the soliton.

2. Central density:

$$\rho_c \simeq 4 \times 10^{-3} \left(\frac{Gm^2}{\hbar^2} \right)^3 M^4 \simeq 2\bar{\rho}_{1/2}, \quad (2.18)$$

with

$$\bar{\rho}_{1/2} = \frac{3(M/2)}{4\pi R_{1/2}^3} \quad (2.19)$$

denoting the mean density within the half-mass radius.

3. Virial velocity:

$$v_{\text{vir}}^2 \simeq 0.1 \left(\frac{GMm}{\hbar} \right)^2 \simeq 0.4 \frac{GM}{R_{1/2}} \simeq -0.3V_{N,c}, \quad (2.20)$$

where $V_{N,c}$ is the central gravitational potential, approximated as

$$V_{N,c} \simeq -0.3 \left(\frac{GMm}{\hbar} \right)^2. \quad (2.21)$$

4. Coherence (de Broglie) length:

$$\lambda_{\text{dB}} = \frac{\hbar}{mv_{\text{vir}}} \simeq 0.8R_{1/2}, \quad (2.22)$$

demonstrating that the characteristic size of the soliton is directly related to the de Broglie wavelength of its constituent particles.

The central density relation can also be interpreted in relation to the boundary condition $\hat{\psi}(0) = 1$. In the system, this condition fixes the central value of the normalised wave function and therefore also the dimensionless central density $\tilde{\rho}_c \propto |\hat{\psi}(0)|^2$. However, the physical central density ρ_c is obtained only after restoring the dimensional scaling of the solution. While $\tilde{\rho}_c$ is fixed by the chosen normalisation, ρ_c depends on how this reference profile is rescaled to a physical soliton of a given mass and size.

An important property of the SP equations is their invariance under a scaling transformation:

$$\{t, x, V_N, \psi, \rho\} \rightarrow \{\lambda^{-2}\hat{t}, \lambda^{-1}\hat{x}, \lambda^2\hat{V}_N, \lambda^2\hat{\psi}, \lambda^4\hat{\rho}\}, \quad (2.23)$$

and therefore

$$\{M, E\} \rightarrow \{\lambda^{-2}\hat{M}, \lambda^3\hat{E}\}, \quad (2.24)$$

where λ is an arbitrary scaling parameter. This symmetry enables rescaling of soliton profiles to different astrophysical contexts, from galactic cores to mini-clusters, while preserving the underlying structure of solutions [46, 47, 22]. This implies that solutions form a one-parameter family related by scaling transformations.

2.1.4 Formation and Growth of Solitonic Cores

In FDM models, the wave nature of ultra-light bosons leads to the formation of gravitationally bound central structures known as solitonic cores. These cores arise from the balance between gravitational attraction and quantum pressure [19]. As can be seen in Equation 1.14, the quantum-pressure term originates from the gradient energy in the Schrödinger–Poisson system and prevents gravitational collapse on small scales. This stabilising effect allows the inner part of a halo to settle in a localised, approximately stationary configuration [19].

The core forms first in the densest regions, which then attract additional mass. As the central density increases, the quantum pressure counteracts further compression, and the system relaxes toward a bound solitonic state. Cosmological simulations show that this behaviour is a robust feature of FDM halos: their innermost regions are well described by soliton-like profiles, while the outer parts remain more extended and are better described by an NFW-like distribution, as shown in Equation 1.19 [21, 48, 19].

FDM halos therefore exhibit a characteristic core–halo structure in which a dense central soliton is embedded in a more diffuse outer halo. For the systems considered in this work, i.e. halos resulting from mergers, the full density distribution is more suitably described by a composite profile

$$\rho_{\text{halo}}(r) = \Theta(r_\epsilon - r)\rho_{\text{sol}}(r) + \Theta(r - r_\epsilon)\rho_{\text{NFW}}(r), \quad (2.25)$$

where Θ is the Heaviside step function and r_ϵ is the transition radius between the solitonic core and the NFW-like outer halo. The solitonic part is commonly approximated by

$$\rho_{\text{sol}}(r) = \frac{\rho_c}{[1 + \alpha(r/r_c)^2]^8}, \quad (2.26)$$

where ρ_c is the central density, r_c is the characteristic radius of the core, and α is a dimensionless parameter that controls the transition from the flat core to the outer decline profile. In this work, the value $\alpha = 0.091$ is adopted [22].

The solitonic core is not independent of its host halo. Its mass is correlated with the halo mass through the so-called core–halo relation, which implies that more massive halos host more massive central solitons. During structure formation, this growth can proceed through mergers with smaller cores as well as through accretion of ambient matter. Using cosmological simulations it was found in [21] that a relation between the mass of the halo M_{halo} and the mass of the core M_{sol} exists:

$$M_{\text{sol}} \propto M_{\text{halo}}^{1/3} m_s^{-1}, \quad (2.27)$$

as was shown in Davies & Mocz [49]. On the other hand simulations of idealised soliton mergers suggest a somewhat different relation,

$$M_c \propto M_h^{5/9}, \quad (2.28)$$

as reported in [36]. Nevertheless, both relations reflect the same general picture: the growth of the central soliton is linked to the assembly history of the host halo. These processes are accompanied by wave interference and relaxation phenomena characteristic of self-gravitating scalar fields, often described as gravitational cooling [19].

2.1.5 Spin Implementation in the Simulation

So far, the discussion has been restricted to the standard scalar-field description of ULDM, but one of the goals of this work was to extend the numerical framework in a way that allows the simulation of ULDM dynamics for fields with higher integer spin. Studying how DM structures evolve for non-zero spin representations may reveal spin-dependent effects relevant for specific theoretical models. For this purpose, it is necessary to introduce a numerical prescription that incorporates spin into the wave-function description. Following [22], this can be done by decomposing the wave function into polarisation states as

$$\Psi(t, \mathbf{x}) = \sum_p \psi_p(t, \mathbf{x}) \epsilon^p, \quad (2.29)$$

where ψ_p is the wave-function component associated with polarisation state p . Individual polarisation states are initialised by taking the solitonic solution as a reference profile, which means that $\psi_p(t = 0, \mathbf{x})$ is set to be equal to the spin-0 case and multiplied by a real polarisation coefficient c_p and a random phase θ_p , such that

$$\psi_p(t = 0, \mathbf{x}) = \psi_{\text{sol}}(\mathbf{x}) c_p e^{-i\theta_p}. \quad (2.30)$$

This choice provides a straightforward generalisation of the spin-0 initial condition while allowing the total wave function to be distributed among different polarisation states. The coefficients c_p determine how the initial solitonic profile is distributed among the individual polarisation components. Their introduction is motivated by the standard angular momentum coupling formalism, in which total angular momentum eigenstates are expanded in an uncoupled tensor-product basis using Clebsch–Gordan (CG) coefficients. In general, this coupling is written as

$$(j_1, m_1) \otimes (j_2, m_2) \rightarrow (J, M), \quad (2.31)$$

where j_i are the angular momenta of the constituent states, m_i are their magnetic quantum numbers, and J, M denote the total angular momentum and its projection. The corresponding coupled state can be expressed as

$$|J, M\rangle = \sum_{m_1, m_2} \langle j_1, m_1; j_2, m_2 | J, M \rangle |j_1, m_1\rangle |j_2, m_2\rangle,$$

where $\langle j_1, m_1; j_2, m_2 | J, M \rangle$ is the Clebsch–Gordan coefficient. In the present construction, this formalism provides the motivation for decomposing the higher-spin wave function into polarisation components and for assigning their relative weights.

We use these coefficients to “lift” the scalar wave function, which solves the Schrödinger equation for the spin-0 case, to a tensor-valued wave function representing a particle with higher spin. In this way, the spin-0 solitonic solution is decomposed into several components and recombined into a multi-component state, with each component associated with a particular spin projection. One advantage of using this Clebsch–Gordan-inspired construction is that it naturally reflects the symmetry of the polarisation states: some of them differ only by complex conjugation. In numerical simulations, this can be exploited to reduce memory requirements, since at each step the key quantity entering the Schrödinger–Poisson system is the density $|\psi|^2$, for which complex-conjugate polarisation components contribute equally. Another practical advantage is that precomputed Clebsch–Gordan coefficients are readily available in existing Python libraries.

To illustrate how this lifting procedure works, we now consider the cases of spin 0, 1, and 2:

Spin 0 This is the simplest case, as the field is described by a single component ψ_p in Equation 2.30, with $c_p = 1$:

$$\psi_0 = \psi_{\text{sol}} e^{-i\theta_0}. \quad (2.32)$$

Spin 1 The basis is represented by the following set of orthonormal vectors associated with three polarisation states $p = \pm 1, 0$:

$$\epsilon(\pm 1) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \\ 0 \end{pmatrix}, \quad \epsilon(0) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \quad (2.33)$$

Two of the coefficients c_p are chosen randomly and the third is then fixed by requiring $\sum_p |c_p|^2 = 1$. This ensures that the resulting state is normalised. In practice, we are creating a random normalised vector in basis shown in Equation 2.33. If we look at the polarisation states $\epsilon(\pm 1)$, we can see that they differ only by the sign of the imaginary part. That means that when we compute physical observables that depend only on the square of the wave function $|\Psi|^2$, these two states contribute equally. That is, instead of treating them separately, we can compute one and count it twice. This effectively reduces the number of components needed from three to two, which saves memory. And that is exactly what we need, since the only observable required to solve the SP system is the density $\rho = |\Psi|^2$, and not the full complex wave function.

This trick is especially handy because we can exploit the natural symmetry of rank- s tensors in 3D space. In three dimensions, a rank- s tensor has 3^s components if no symmetry is assumed. But when the tensor is fully symmetric and therefore is unchanged under permutation of its indices, the number of unique components drops to only $\binom{s+2}{s}$. So instead of needing to store all components, we can just keep the symmetric ones and reconstruct the full tensor from them.

When we decompose a spin- s state using the Clebsch–Gordan-inspired construction, the resulting tensor can be written as a linear combination of products of the polarisation vectors. I will describe this in more detail in Section 3.2.1. This means that many of the components in this decomposition are related by symmetry and do not need to be computed explicitly, but can instead be taken into account through the multiplicity of their symmetric counterparts.

Spin 2 For spin-2, there are five independent polarisation states, corresponding to $p = \pm 2, \pm 1, 0$. These states form an orthonormal basis and are represented by the following maximally polarised symmetric tensors:

$$\epsilon(\pm 2) = \frac{1}{2} \begin{pmatrix} 1 & \pm i & 0 \\ \pm i & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (2.34)$$

$$\epsilon(\pm 1) = \frac{1}{2} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & \pm i \\ 1 & \pm i & 0 \end{pmatrix}, \quad (2.35)$$

$$\epsilon(0) = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}. \quad (2.36)$$

As before, the five coefficients c_p are randomly assigned, with the normalisation constraint $\sum_p |c_p|^2 = 1$, for each soliton. For example, in this spin-2 case, we do not distinguish between T_{01} and T_{10} , we just need to store one of them and factor in how many times it appears when computing the density. Using this symmetry, we only need six independent tensor components instead of all nine components of a general 3×3 tensor.

This pattern becomes even more useful for higher spins. For spin-3, we reduce from 27 components to just 10, and for spin-4, from 81 to 15. In general, the number of components drops from 3^s to $\binom{s+2}{s}$, which follows from the counting of combinations with replacement of the three spatial indices. Instead of tracking every component of the full tensor, we only need to store the unique symmetric ones and their multiplicity.

This approach preserves the information needed to compute the density relevant for gravitational evolution while significantly reducing the memory required to store the wave function. More on how these coefficients are used to construct spin states will be covered in the next chapter, especially in Section 3.2.1.

2.2 Simulating Gas

Another important part of the simulation framework developed in this work is a grid-based compressible hydrodynamics solver for an ideal adiabatic gas. Its implementation was inspired by the Jaxion¹ framework developed by Philip Mocz, in which the gas component is modelled by the compressible Euler equations solved with a finite-volume method [50].

2.2.1 Modelling gas

Dark matter does not exist in isolation. In astrophysical environments where ULDM is expected to be dynamically important, such as dwarf galaxies, early stages of galaxy formation, or regions of strong gravitational collapse [15], there is also an ordinary baryonic gas present. This gas is gravitationally attracted to any overdense structure that forms in the dark matter field, and in turn it contributes its own gravity back to the system.

2.2.2 Physical Description of the Gas

To model the gas component, it is necessary to evolve the local mass, momentum, and energy of a compressible fluid on a spatial grid. In the absence of viscosity and other dissipative effects, this is described by the compressible Euler equations, which express the conservation of mass, momentum, and total energy. In the present case, the gas is additionally coupled to gravity through the gravitational potential shared with the other simulation components [50, 51].

In conservative form, the compressible Euler equations read as follows:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \quad (2.37)$$

$$\frac{\partial(\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \mathbf{v} + P \mathbf{I}) = -\rho \nabla \Phi, \quad (2.38)$$

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + P) \mathbf{v}] = -\rho \mathbf{v} \cdot \nabla \Phi, \quad (2.39)$$

where ρ is the gas density, $\mathbf{v}$ is the velocity field, P is the pressure, E is the total energy density (kinetic plus internal), and Φ is the gravitational potential [51].

The physical content is straightforward. Equation (2.37) says that mass cannot be created or destroyed: any change in the local density must be accounted for by a net flux of mass through the cell boundaries. Equation (2.38) is Newton's second law for a fluid element: the momentum changes because of pressure gradients (gas pushes harder from the high-pressure side) and because gravity pulls the gas toward regions of lower potential. Equation (2.39) is the same idea for energy: both the pressure work done at cell interfaces and the work done by gravity on the moving gas must be tracked.

To close the system, an equation of state must be specified. For an ideal adiabatic gas, the pressure is related to the conserved variables through

$$P = (\gamma - 1) \left(E - \frac{1}{2} \rho |\mathbf{v}|^2 \right), \quad (2.40)$$

¹Accessible at: Jaxion

where γ is the adiabatic index. In our implementation $\gamma = 5/3$, which corresponds to a monatomic ideal gas. The term in parentheses is simply the internal energy density. Subtracting the kinetic part $\frac{1}{2}\rho|\mathbf{v}|^2$ from the total energy E leaves only the thermal contribution, and the factor $(\gamma - 1)$ converts it into pressure [51, 50].

Together, Equations (2.37)–(2.40) form a closed system that can, in principle, evolve forward in time given the appropriate initial conditions.

The main difficulty in simulating gas on a grid is that the Euler equations are hyperbolic conservation laws. Even if the initial conditions are perfectly smooth, the nonlinear terms can cause gradients to steepen until they become shocks or contact discontinuities in a finite time. A naive finite-difference scheme applied pointwise to the differential form of the equations will produce unphysical oscillations near such features, or simply blow up. The standard remedy is to switch to a finite-volume formulation, where the unknowns are cell-averaged quantities rather than pointwise values, and where the time evolution is driven by numerical fluxes at cell interfaces [51, 52, 53].

The key property of a finite-volume scheme is that it is conservative by construction: the flux that leaves one cell enters the neighbouring cell with the same sign, so the total amount of any conserved quantity (mass, momentum, energy) in the domain can only change through the boundary. This mirrors what the continuous equations guarantee, and it means that the scheme remains physically meaningful even when discontinuities are present.

2.2.3 How is the Numerical Scheme Implemented?

The code solves the Euler equations on a uniform Cartesian grid using a dimensionally split MUSCL scheme with a Rusanov (local Lax–Friedrichs) flux. This is a standard finite-volume approach for compressible flows [51]. The full multi-dimensional update is broken into a sequence of one-dimensional sweeps, one per spatial axis, each of which updates all conserved variables: ρ , $\rho\mathbf{v}$, and E along that direction.

Each one-dimensional sweep proceeds in three steps.

Step 1: Reconstruction (MUSCL). The scheme stores a cell-averaged value per conserved variable per cell. To achieve better than first-order accuracy, a piecewise-linear reconstruction is performed within each cell before computing the flux. The slope of the linear profile inside the cell i is estimated using the minmod limiter,

$$\sigma_i = \text{minmod}(U_i - U_{i-1}, U_{i+1} - U_i), \quad \text{minmod}(a, b) = \frac{1}{2}(\text{sgn } a + \text{sgn } b) \min(|a|, |b|), \quad (2.41)$$

where U_i stands for any conserved variable. The minmod function returns zero if the two finite differences have opposite signs (i.e. a local extremum), which prevents the reconstruction from introducing new extrema and prevents the scheme from generating spurious oscillations near discontinuities. From this slope the left and right interface states at the boundary between cells i and $i + 1$ are

$$U_{i+1/2}^L = U_i + \frac{1}{2}\sigma_i, \quad U_{i+1/2}^R = U_{i+1} - \frac{1}{2}\sigma_{i+1}. \quad (2.42)$$

This method was first introduced in [54].

Step 2: Flux Computation (Rusanov Solver). Given the reconstructed left and right states at each interface, the intercell flux is computed with the Rusanov (or local Lax–Friedrichs) approximate Riemann solver. For any conserved variable U with associated physical flux $F(U)$, the numerical flux at the interface $i + 1/2$ is

$$\hat{F}_{i+1/2} = \frac{1}{2} (F(U_{i+1/2}^L) + F(U_{i+1/2}^R)) - \frac{1}{2} s_{\max} (U_{i+1/2}^R - U_{i+1/2}^L), \quad (2.43)$$

where the maximum signal speed is estimated locally as

$$s_{\max} = \max(|v_k^L| + c_s^L, |v_k^R| + c_s^R). \quad (2.44)$$

Here, v_k is the velocity component along the sweep direction and $c_s = \sqrt{\gamma P/\rho}$ is the local sound speed. The first term in equation (2.43) is the arithmetic average of the two physical fluxes (which would be exact if there were no discontinuity), and the second term is a diffusive correction proportional to the jump in the conserved variable that stabilises the scheme across shocks and contact discontinuities [51, 55].

The physical fluxes used in the code are directly those appearing in the Euler equations. For a sweep along the axis k , the mass flux is $F^\rho = \rho v_k$, the momentum flux in the direction i is $F^{m_i} = \rho v_i v_k + P \delta_{ik}$, and the energy flux is $F^E = (E + P) v_k$, with all quantities evaluated from the reconstructed interface states [55].

Step 3: Conservative Update. Once the fluxes at all interfaces are known, the cell-averaged values are updated as

$$U_i^{n+1} = U_i^n - \frac{\Delta t}{\Delta x} (\hat{F}_{i+1/2} - \hat{F}_{i-1/2}), \quad (2.45)$$

which is the standard finite-volume update formula. Because each flux appears with opposite signs in the two adjacent cells it couples, mass, momentum, and energy are conserved across the grid up to machine precision. After each full-dimensional update, a density and internal energy floor is applied to prevent unphysical negative values in low-density regions.

This computational paradigm was inspired by [53]

2.2.4 Coupling to Gravity: Strang Splitting

The gravitational source terms on the right-hand sides of equations (2.38) and (2.39) are handled separately from the hydrodynamic fluxes by using operator splitting [56]. In practice, this means that each subcycle is structured as a Strang-split sequence:

1. **Half gravity kick:** update all velocity components due to gravitational acceleration for $\Delta t/2$.
2. **Hydrodynamic sweep:** perform the complete MUSCL–Rusanov finite-volume update across all dimensions for Δt .
3. **Half gravity kick:** apply another half-step gravity kick for $\Delta t/2$.

The gravitational acceleration is obtained from the same potential Φ that drives the ULDM and all other matter components, ensuring that the gas responds to self-consistently updated gravity at every step. Strang splitting is second-order accurate in time and is the standard approach to incorporate source terms into hyperbolic solvers [56, 53].

2.2.5 Optional Cooling

The code includes a simple exponential cooling model for the internal energy. When a cooling time scale t_{cool} is specified, the internal energy density e_{int} is reduced at the end of each substep according to

$$e_{\text{int}}^{\text{new}} = e_{\text{int}} \exp\left(-\frac{\Delta t}{t_{\text{cool}}}\right), \quad (2.46)$$

while the kinetic energy is left unchanged. The total energy radiated away is tracked cumulatively for diagnostic purposes. This cooling prescription models the radiative losses without the need for detailed atomic physics and is mainly useful for exploring how energy dissipation affects the coupled ULDM–gas dynamics when a more physical cooling function is not required.

2.2.6 Time Step Constraint and CFL Condition

The stability of the gas solver is constrained by the Courant–Friedrichs–Lewy (CFL) condition. The physical meaning of this requirement is that the information carried by the fastest characteristic waves of the system should not travel farther than approximately one grid cell during a single numerical time step. For the Euler equations, the relevant signal speed is given by the sum of the local fluid velocity and the sound speed [53].

In the present implementation, the hydrodynamic time step is therefore limited by

$$\Delta t_{\text{hydro}} = C_{\text{CFL}} \frac{\Delta x}{v_{\text{max}} + c_{s,\text{max}}}, \quad (2.47)$$

where Δx is the grid spacing, v_{max} is the maximum gas velocity on the grid, $c_{s,\text{max}}$ is the maximum sound speed, and $C_{\text{CFL}} < 1$ is a so called safety factor. For an ideal gas, the sound speed is given by its adiabatic index γ , pressure P and density ρ :

$$c_s = \sqrt{\frac{\gamma P}{\rho}}. \quad (2.48)$$

This condition ensures that the update remains stable even in the presence of shocks or steep gradients. If the global simulation time step Δt is larger than Δt_{hydro} , the gas solver is split into sub-steps, meaning that the gas evolves through

$$n_{\text{sub}} = \left\lceil \frac{\Delta t}{\Delta t_{\text{hydro}}} \right\rceil \quad (2.49)$$

smaller hydrodynamic steps of size $\Delta t/n_{\text{sub}}$ within one global step.

2.3 N-body Stellar Component

In addition to the dark matter field and the gas component, the simulation framework includes a collisionless N-body component intended to represent stars. Unlike gas, stars do not experience pressure forces or shocks; they interact with the rest of the system only through gravity. This makes a collisionless particle description appropriate, since on the scales relevant for our simulations, stellar dynamics are dominated by the mean gravitational field of the host system rather than by two-body encounters [57, 58].

2.3.1 Particle Representation

Each star is represented as a collisionless particle whose mass is distributed onto the grid with position $\mathbf{x}_i$ and velocity $\mathbf{v}_i$. All particles share the same mass $m_\star = M_{\text{tot}}/N_\star$, where M_{tot} is the total stellar mass and $N_\star$ is the number of particles. In the simulations of our interest the particles are initialised with a distribution drawn from the so - called Hernquist profile [59], a spherically symmetric density distribution given by

$$\rho_{\text{H}}(r) = \frac{M_\star}{2\pi} \frac{a}{r} \frac{1}{(r+a)^3}, \quad (2.50)$$

where a is the scale radius and $M_\star$ is the total mass. This profile was originally introduced as a computationally convenient analytic model for elliptical galaxies and bulges. Its cumulative mass function,

$$M(r) = M_\star \frac{r^2}{(r+a)^2}, \quad (2.51)$$

can be analytically inverted, which allows the positions of the particles to be drawn directly from it using the inverse transform sampling without rejection steps [59].

2.3.2 Mass Deposition onto the Grid

The simulation framework is formulated on a uniform grid with a fixed spacing between neighbouring grid points. This representation is well suited for solving the Schrödinger–Poisson system for ULDM, as well as the compressible Euler equations for the gas component. However, it introduces an additional step when collisionless N -body particles are included, since particle positions are continuous and, in general, do not coincide with grid nodes. Their mass must therefore be mapped onto the mesh in order to contribute to the density field used in the Poisson solver.

In particle–mesh methods, several mass-assignment schemes are possible. The simplest is the nearest-grid point (NGP) scheme, in which the full particle mass is assigned to the single closest grid point. This approach is straightforward to implement and computationally cheap, but produces a relatively noisy density field and is generally too inaccurate for the present application. At the other extreme, one may use higher-order schemes such as the triangular-shaped-cloud (TSC) method, in which the particle mass is distributed over a wider stencil with smoother weights. Such schemes reduce discretization noise even further, but they are more complicated to implement and increase the numerical overhead. In this work, a compromise between these two extremes is provided by the Cloud-in-Cell (CIC) scheme, where each particle is treated as a finite-sized cloud occupying one grid cell, and its mass is distributed to the eight surrounding grid vertices of the cell that contains the particle. The corresponding weights are determined by trilinear interpolation, i.e. by the particle’s relative distance from the neighbouring grid points. Compared to the nearest-grid-point assignment, this produces a smoother density field and reduces discretization noise. Moreover, when the same CIC weighting is used consistently both for depositing mass onto the grid and for interpolating the gravitational field back to particle positions, spurious self-forces are avoided [60, 61].

Concretely, if a particle lies in fractional grid coordinates (t_x, t_y, t_z) within its host cell, the weight assigned to the corner $(i + \delta_x, j + \delta_y, k + \delta_z)$, where $\delta \in \{0, 1\}$, is

$$w = W_x^{\delta_x} W_y^{\delta_y} W_z^{\delta_z}, \quad W^\delta = \begin{cases} 1 - t & \delta = 0 \\ t & \delta = 1 \end{cases}$$

and the density contribution to each cell is $m_\star w / \Delta V$, where ΔV is the volume of the cell [60, 62].

The total particle density deposited at a grid node $\mathbf{x}_{ijk}$ is therefore the sum over all particles p :

$$\rho_{\text{part}}(\mathbf{x}_{ijk}) = \sum_p \frac{m_p}{\Delta V} w_{ijk}^{(p)},$$

where m_p is the mass of the particle p .

2.3.3 Force Interpolation

Once the total density has been assembled, Poisson's equation is solved on the grid to obtain the total gravitational potential Φ . The gravitational field is obtained from the gradient of the potential,

$$\mathbf{g}(\mathbf{x}) = -\nabla\Phi(\mathbf{x}).$$

In the code, this gradient is computed spectrally. Taking the gradient in Fourier space gives

$$\tilde{g}_\alpha(\mathbf{k}) = -ik_\alpha \tilde{\Phi}(\mathbf{k}),$$

for each Cartesian component $\alpha \in \{x, y, z\}$, which is then transformed back to real space. The force experienced by each particle is subsequently obtained by interpolating these three force grids back to the particle position using the same CIC weights as in the deposition step. Using the same assignment function for both deposition and interpolation is essential: it ensures that the force on a particle is the exact gradient of the potential energy that the particle contributes to the grid, which in turn guarantees Newton's third law on the mesh and prevents the introduction of self-forces [60]. This gives the particle acceleration

$$\mathbf{a}_p = \mathbf{g}(\mathbf{x}_p),$$

which is simply the gravitational force per unit mass.

2.3.4 Time Integration

The equations of motion for each particle are integrated using a Drift-Kick-Drift (DKD) leapfrog scheme [63]. Within each global time step, the integrator first advances positions by half a step (drift), then updates velocities using the forces evaluated at the new positions (kick), and finally advances positions by another half step (drift). This can be written for one particle and one sub-step Δt_{sub} as follows:

$$\mathbf{x}_p^{n+1/2} = \mathbf{x}_p^n + \frac{\Delta t_{\text{sub}}}{2} \mathbf{v}_p^n, \quad (2.52)$$

$$\mathbf{v}_p^{n+1} = \mathbf{v}_p^n + \Delta t_{\text{sub}} \mathbf{a}_p, \quad (2.53)$$

$$\mathbf{x}_p^{n+1} = \mathbf{x}_p^{n+1/2} + \frac{\Delta t_{\text{sub}}}{2} \mathbf{v}_p^{n+1}. \quad (2.54)$$

The leapfrog scheme was adopted for its stability in long running simulations [62].

To maintain numerical stability, the time step for the stellar particles is adaptively subdivided based on two criteria: a velocity criterion $\Delta t_v = f_v \Delta x / v_{\text{max}}$ and an acceleration criterion $\Delta t_a = f_a \sqrt{\Delta x / a_{\text{max}}}$, where Δx is the cell size, v_{max} and a_{max} are the maximum particle velocity and acceleration, and f_v, f_a are safety prefactors. The number of substeps within one global step is chosen as the smallest integer such that both criteria are satisfied [62, 60].

2.4 Sink Particles as Black Hole Proxies

Directly simulating a black hole within our framework is not feasible simply because the simulations cover regions of the order of tens of kiloparsecs, whereas even the largest supermassive black holes (SMBHs) have Schwarzschild radii many orders of magnitude smaller. These scales are therefore completely unresolved on our grid. The standard way to handle this is to represent the black hole as a sink particle, i.e. a massive point-like particle that moves under gravity and removes surrounding matter once it falls within a prescribed capture radius r_{acc} [64].

The dark matter field itself is not accreted by the sink. Because of the quantum pressure of the scalar field, ULDM is stabilised against gravitational collapse below roughly the de Broglie scale, and is therefore not expected to accrete onto the black hole in the same way as ordinary baryonic matter, at least on the resolved scales of the simulations [19].

Within a simulation, one or more sink particles are created with prescribed positions, velocities, and masses. They are evolved using the same CIC deposition, leapfrog integration, and force interpolation scheme as the stellar particles described in Subsection 2.3.2, but they are excluded from the density field entering the Poisson solver. Instead, the gravitational potential of the sinks is treated separately, in order to avoid the numerical inaccuracies that would arise from a very massive point-like object being deposited directly onto the grid, where it would effectively appear as a delta-like peak in the density field.

2.4.1 Accretion Criteria

The only additional physics relative to the stellar particles is accretion into the sink particles. At each time step, any stellar particle or gas cell that simultaneously satisfies three prescribed criteria is removed from the simulation, and its mass and momentum are added to the sink particle. The three criteria are

$$r < r_{\text{acc}}, \quad v_{\text{rel}} < v_{\text{esc}} = \sqrt{\frac{2GM_{\text{BH}}}{r}}, \quad |\mathbf{L}|^2 < GM_{\text{BH}} r_{\text{acc}}, \quad (2.55)$$

requiring that the particle distance from the sink r is smaller than the capture radius r_{acc} , its velocity v_{rel} relative to the sink is smaller than the escape velocity v_{esc} of the sink, and carries angular momentum low enough L that it could not sustain a stable orbit. These criteria follow the original formulation of [64] and ensure that only genuinely infalling matter is accreted, avoiding spurious capture of particles on hyperbolic or wide-orbit trajectories. The sink mass and velocity are updated by conservation of momentum after each accretion event, and the accreted matter is removed from the simulation entirely, which is why these objects are called sinks.

Gas accretion is treated separately because the gas is not represented by individual particles but by cell-averaged hydrodynamic fields. For an existing sink, the local gas density is sampled from the neighbouring grid cells using the same CIC weights as in the particle–mesh coupling. The accreted gas mass is then estimated by using a Bondi-like prescription.

2.4.2 Accretion Model

The rate at which gas is accreted onto the sink follows the Bondi–Hoyle–Lyttleton (BHL) formula [65], which describes the steady-state spherical infall of a uniform gas onto a point mass. For a black hole of mass M_{BH} embedded in gas of density ρ_∞ and sound speed c_s , the accretion rate is

$$\dot{M}_{\text{Bondi}} = \frac{4\pi\lambda G^2 M_{\text{BH}}^2 \rho_\infty}{c_s^3}, \quad (2.56)$$

where $\lambda \approx e^{3/2}/4 \approx 1.12$ is a dimensionless factor that depends on the adiabatic index of the gas [65]. This formula is the standard sub-grid prescription for black hole accretion in galaxy-scale simulations [66], and it is adopted here for the same reason: the actual accretion flow occurs far below the resolution of our grid, and the Bondi rate provides a physically motivated estimate of the mass supply to the black hole based solely on the resolved large-scale gas properties.

In practice, the gas density is sampled at the sink position using the same CIC interpolation scheme as described in Subsection 2.3.2. The mass removed from the surrounding grid cells per time step is then $\Delta M = \dot{M}_{\text{Bondi}} \Delta t$, distributed among the eight neighbouring cells with their CIC weights, so that only cells above a density floor contribute and the removal is evenly distributed over the local gas distribution.

Rather than transferring the accreted mass directly to the black hole, the code uses a two-stage reservoir model inspired by the Jaxion framework [50]. The mass removed from the gas grid is first deposited in a reservoir M_{res} associated with each sink, and then drained into the actual black hole mass M_{BH} on an exponential timescale τ :

$$\frac{dM_{\text{BH}}}{dt} = \frac{M_{\text{res}}}{\tau}, \quad (2.57)$$

which in discrete form gives $\Delta M_{\text{BH}} = M_{\text{res}}(1 - e^{-\Delta t/\tau})$ per step. The total mass seen by the gravitational solver at any time is $M_{\text{sink}} = M_{\text{BH}} + M_{\text{res}}$. The physical motivation for this two-stage approach is that the gas falling through the capture radius does not reach the event horizon instantaneously. It first loses angular momentum and radiates energy away in an accretion disc. The reservoir mimics this delay, preventing the black hole mass from jumping discontinuously at each accretion event and providing a smoother coupling between the sink and the surrounding gas dynamics. This approach also greatly improves the energy conservation throughout the simulation run.

The sink particle should only be understood as an effective way of modelling unresolved accretion below the resolution limit of the simulation. It is not meant to describe the detailed physics near the event horizon, but only the large-scale effect of a growing central compact object [66].

2.5 Astrophysical Scaling Estimates for Composite Systems

So far, this chapter has introduced the individual components of the simulation framework. I now combine them into a simplified galaxy-like central system consisting of (i) an ULDM soliton evolved with the Schrödinger–Poisson solver, (ii) a central sink particle representing an unresolved compact object, and (iii) baryons represented by a collisionless Hernquist component and a gas component. The aim of this setup is not to reproduce a fully self-consistent Milky-Way galaxy, but to construct a controlled multi-component configuration in which the mutual gravitational coupling of all implemented components can be studied.

2.5.1 Initial Conditions

The model which will be studied in Section 5.5 is initialised as a simplified galaxy-like system consisting of a gas component distributed as an exponential disc and a stellar component distributed according to a Hernquist profile, Equation 2.50, a central solitonic ULDM core obtained by solving the static Schrödinger–Poisson system shown in Equation 2.15, and a supermassive black hole proxy placed at the centre of the system. The relations discussed below provide an order-of-magnitude astrophysical reference for choosing component masses. However, in simulations, the final masses used in Section 5.5 will be adjusted to obtain a numerically stable and well-resolved composite system.

Central SMBH Mass

A useful reference point is provided by the SMBH mass inferred from measurements of stellar orbits around Sagittarius A*,

$$M_{\bullet} \simeq 4 \times 10^6 M_{\odot}. \quad (2.58)$$

Following, e.g. the analyses in [67, 68], this value can be used as an astrophysical reference scale for estimating the masses of the other components.

Halo Mass Estimate from the SMBH Mass

To obtain an order-of-magnitude estimate of the host halo mass, we use an empirical SMBH–halo (total gravitating mass) correlation (originally calibrated using strong lensing constraints).

$$\log_{10} \left(\frac{M_{\bullet}}{M_{\odot}} \right) = (8.18 \pm 0.11) + (1.55 \pm 0.31) \left[\log_{10} \left(\frac{M_{\text{halo}}}{M_{\odot}} \right) - 13 \right], \quad (2.59)$$

from Bandara et al. [69]. Inverting Eq. (2.59) for $M_{\bullet} \approx 4 \times 10^6 M_{\odot}$ yields $M_{\text{halo}} \sim 10^{12} M_{\odot}$, i.e. a Milky-Way-like halo mass scale.

Soliton Mass

For ULDM models, the soliton mass is expected to scale with the host halo mass and with the ULDM particle mass m_s . Following Equation 2.27 we can estimate the soliton mass for the halo mass scale considered here. The corresponding soliton profile is obtained by solving the stationary Schrödinger–Poisson problem, shown in Equation 2.15, with the SMBH included as a point-mass perturbation [49]. In the code, this profile is generated in tabulated form and then rescaled to the desired physical soliton mass using Equation 2.23.

Baryon Budget vs. Simulated Baryons

A cosmological upper bound on the total baryonic mass associated with a halo is given by the cosmic baryon fraction.

$$f_b \equiv \frac{\Omega_b}{\Omega_m} \approx 0.156, \quad (2.60)$$

taken from the Planck 2020 cosmological parameters [12]. For a halo of mass $M_{\text{halo}} \sim 10^{12} M_{\odot}$, this would correspond to a maximum baryon budget of

$$M_{b,\text{max}} \sim f_b M_{\text{halo}} \sim 1.6 \times 10^{11} M_{\odot}. \quad (2.61)$$

However, this value should be understood only as a global upper limit. A substantial fraction of baryons is expected to reside in the warm-hot circumgalactic medium extending to large radii [70], while the present framework only follows the resolved central region and does not model a multi-phase CGM.

For this reason, the baryonic mass used in the composite simulations is not taken as $f_b M_{\text{halo}}$. Instead, the cosmic baryon fraction is used only as a reference value to choose a local baryonic loading relative to the resolved ULDM soliton. In the simulations presented in Section 5.5, therefore, I use

$$M_{\text{bar}} = 0.15 M_{\text{sol}}, \quad (2.62)$$

which keeps the total baryonic mass close to the cosmological baryon fraction in relative terms, but applies it only to the resolved central composite system rather than to the full host halo.

The baryonic mass is then split between a collision-less Hernquist stellar component and a gaseous disk,

$$M_{\text{H}} = 0.6 M_{\text{bar}}, \quad M_{\text{gas}} = 0.4 M_{\text{bar}}. \quad (2.63)$$

This split is not intended to reproduce the detailed stellar-to-gas mass ratio of the present-day Milky Way. It is chosen so that both baryonic components remain dynamically visible in the simulations, while the total baryonic contribution remains small enough not to dominate the gravitational potential of the resolved ULDM soliton.

Chapter 3

Code Implementation

As the physical framework of the model was introduced in Chapter 2, the aim of this chapter is to describe its numerical realisation. The simulation evolves a coupled ULDM–gas–N-body–sink system on a Cartesian grid, using an FFT-based Poisson solver for the gravitational potential and operator-splitting methods for the time evolution of the individual components. The implementation is built as a modular object-oriented framework in Python, which makes it possible to treat the different physical sectors separately within the code, as can be seen in Figure 3.1, while preserving their mutual coupling through the gravitational field.

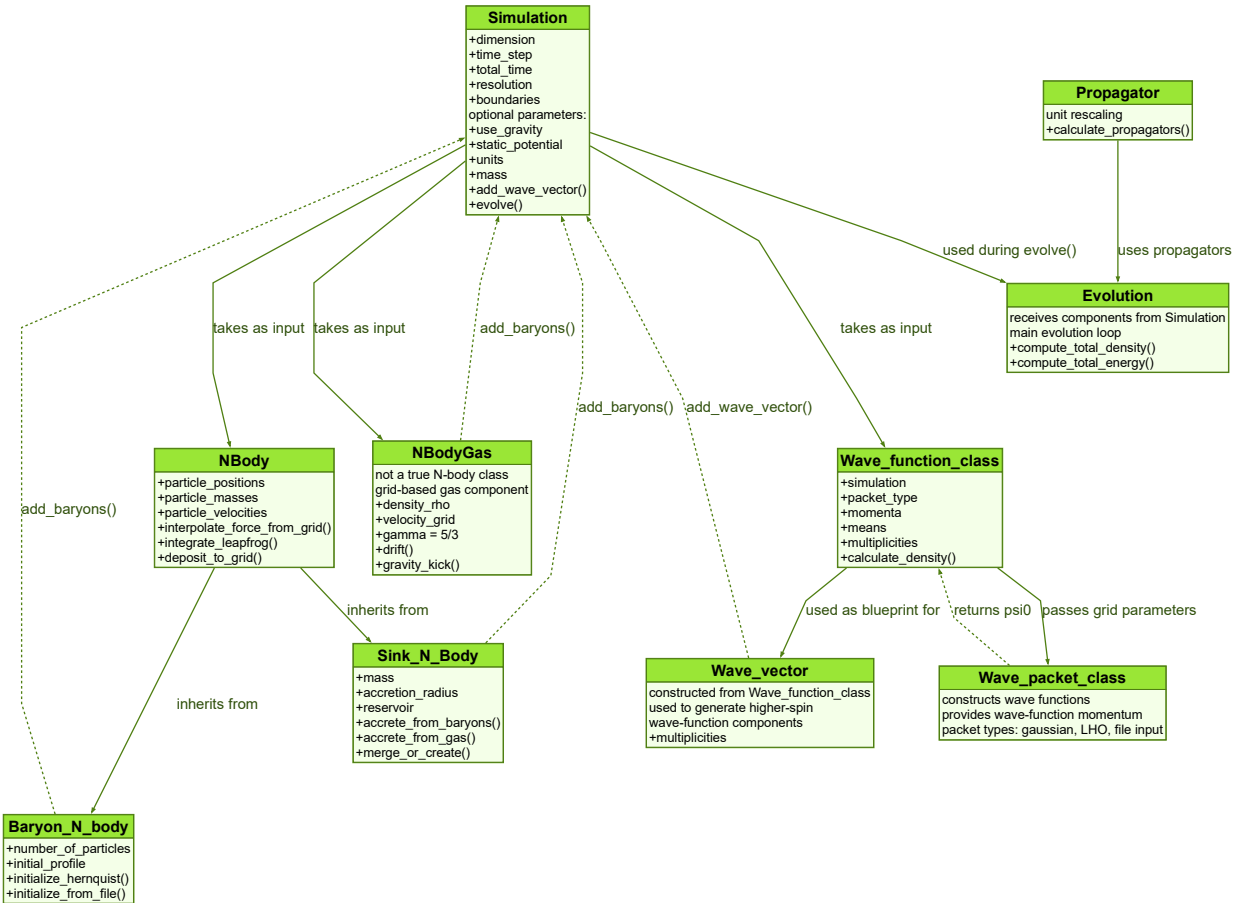


Figure 3.1: A schematic structure of all the components that make up the simulation framework and how they communicate.

Python was chosen mainly because of its simplicity, broad ecosystem of scientific libraries, and the fact that it is the only programming language I know. This project heavily uses libraries

that aid with numerical work, data handling, and visualisation, namely NumPy [71], SciPy [72], Astropy [73] and Matplotlib [74].

Most importantly, the implementation relies heavily on the CuPy library [75], which enables GPU-accelerated execution of large array operations and Fourier transforms. CuPy provides a NumPy-compatible interface for CUDA programming, which makes it possible to transfer array-based computations to the GPU with only relatively small changes to the code. This is particularly useful for the present framework.

While CPUs excel at sequential processing with typically 4-16 cores, modern GPUs contain thousands of smaller cores optimised for parallel computation. The Fourier split-step method involves operations on large arrays and Fast Fourier Transforms (FFTs), both of which are highly parallelizable operations that benefit enormously from GPU architecture. Basically, instead of trying to solve many simple tasks with "smart" CPU cores, I was advised to attack the tasks with an armada of thousands (5120) of "dumb" GPU cores.

For this project, development was carried out primarily using an NVIDIA GeForce RTX 3070 laptop GPU with 8 GB of VRAM, which provided sufficient memory for smaller simulation grids. Additional options for running and testing the simulation framework included an NVIDIA Quadro P4000 with 1792 CUDA cores, acquired from Facebook Marketplace, and, most importantly, access to Phoebe computational cluster at FZU in the Czech Republic, which uses The NVIDIA A100 80GB PCIe GPU. Comparison of their respective computational abilities will be shown in Section 4.6.

3.1 Discretization and computational representation

The simulation framework is built on a uniform Cartesian grid, which provides the common numerical domain for all grid-based components of the system. In a simulation of dimension d , the computational domain is defined by a set of boundary intervals

$$[x_{\min}^{(1)}, x_{\max}^{(1)}] \times [x_{\min}^{(2)}, x_{\max}^{(2)}] \times \cdots \times [x_{\min}^{(d)}, x_{\max}^{(d)}], \quad (3.1)$$

which determines the physical size of the box in each spatial direction. Each axis is then discretised into N uniformly spaced points, so that the grid spacing in the i -th direction is

$$\Delta x_i = \frac{x_{\max}^{(i)} - x_{\min}^{(i)}}{N - 1}. \quad (3.2)$$

In the present implementation, the same number of grid points is used along each axis, to simplify the Fourier treatment of spatial derivatives used to resolve the SP system of equations shown in Equation 1.10.

Once the one-dimensional coordinate arrays are constructed, the full multidimensional spatial grid is generated using the `meshgrid` method. This procedure combines the coordinate values along each axis into arrays of identical shape such that each element of the resulting tensors corresponds to one point in the computational domain. In this way, scalar fields such as the ULDM wave function, gas density, pressure, or gravitational potential can be stored directly as multidimensional arrays defined over the entire simulation volume. This construction is schematically illustrated in Figure 3.2.

This representation is particularly convenient for vectorised GPU computation, since point-wise algebraic operations can then be applied simultaneously over the entire grid.

The same spatial discretization also defines the resolution of the simulation. Increasing N decreases the grid spacing and therefore improves the ability of the code to resolve steep density gradients, compact solitonic cores, and localised gravitational structures. At the same time, the

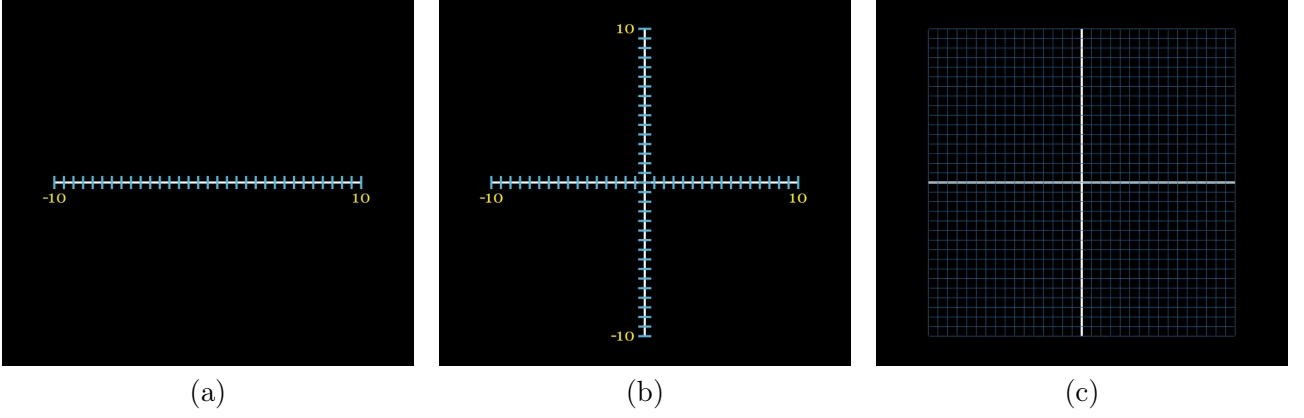


Figure 3.2: Schematic illustration of the construction of the computational grid. (a) A single spatial direction is discretized into uniformly spaced points. (b) The procedure is extended to multiple coordinate directions. (c) The resulting coordinate arrays are combined using the `meshgrid` method to form the Cartesian simulation domain on which grid-based fields are stored and evolved.

total number of grid cells scales as N^d , so the computational cost and memory requirements grow rapidly with increasing resolution. The choice of N therefore represents a compromise between numerical precision and computational feasibility.

In addition to the real-space grid, the framework constructs the corresponding Fourier-space grid, which is required for the split-step evolution of the Schrödinger equation and for the FFT-based solution of Poisson’s equation, as it is currently implemented. For each spatial direction, a discrete set of wave numbers is defined, and the full d -dimensional k -space grid is again assembled using the same `meshgrid` construction. Spatial derivatives are then evaluated spectrally by replacing

$$\partial_i \longrightarrow ik_i, \quad (3.3)$$

and the Laplacian by

$$\nabla^2 \longrightarrow -k^2 = -\sum_{i=1}^d k_i^2. \quad (3.4)$$

This makes the Fourier representation a natural computational setting both for the kinetic part of the Schrödinger evolution and for the Poisson solver.

Because the framework relies on the Fast Fourier Transform, periodic boundary conditions are assumed throughout the simulation domain. Numerically, this means that opposite faces of the computational box are identified, so that fields leaving one side of the domain re-enter from the other. This assumption is particularly convenient for spectral methods, since it allows Fourier modes to form a complete basis for the numerical representation of the evolved fields. At the same time, it represents a physical limitation, since any unbound stars or any other matter within the system that leaves the simulation box will simply reappear on the other side. Nevertheless, this non physical effect of unbound mass being unable to escape the gravitational pull causes only negligible errors and periodic boundaries are convenient for cosmological simulations, to mimic the vast scale of the universe.

The creation of the simulation grid and its adjunct Fourier space is handled by the `Simulation_class`, which acts as the main container of the simulation. It stores the dimension of the problem, the simulation boundaries, the grid resolution, the total evolution time, the global time step, and the chosen order of the evolution scheme. It also keeps track of the active physical components together with global settings such as self-gravity, optional static potentials, and the adopted unit system. This makes it the central object through which the different parts of the framework are coupled within one common computational domain.

3.2 Component representation

Within this simulation grid, individual physical components are represented in different ways according to their nature.

The ULDM component is represented directly on the grid by a complex wave function $\psi(\mathbf{x}, t)$, or by several of such components in the case of the higher-spin related by using the mechanism outlined in Subsection 2.1.5. Its modulus determines the density field, while its phase carries velocity information through the Madelung interpretation shown in Equation 1.14.

The gas component is also represented on the same Cartesian grid, but in terms of cell-averaged hydrodynamic variables, as was explained in Section 2.2. These include the gas density, the momentum density, and the total energy density. In contrast to the ULDM field, which evolved as a complex wave function, the gas evolved as a set of conserved real-valued fields using a finite-volume method. Storing both ULDM and gas on the same mesh simplifies their coupling through the gravitational potential.

The stellar component is represented differently. Since stars are treated as collisionless bodies, they are modelled as discrete N-body particles with continuous positions and velocities rather than as fields defined on the grid. Their masses are deposited onto the mesh only when constructing the total density entering the Poisson equation, and the resulting gravitational forces are then interpolated back from the grid to the particle positions. The deposition of stellar masses onto the grid was discussed in Subsection 2.3.2.

Sink particles, which serve as an approximation for a central compact object, in our case a SMBH, are represented in the same particle-based way. Each sink carries its own position, velocity, and mass and interacts gravitationally with the rest of the system. As in the stellar case, sink particles do not evolve as grid fields, but as discrete objects coupled to the mesh through CIC deposition and interpolation procedures.

All components require an instance of `Simulation_class` to be passed into them during initialisation so that they can inherit the parameters needed for their setup, such as the simulation dimensionality and the grid size and resolution. After the creation, the user passes the components to the `Simulation_class` instance using the methods `add_wave_vector` for ULDM wave vectors and `add_baryons` for any baryonic component.

3.2.1 The ULDM Sector

The ULDM sector is represented in the code through a hierarchy of three objects.

The Packet Class

is used for the creation of the initial profile. The `packet` class object creates an array filled with numerical values in response to the Cartesian grid associated with the simulation. Depending on the chosen initialisation, this profile may be a Gaussian packet, a harmonic-oscillator state, or a solitonic profile imported from tabulated spherical data, obtained by solving the Equation 2.15 and interpolated onto the simulation grid.

The Wave_Function_class

Stores this grid-based profile representing one complex scalar field component $\psi(\mathbf{x}, t)$. In addition to storing the field itself, this class is responsible for operations directly associated with a single component, most importantly the conversion to physical units and the computation of the density field from the absolute square of the wave function. When reduced higher-spin

representations are used, the corresponding multiplicity factor is included at this stage to recover the correct total density contribution.

The Wave_vector_class

To finalise the part concerned with soliton creation within the simulation framework, it is necessary to discuss the `wave_vector_class` in more detail. This is the class with which the user interacts when creating solitons for the simulation. During its initialisation, the received attributes are passed into a wave-function creator, which uses the `Packet` class to construct a `Wave_function` instance saved as `wave_blueprint`. This blueprint is then used to generate all components.

Construction of the wave vector relies on a recursive helper function named `build`, which iterates over all possible sequences of spin projections $(m_1, m_2, \dots, m_s)$ that satisfy the constraint

$$m_1 + m_2 + \dots + m_s = m$$

for a given total magnetic quantum number $m \in \{-s, \dots, +s\}$.

Each valid sequence corresponds to a tensor product

$$\mathbf{e}_{m_1} \otimes \mathbf{e}_{m_2} \otimes \dots \otimes \mathbf{e}_{m_s},$$

which contributes to the construction of the spin-dependent tensor associated with the selected projection m . This mirrors the angular-momentum-inspired procedure introduced in Subsection 2.1.5, where the higher-spin state is built from products of the spin-1 polarization basis vectors $\mathbf{e}_m$.

1. For each valid tuple of spin projections `ms = (m1, m2, ..., ms)`, which represents one specific combination of single-spin basis states entering the tensor product, the corresponding basis vectors are gathered by selecting from the full set of available spin-1 eigenvectors `e_m`:

```
vecs = [e_m[mi] for mi in ms]
```

This selects and orders only the basis vectors relevant to that particular combination, rather than using all available eigenvectors. **For example**, for $m_s = (-1, 1)$ in the **spin-2 construction**, we obtain

$$\text{vecs} = \left[\frac{1}{\sqrt{2}}(1, -i, 0), \quad \frac{1}{\sqrt{2}}(-1, -i, 0) \right].$$

2. The tensor product is initialized with the first vector:

```
t = vecs[0]
```

3. The remaining vectors are combined iteratively using `np.tensordot` with `axes=0`, which computes the outer product:

```
for v in vecs[1:]:
    t = np.tensordot(t, v, axes=0)
```

The shape of the tensor grows with each multiplication:

- $(3,) \otimes (3,)$ yields $(3,3)$,
- $(3,3) \otimes (3,)$ yields $(3,3,3)$, and so on.

Again, for example:

- $\text{ms} = (-1, 1)$ (spin-2): $t = \mathbf{e}_{-1} \otimes \mathbf{e}_{+1}$, shape $(3,3)$,
- $\text{ms} = (-1, 0, +1)$ (spin-3): $t = \mathbf{e}_{-1} \otimes \mathbf{e}_0 \otimes \mathbf{e}_{+1}$, shape $(3,3,3)$.

4. The resulting tensor is added to an array:

```
tensor[:,] += t
```

This step sums over all valid combinations for a given magnetic quantum number m , producing the tensor structure associated with that projection in the implemented construction.

For example, for spin-2 and $m = 0$, the following combinations contribute:

- $(-1, +1): \mathbf{e}_{-1} \otimes \mathbf{e}_{+1}$,
- $(+1, -1): \mathbf{e}_{+1} \otimes \mathbf{e}_{-1}$,
- $(0, 0): \mathbf{e}_0 \otimes \mathbf{e}_0$.

giving

$$\text{tensor} = \mathbf{e}_{-1} \otimes \mathbf{e}_{+1} + \mathbf{e}_{+1} \otimes \mathbf{e}_{-1} + \mathbf{e}_0 \otimes \mathbf{e}_0.$$

After this sum is constructed, the tensor is normalised numerically. Repeating the procedure across all $m \in \{-s, \dots, +s\}$ produces the complete set of $2s + 1$ tensorial basis objects used by the code.

Once the full spin tensor is constructed, the method `_create_combined_wave_function` selects the individual tensor components corresponding to specific spatial directions. This is done using combinations with replacement of `range(3)` taken `spin` times, which produces all unique symmetric index combinations, e.g.:

- Spin-1: $(0), (1), (2)$,
- Spin-2: $(0,0), (0,1), (1,1)$, etc.

The function `get_index_multiplicities` computes the number of equivalent arrangements. This is used to reduce the necessary number of wave functions stored in memory by exploiting the symmetry discussed in Subsection 2.1.5. The corresponding wave function is then normalised by dividing by the square root of this multiplicity:

$$\psi \mapsto \frac{\psi}{\sqrt{N}}.$$

This normalisation accounts for the multiplicity of equivalent symmetric tensor components in the reduced representation. Using this method to create a wave vector leaves us with an object that contains a list of all the components necessary to represent a spin soliton s . During the creation of the wave vector in the method `_create_combined_wave_function` the momentum is applied to all wave functions using the `Packet.momentum_propagator`, which is computed from the input momentum and applied to the wave values as

$$e^{i\frac{\mathbf{p}\cdot\mathbf{x}}{\hbar}}\psi.$$

Storing Solitons

The simulation allows for adding multiple wave vectors, either with the same spin or different ones, to allow simulations of more complex systems. If two or more wave vectors with the same spin are added to the simulation, they are stored in the dictionary with the same key (their spin). In this case, their corresponding components are summed together (e.g., the first component of wave vector one + the first component of wave vector two) and then each of these wave functions instances is evolved as a single, combined wave function.

On the other hand, wave vectors with different spins are stored under separate keys in the dictionary, so they are not mixed together. If there are multiple same-spin ULDM components in one simulation, their components are summed, otherwise, they are treated as separate wave vectors. Then all the wave functions in the `wave_vectors` dictionary are collected in the `wave_functions` list. This list allows the simulation to handle each wave function it received individually during evolution. For example, all the first components of spin-1 wave vectors are grouped and evolved together, separately from the first component of a spin-2 wave vector.

3.2.2 The Baryonic Sector

The baryonic sector is represented in the code through three objects, each corresponding to one of the baryonic subsystems. Unfortunately, since the methods used to evolve these systems are dimension dependent, they are hardcoded for a three dimensional space.

The NBodyGas Class

The unfortunately named `NBodyGas` class is in reality not a N body system but is instead stored directly on the simulation grid. Its main fields are the gas density `rho`, the velocity components `vx`, `vy`, `vz`, and the total energy density `E`. During initialisation, these quantities are provided explicitly or constructed from the total gas mass and the simulation box parameters inherited from the `Simulation_class` instance. The class also stores auxiliary parameters needed for the hydrodynamic evolution, such as the adiabatic index, CFL factor, density floor, optional cooling time, and internal-energy floor. In this way, the `NBodyGas` object acts as a self-contained representation of the gas state on the mesh, while its numerical evolution is carried out later through its drift method and the hydrodynamic routines described in Section 3.4.

The Baryonic_N_Body Class

The collisionless stellar component is built on the `NBody` class, which serves as the generic particle-based representation used throughout the framework. This class stores the number of particles, their individual positions and velocities, and the particle mass associated with a given total stellar mass. In contrast to the gas and ULDM sectors, the particles are not stored as fields on the simulation mesh. Instead, the `NBody` class provides the numerical interface through which particle systems communicate with the grid. Most importantly, it implements the deposition of particle mass onto the mesh through the `deposit_to_grid` method, force interpolation from the grid back to particle positions, and leapfrog-type particle integration shown in more detail in Section 3.4.

The SinkNBody Class

Sink particles are represented by the `SinkNBody` class, which is implemented as a specialised subclass of `NBody`. It therefore inherits the same basic particle representation, namely positions, velocities, and coupling to the grid through CIC-type deposition and force interpolation, but extends it by additional attributes associated with accretion and mass growth. In particular, each sink stores both a black-hole mass component and a reservoir mass, whose sum defines the total gravitating sink mass. The class further contains parameters such as the capture radius, softening lengths, and the reservoir-drain timescale. Beyond ordinary particle motion, `SinkNBody` includes methods for accretion from baryonic particles and from the gas grid, for gradual transfer of mass from the reservoir to the physical sink, and for merging nearby sink particles.

3.3 Time Integration and Time-Step Control

After creating the simulation grid, the system can start to evolve. This consists of advancing through the declared `total_time` step by step. These time steps are within the simulation called `h`. The number of global steps carried out is, therefore:

$$N_{\text{steps}} = \frac{\text{total_time}}{h}. \quad (3.5)$$

Since the framework combines several components evolved by different numerical methods, a single time-step criterion is not sufficient for all of them. The global step h is chosen at the level of the complete simulation, while individual components may subdivide it internally to satisfy their own stability or accuracy requirements.

3.3.1 Split-Step Evolution of the ULDM Field.

The wave-function sector is evolved by operator splitting, where the Hamiltonian is decomposed into potential and kinetic parts. In the simplest second-order case, the evolution over one global step h follows the usual kick–drift–kick structure shown in Equation 2.11.

To improve the long-term accuracy of evolution, the framework also supports fourth- and sixth-order splitting schemes. These offer the same time advancement as the second order, it just splits the time step into more stages by composing kick and drift operators with precomputed coefficients following [76],

$$e^{-ih(\hat{T}+\hat{V})} \approx \prod_j e^{-ia_j h \hat{V}} e^{-ib_j h \hat{T}}. \quad (3.6)$$

In practice, the second-order method is the most numerically economical, while the higher-order variants provide improved energy behaviour at fixed h in runs where the accumulated splitting error would otherwise become significant, this will be shown in more detail in Section 4.5.

3.3.2 Time-Step Constraint for the Schrödinger Solver

The split-step Fourier method used to evolve ULDM wave-functions is used to determine the global simulation step `h`. This is done by requiring the time step to satisfy the following constraint:

$$h \leq \min(h_1, h_2), \quad (3.7)$$

with

$$h_1 = \frac{4}{3\pi} \frac{\Delta x_{\min}^2}{\tilde{h}}, \quad (3.8)$$

and

$$h_2 = 2\pi \tilde{\hbar} \frac{1}{|\Phi|_{\max}}. \quad (3.9)$$

Where, $\Delta x_{\min}$ denotes the smallest grid spacing used in the simulation, $\tilde{\hbar}$ is the rescaled Planck constant appearing in the dimensionless Schrödinger equation and will be mentioned more in Section 3.5, and $|\Phi|_{\max}$ is the maximal absolute value of the potential on the grid. The time step used must be smaller than this constraint for the simulation to be stable. In the code, this is enforced by requiring that the user-defined time step be at least a factor of two smaller than the value h given by Equation 3.7.

The first condition controls the kinetic part of the evolution and reflects the requirement that the phase advance associated with the drift operator remain sufficiently well resolved on the grid. The second condition limits the phase rotation generated by the potential part of the evolution. Together, these bounds ensure that the phase factors appearing in the kick and drift operators do not vary too rapidly across neighbouring grid cells or Fourier modes, which would otherwise lead to loss of accuracy. This prescription follows [22, 35].

3.3.3 Hydrodynamic and Particle Time Stepping

The paradigm for the evolution of the gas component has already been outlined in Section 2.2 together with the necessary constraint for the stability of the gas component Equation 2.47. This is computed for every step within the simulation and if the global time step is internally divided into sub-steps for the gas component to ensure proper evolution.

Particle time stepping and sub-stepping The stellar and sink particles are integrated by a leapfrog scheme, whose implementation is explained in more detail in Section 3.4. The motion of the particle components is advanced over the same global interval h , but, similarly to the gas component, the code may further subdivide this interval internally to maintain sufficient time resolution to ensure correct evolution of the system. The substepping criterion is based on the velocities and accelerations of the particles. If these become too large relative to the grid scale, two time scales are introduced,

$$\Delta t_v = f_v \frac{\Delta x}{v_{\max}}, \quad \Delta t_a = f_a \sqrt{\frac{\Delta x}{a_{\max}}}, \quad (3.10)$$

where $v_{\max}$ and $a_{\max}$ are the maximal particle velocity and acceleration in the current particle system, and f_v , f_a are safety prefactors. This prevents particles from moving over a significant fraction of a grid cell or from undergoing too large a velocity change, within a single particle time step. The number of particle sub-steps is then chosen as

$$n_{\text{sub,part}} = \max \left(\left\lceil \frac{h}{\Delta t_v} \right\rceil, \left\lceil \frac{h}{\Delta t_a} \right\rceil \right). \quad (3.11)$$

In this way, fast moving particles or particles with large accelerations can still be evolved accurately, even when the global simulation step is comparatively large.

3.4 One Global Simulation Step

This section outlines what happens during each global step of the main evolution loop. In short, this can be summarised by the following sequence of operations:

1. computation of the densities of all active components,
2. deposition of particle masses onto the grid,
3. solution of Poisson’s equation for the gravitational potential,
4. application of the potential step to the ULDM field,
5. update of the baryonic components using the gravitational forces computed from the current density field; for the gas component, this includes hydrodynamic subcycling when required by the CFL condition,
6. drift step of the ULDM field by the Fourier split-step method,
7. application of sink accretion and evaluation of diagnostics and output.

This update should be understood as a sequentially split scheme rather than as a fully time-centred global kick–drift–kick update of all components. In a more strictly time-centred formulation, the ULDM field, particles, and gas would be advanced within a common global KDK structure, with the gravitational potential recomputed between the drift and the final kick. Such a formulation would improve the consistency of the coupled evolution, but it is not the scheme implemented in the present version of the code.

3.4.1 Density Assembly

At the beginning of each step, the code constructs the total density field which is used for the gravitational solver. For the ULDM sector, this is obtained from the wave function as

$$\rho_{\text{ULDM}}(\mathbf{x}, t) = |\psi(\mathbf{x}, t)|^2, \quad (3.12)$$

or, in the higher-spin case, as the sum of the densities of all the components of the evolved wave-function, including their multiplicity factors, which are calculated and used to reduce the necessary number of components of the wave-function in the memory of the computer as outlined in Subsection 2.1.5. If gas, stellar, or sink particles are present, their densities are added to the total density:

$$\rho_{\text{tot}}(\mathbf{x}, t) = \rho_{\text{ULDM}}(\mathbf{x}, t) + \rho_{\text{gas}}(\mathbf{x}, t) + \rho_{\star}(\mathbf{x}, t) + \rho_{\text{sink}}(\mathbf{x}, t), \quad (3.13)$$

so that the gravitational potential is computed with the full matter distribution rather than only the ULDM component alone.

3.4.2 Poisson Solver

Once the total density field has been assembled, the gravitational potential is obtained by solving Poisson's equation in Fourier space. In the periodic domain used here, this is done by Fourier transforming the density contrast,

$$\tilde{\rho}(\mathbf{k}, t) = \mathcal{F}[\rho(\mathbf{x}, t) - \langle \rho \rangle], \quad (3.14)$$

and then computing

$$\tilde{\Phi}(\mathbf{k}, t) = -\frac{4\pi G}{k^2} \tilde{\rho}(\mathbf{k}, t), \quad (3.15)$$

followed by the inverse transform

$$\Phi(\mathbf{x}, t) = \mathcal{F}^{-1}[\tilde{\Phi}(\mathbf{k}, t)]. \quad (3.16)$$

As usual, the $k = 0$ mode is set to zero, which is consistent with solving for the potential generated by the density contrast in a periodic box.

3.4.3 ULDM Update

With the gravitational potential known, the ULDM field is advanced by one time step using the split-step Fourier method explained in Subsection 2.1.1. In its second-order form, the update may be written schematically as

$$\psi_c^{(1)} = \exp\left(-i\frac{m}{\hbar}\frac{1}{a(t)}\frac{\Delta t}{2}\Phi_c(t, x)\right)\psi_c(t, x) \quad (3.17)$$

$$\tilde{\psi}_c^{(2)} = \exp\left(-i\frac{\hbar}{m}\frac{1}{a(t)^2}\frac{\Delta t}{2}k^2\right)\mathcal{F}[\psi_c^{(1)}] \quad (3.18)$$

$$\psi_c^{(3)} = \mathcal{F}^{-1}[\tilde{\psi}_c^{(2)}] \quad (3.19)$$

$$\psi_c(t + \Delta t, x) = \exp\left(-i\frac{m}{\hbar}\frac{1}{a(t)}\frac{\Delta t}{2}\Phi_c(t + \Delta t, x)\right)\psi_c^{(3)} \quad (3.20)$$

3.4.4 Particle Update

The stellar and sink components are represented by particles with continuous positions $\mathbf{x}_p(t)$ and velocities $\mathbf{v}_p(t)$. At the beginning of the update step, because periodic boundaries are assumed, the particle coordinates are first wrapped back into the simulation box. The particle positions are then deposited onto the mesh to assemble the density field. Their mass is distributed to the eight neighbouring vertices using the trilinear Cloud-in-Cell (CIC) weights introduced in Subsection 2.3.2.

In the stellar case, all particles carry the same mass, whereas sink particles deposit their individual instantaneous masses. Once this density enters the Poisson solver and the gravitational potential is obtained, the force field is computed spectrally and transformed back to real space. This force field is then interpolated back to the particle positions using the exact same CIC weights to obtain the particle acceleration, ensuring that no spurious self-forces are generated.

Finally, particle trajectories are advanced in time using the leapfrog drift-kick-drift scheme shown in Equation 2.54. The particle positions are shifted by a half-step drift, the velocities are updated via a full kick using the interpolated acceleration, and the positions are advanced by a final half-step drift to conclude the update.

3.4.5 Gas Update

The gas component first checks the CFL condition shown in Equation 2.47. If this condition is violated, the global time step is internally split into several sub-steps. In each sub-step, the main steps listed in Subsection 2.2.4 are repeated until the gas component is stably evolved throughout the time step.

3.4.6 Sink Formation and Accretion

After the regular field and particle updates, the code applies the sink routines. If sink formation is enabled, grid cells whose baryonic density remains above a prescribed threshold for a selected number of consecutive steps are identified as candidates for the creation of a new sink. The threshold may be set using a Truelove-type density criterion, which is intended to detect when the local Jeans length is no longer sufficiently resolved on the grid. In that case, the density threshold is taken as

$$\rho_{\text{thr}} = \frac{\pi c_{\text{eff}}^2}{G (N_J \Delta x)^2}, \quad (3.21)$$

where c_{eff} is the effective sound speed, G is the gravitational constant, Δx is the smallest grid spacing in the simulation, and N_J is the number of grid cells required to resolve the Jeans length [77]. The physical idea behind this criterion is that once the local Jeans scale becomes smaller than $N_J \Delta x$, the collapse can no longer be reliably followed by the grid-based solver and the unresolved collapsing region is therefore replaced by a sink particle.

The simulation framework currently supports two sink-formation mechanisms that can be turned on and off independently. In the particle-based formation mode, nearby baryonic particles inside an aggregation region of radius r_{agg} are absorbed into the new sink, and the sink velocity is given by the mass-weighted mean velocity of the absorbed particles,

$$\mathbf{V}_{\text{sink}} = \frac{\sum_i m_i \mathbf{v}_i}{\sum_i m_i}. \quad (3.22)$$

In the gas-based formation mode, excess gas mass above the density threshold is removed from neighbouring cells inside the chosen capture region and converted into a sink with the same centre-of-mass prescription.

To alleviate the issues with the energy conservation that this "disappearing" of velocity and mass below the resolution threshold inevitably causes, the difference between the kinetic energy before and after the merge is computed. This is simply computed as:

$$\Delta E_{\text{diss}} = K_{\text{before}} - K_{\text{after}}, \quad (3.23)$$

tracked as dissipated formation energy. The way to obtain the kinetic energy will be shown in Subsection 3.4.7.

After creation of new sink particles, existing sink particles then accrete both discrete baryonic particles and gas. For particle accretion, the code checks the physical criteria introduced in Equation 2.55. With all conditions satisfied, the particle mass and momentum are added to the sink. The post-accretion sink velocity again follows from momentum conservation,

$$\mathbf{V}_{\text{new}} = \frac{M_{\text{old}} \mathbf{V}_{\text{old}} + \sum_i m_i \mathbf{v}_i}{M_{\text{old}} + \sum_i m_i}. \quad (3.24)$$

For gas accretion, the implementation follows a BHL model shown in Equation 2.56. If the reservoir model is enabled, the newly accreted mass is first stored in a reservoir and then drained into the physical sink mass on a timescale τ ,

$$\Delta M_{\text{BH}} = M_{\text{res}} (1 - e^{-\Delta t / \tau}). \quad (3.25)$$

This is implemented once again for smoother growth of the sink and avoids discontinuous jumps of the central mass during the evolution, leading to better energy conservation within the simulation run.

3.4.7 Diagnostics and Output

In addition to the dynamical update itself, the evolution loop computes a set of interesting quantities used to monitor the stability, conservation properties, and structural evolution of the system. The most important of these is the total energy, which is decomposed into kinetic and potential contributions and is written to file at every time step. In the present implementation, the kinetic part is further split into a flow contribution from the wave field, a quantum contribution associated with its density gradients, and the kinetic energy of the baryonic sector:

$$K_{\text{tot}} = K_{\text{flow}} + U_{\text{quantum}} + K_{\text{baryons}}. \quad (3.26)$$

The quantum contribution is evaluated from the gradient of the wave-field amplitude,

$$U_{\text{quantum}} = \frac{\tilde{\hbar}^2}{2} \int |\nabla \sqrt{\rho}|^2 dV, \quad (3.27)$$

while the full kinetic energy of a given wave-function component is computed as

$$K_{\psi} = \frac{\tilde{\hbar}^2}{2} \int |\nabla \psi|^2 dV. \quad (3.28)$$

The flow part is then obtained as the difference

$$K_{\text{flow}} = K_{\psi} - U_{\text{quantum}}, \quad (3.29)$$

summed over all evolved wave-function components. For the particle and gas sectors, the baryonic kinetic energy is evaluated separately and added to the total:

$$K_{\text{baryons}} = \sum_{\text{systems}} K_{\text{sys}}. \quad (3.30)$$

For particle-based systems, this contribution is computed in the standard classical form as the sum over all particles,

$$K_{\text{part}} = \sum_p \frac{1}{2} m_p |\mathbf{v}_p|^2. \quad (3.31)$$

For a gas component, the kinetic contribution is obtained from the grid velocity field as

$$K_{\text{gas}} = \int \frac{1}{2} \rho |\mathbf{v}|^2 dV. \quad (3.32)$$

The internal energy of the gas is computed by:

$$U_{\text{gas}} = \int e_{\text{int}} dV, \quad (3.33)$$

where e_{int} denotes the internal energy density of the gas. This quantity is stored separately from the mechanical energy budget. In runs with cooling enabled, the cumulative radiated energy is also tracked by quantity E_{radiated} .

Similarly, if sink particles are active, the code keeps track of the total dissipated energy associated with sink formation and inelastic accretion events,

$$E_{\text{diss}} = E_{\text{diss}}^{\text{form}} + E_{\text{diss}}^{\text{acc}}, \quad (3.34)$$

so that non-conservative processes introduced by the sub-grid sink treatment can be monitored explicitly.

The potential energy is also decomposed into several physically distinct contributing terms. The wave self-gravity, sink self-gravity, and their mutual coupling are all included in the accounting, together with any externally imposed static potential. This can be written as:

$$W_{\text{tot}} = \frac{1}{2} \int \rho_{\text{w}} \Phi_{\text{w}} dV + \frac{1}{2} \int \rho_{\text{s}} \Phi_{\text{s}} dV + \int \rho_{\text{w}} \Phi_{\text{s}} dV + \int \rho_{\text{s}} \Phi_{\text{w}} dV + \int (\rho_{\text{w}} + \rho_{\text{s}}) \Phi_{\text{static}} dV, \quad (3.35)$$

where the subscripts w and s denote the wave and sink sectors, respectively. Here, ρ_{w} and Φ_{w} are the density and gravitational potential associated with the ULDM wave component, while ρ_{s} and Φ_{s} denote the corresponding density and potential of the sink sector. The externally imposed potential is denoted by Φ_{static} . The factors of 1/2 in the self-energy terms avoid double counting, while the remaining terms describe interaction energies between distinct components or with the prescribed static potential.

The total energy at a time t recorded during the run is therefore given by the sum of all contributions:

$$E_{\text{tot}}(t) = K_{\text{tot}}(t) + W_{\text{tot}}(t) + E_{\text{diss}}(t). \quad (3.36)$$

3.5 Manipulation with Small Numbers

Since this project focuses on simulating extremely light dark matter with $m \sim 10^{-22}$ eV, and we work in astrophysical units such as $M_{\odot}$, the relevant physical quantities, especially those involving the reduced Planck constant $\hbar$, can become extremely small. These small values may lead to numerical precision issues, as they approach the limits of floating-point accuracy in standard computational implementations.

To address this, we introduce a rescaled constant defined as $\tilde{\hbar} = \hbar/m$. Some of the relevant values for this quantity are shown in Table 3.1. These values are significantly more suitable for

m_{scalar} [eV]	$\tilde{\hbar}$ [kpc ² Gyr ⁻¹]
2.50×10^{-22}	7.84 (± 0.01)
1.00×10^{-22}	19.60 (± 0.01)
5.00×10^{-22}	3.92 (± 0.01)
2.50×10^{-23}	78.42 (± 0.01)

Table 3.1: Values of $\tilde{\hbar} = \hbar/m_{\text{scalar}}$ for selected scalar-field masses.

numerical computation, mitigating precision errors associated with extremely small numbers.

Using this constant, we rescale the variables as:

$$\hat{\psi}_c = \frac{\sqrt{G}}{\tilde{\hbar}} \psi_c, \quad \hat{\Phi} = \frac{\Phi}{\tilde{\hbar}^2},$$

which improves numerical stability and precision. This approach follows the rescaling procedure adopted in [22].

This mechanism also introduces a physically meaningful set of units within the simulation framework. For our purposes, by default the units of Gyr, kpc, eV, $M_{\odot}$ are used.

3.6 Higher Order Methods

Numerical precision issues are alleviated by using higher order evolution schemes. While the second-order symmetric split step method presented in Subsection 2.1.1 is widely used due to being computationally inexpensive, it is possible to employ higher-order methods. This can provide improved accuracy for long-term simulations at the cost of additional computational steps. The fourth-order and even one step further going sixth-order split step method developed by Yoshida [76] offers significantly enhanced precision through a more complex decomposition of the time evolution operator. This method uses a sequence of eight alternating kick and drift steps with carefully optimized coefficients, achieving $\mathcal{O}(\Delta t^7)$ accuracy while maintaining the desirable properties of unitarity and time-reversibility. This is very desirable for long-term simulations requiring stability in gravitational wave simulations [48]. Higher-order methods become particularly valuable when there is a demand for increased computational precision. However, for many simulations of structure formation on cosmological scales, the second-order method provides a great balance between computational cost and accuracy [78].

Chapter 4

Code Validation

To verify the correct functionality of the simulation framework, several tests along its development were conducted. These included comparisons of code output with theoretical predictions and tests for numerical stability. In this chapter, I shall discuss how each test was conducted and its results.

4.1 Quantum Harmonic Oscillator

The first test was performed to verify the correct functionality of the Schrödinger equation solver using the Fourier method explained in Subsection 2.1.1. This test aimed to reproduce the results that can be obtained from the analytical solution of the quantum linear harmonic oscillator in a quadratic potential using the simulation framework. This was done first by computing the analytical solution.

4.1.1 Analytical Solution of Quantum Harmonic Oscillator

The one-dimensional time-independent Schrödinger equation for a particle of mass m in a harmonic potential

$$V(x) = \frac{1}{2}m\omega^2x^2 \quad (4.1)$$

is given by

$$-\frac{\hbar^2}{2m} \frac{d^2\psi_n(x)}{dx^2} + \frac{1}{2}m\omega^2x^2\psi_n(x) = E_n\psi_n(x), \quad (4.2)$$

where ω is the angular frequency of the oscillator and $n \in \mathbb{N}_0$ is the quantum number that indexes the eigenstates.

Normalised eigenfunctions $\psi_n(x)$ are as follows:

$$\psi_n(x) = \left(\frac{1}{\sqrt{2^n n!}} \right) \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}} H_n \left(\sqrt{\frac{m\omega}{\hbar}} x \right), \quad (4.3)$$

and the corresponding energy eigenvalues are:

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right). \quad (4.4)$$

The Hermite polynomials $H_n(\xi)$ were obtained from the Python library called *scipy* [72]. The description of the LHO model here is shown for a one-dimensional case but can be trivially extended to three dimensions [79].

4.1.2 Test Results

In this validation test, two ground states corresponding to $\psi_0(x)$ from Equation 4.3 were initialised and evolved. One was evolved using the simulation framework, and the other was evolved using the analytical solution via multiplication by the time-dependent exponential:

$$\psi_0(x, t) = \psi_0(x) e^{-iE_0 t/\hbar}. \quad (4.5)$$

To verify the correctness of the implementation, the real and imaginary parts of both the analytical and simulated wave functions were compared at multiple time steps across all spatial dimensions. Because the simulation operates in a fully three-dimensional domain, it is essential to demonstrate accuracy independently along each Cartesian axis to ensure that no directional bias or multi-dimensional grid artefacts are introduced. The numerical results (dashed lines) presented in Figure 4.1, showing the profiles along the x - Figure 4.1a, y - Figure 4.1b, and z - axes Figure 4.1c, respectively, are in excellent agreement with the analytical solutions (solid lines). This confirms the robust and correct implementation of the Fourier-based time evolution method across the entire spatial grid.

Because this solution remains stationary up to a global phase, the modulus $|\psi(x, t)|$ should not change in time. For that, the probability density $|\psi(x, t)|^2$ was plotted and is shown in Figure 4.2 for both the numerical (blue line) and analytical (pink line) solutions. Since it's not changing and the two solutions are the same, the blue line is not visible on the plot due to the lines overlapping. Next, the plot shows the time evolution of the spatial averages of real and imaginary parts of the wave functions. The light green and dark blue curves represent the real and imaginary parts of the numerically evolved wave function, respectively. These are compared with the corresponding analytical values, plotted as light green and dark green scatter points. As expected for the harmonic oscillator, the real and imaginary parts oscillate as a sine function and remain in phase throughout the simulation.

4.2 Stability Testing

The next tests focused on verifying the stability of the code and the eigenvalue problem shown in Equation 2.15, that corresponds to the spherically symmetric case. Solving the system of equations yields the spherically symmetric solitonic solution generated by the SP system, which is static under self-gravity.

To test the numerical stability of the simulation framework, we tracked the maximal value of the density on the grid during time evolution. This quantity serves as an indicator of numerical errors, as the soliton profile is expected to remain stationary in the ideal case. In practice, due to discretization effects and numerical noise, small oscillations of the peak density around its equilibrium value are expected. These arise primarily from the finite resolution of the simulation grid. As the spatial resolution increases, the discretization should become more accurate. The amplitude of these oscillations should decrease with an increase in resolution, reflecting convergence toward the continuous limit.

Test Results

In Figure 4.3, one can observe the time evolution of the normalized maximal value of the density on the simulation grid. The values are normalized to their initial value so that all plots begin at 1. A clear trend can be seen where the amplitude of oscillations in the maximal density decreases as the spatial resolution N increases. This behaviour is consistent with the expectation that numerical artifacts diminish with finer discretization. The simulations were

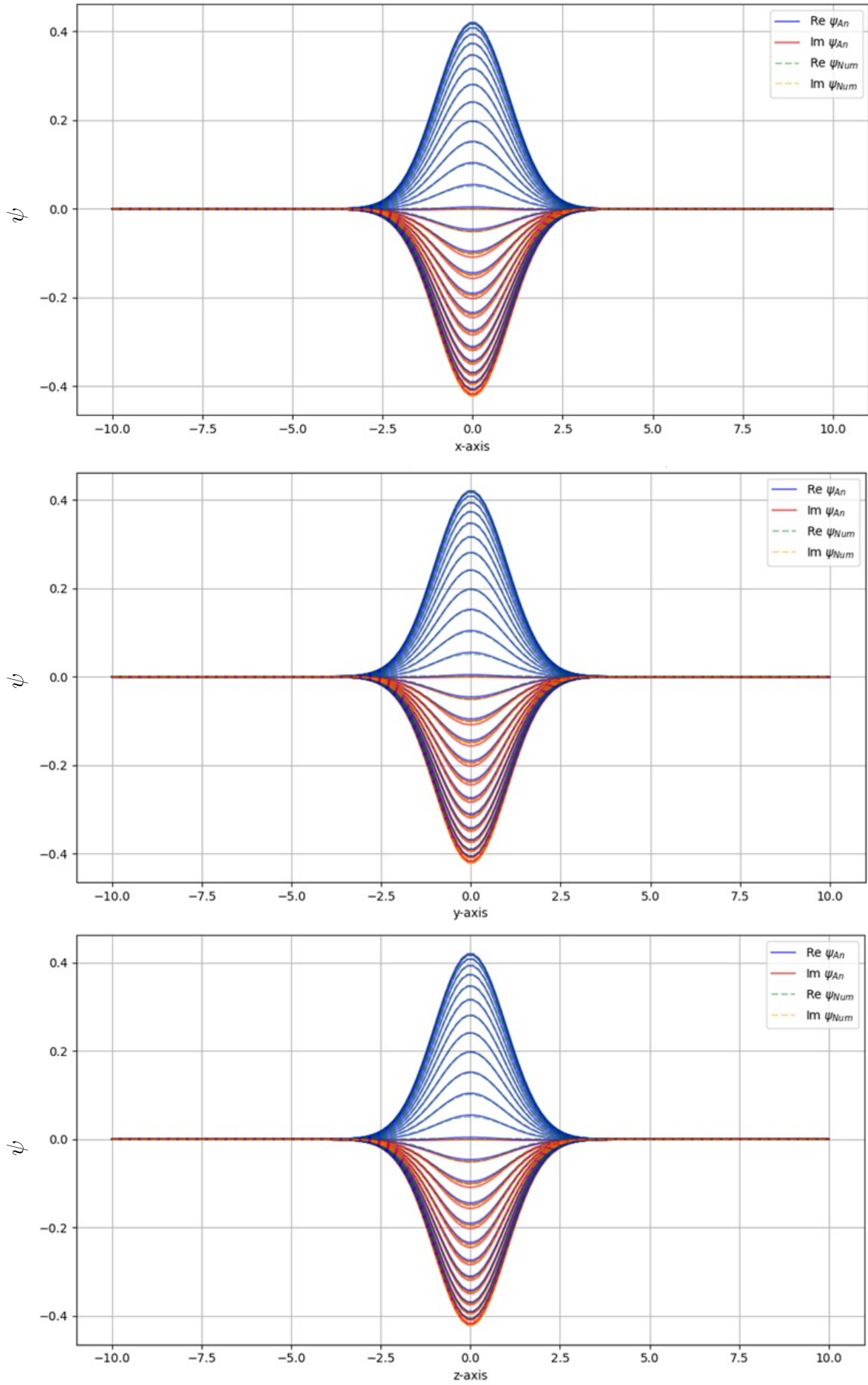


Figure 4.1: Comparison of analytical (full lines) and numerical solutions (dashed lines) of the ground-state wave function of the quantum harmonic oscillator at several time steps in three spatial directions. Both the real and imaginary parts are shown.

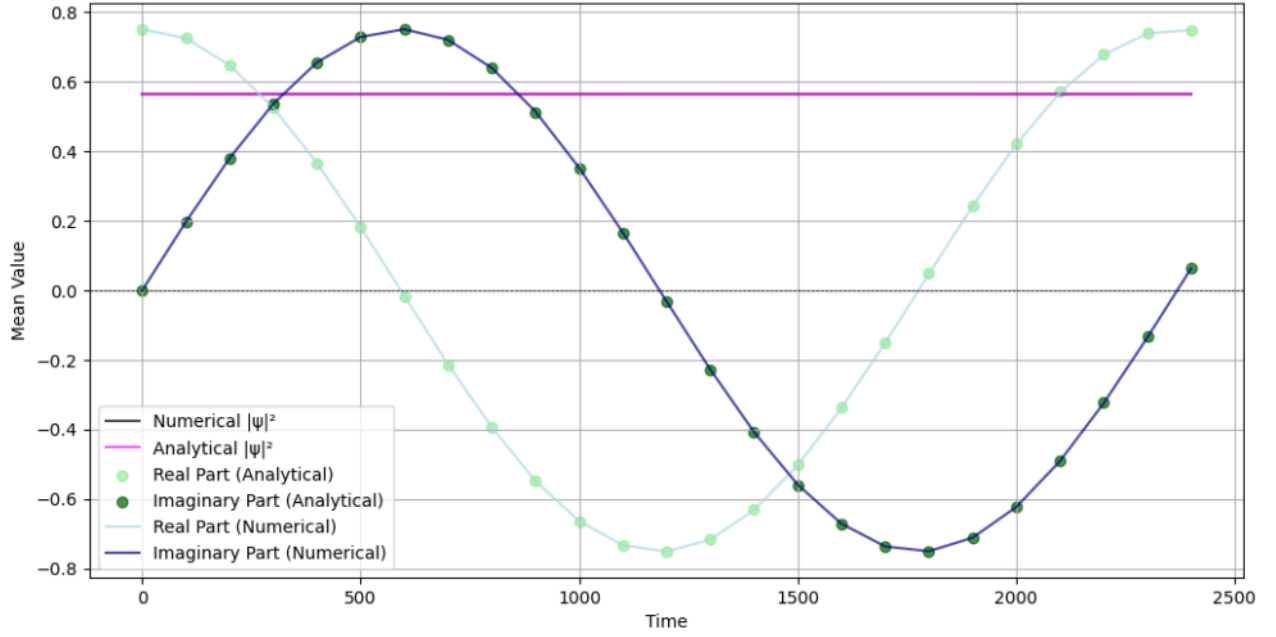


Figure 4.2: Time evolution (shown in number of steps) of the mean values of the real and imaginary parts of the ground-state wave function $\psi_0(x, t)$, with analytical results shown as scatter points. The mean density $|\psi(x, t)|^2$ is shown for both the simulation (solid blue line) and the analytical calculation (solid pink line).

performed with a box size of 20 kpc and a total evolution time of 10 Gyr. Although the time step constraint from Equation 3.7 was not satisfied at the highest resolutions, the simulations remained stable.

The stability of the soliton can also be observed in Figure 4.4, which shows the wave-function probability density $|\psi(x, y, z)|^2$ of a soliton located at $(0, 0, 0)$ at different times denoted by t in Gyr. The plots display a cross-section through the soliton at $z = 0$ and show that, although the density profile exhibits small oscillations, its overall spatial structure is preserved throughout the simulation, indicating numerical stability over time.

4.3 Movement Testing

4.3.1 Linear Motion

To test the correct implementation of momentum in different components of the simulation, several simple tests have been carried out. First, we simply set the initial position of a clump of simulated matter to $(x, y, z) = (5, 0, 0)$ kpc and assigned it a velocity of $(-5, 0, 0)$ kpc/Gyr, then verified that after one unit of time its final position was indeed at the origin of the grid. The test yielded the expected results for the N-body and ULDM components, which will not be included as they are trivial and will be left to the reader's imagination.

Where this test becomes more interesting is in the gas component, since for grid-based methods it can also help reveal possible errors in the implementation of spatial derivatives along individual axes. In particular, one may check whether equivalent displacements along the x -, y -, and z -directions produce the same result, which is especially useful in finite-difference-based schemes. For simplicity, however, the present validation was performed only in 1D. A localized density cluster was initialized at $x = 5$ kpc with velocity $v_x = -5$ kpc/Gyr (blue line), and it was verified that after one unit of time it reached the expected position at $x = 0$ kpc at

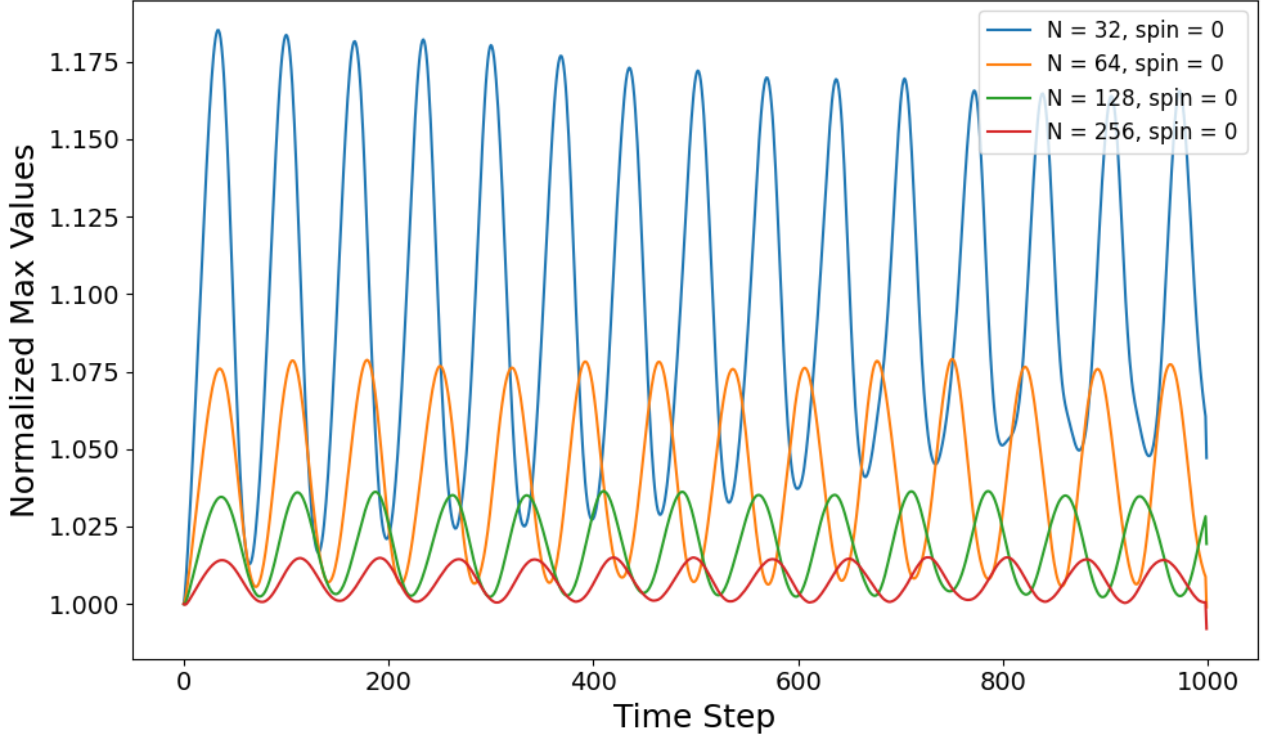


Figure 4.3: Time evolution of the normalized maximal density values on the simulation grid for different resolutions: $N = 32$ (blue), $N = 64$ (orange), $N = 128$ (green), and $N = 256$ (red). All simulations used a spin-0 soliton.

$t = 1$ Gyr (orange line). The result of this test is shown in Figure 4.5. A slight deformation of the profile is visible, reflecting both the intrinsic numerical diffusion and the fact that the gas density cluster does not evolve as a perfectly rigid body.

Another set of tests included testing of the SP system and gravity implementation within the leapfrog DKD scheme explained in Subsection 2.3.4. The test aimed to reproduce an orbital motion around a massive point-like source of gravity.

4.3.2 Rotational Motion Around a Point like Source

The gravitational potential of a point-like mass M located at the origin is given by:

$$V(r) = -\frac{GM}{r}, \quad (4.6)$$

where G is the gravitational constant and r is the radial distance from the mass.

For a stable circular orbit with radius r , the required orbital velocity is determined from the balance of centripetal and gravitational force:

$$v = \sqrt{\frac{GM}{r}}. \quad (4.7)$$

The corresponding orbital period is then:

$$T = \frac{2\pi r}{v} = 2\pi \sqrt{\frac{r^3}{GM}}. \quad (4.8)$$

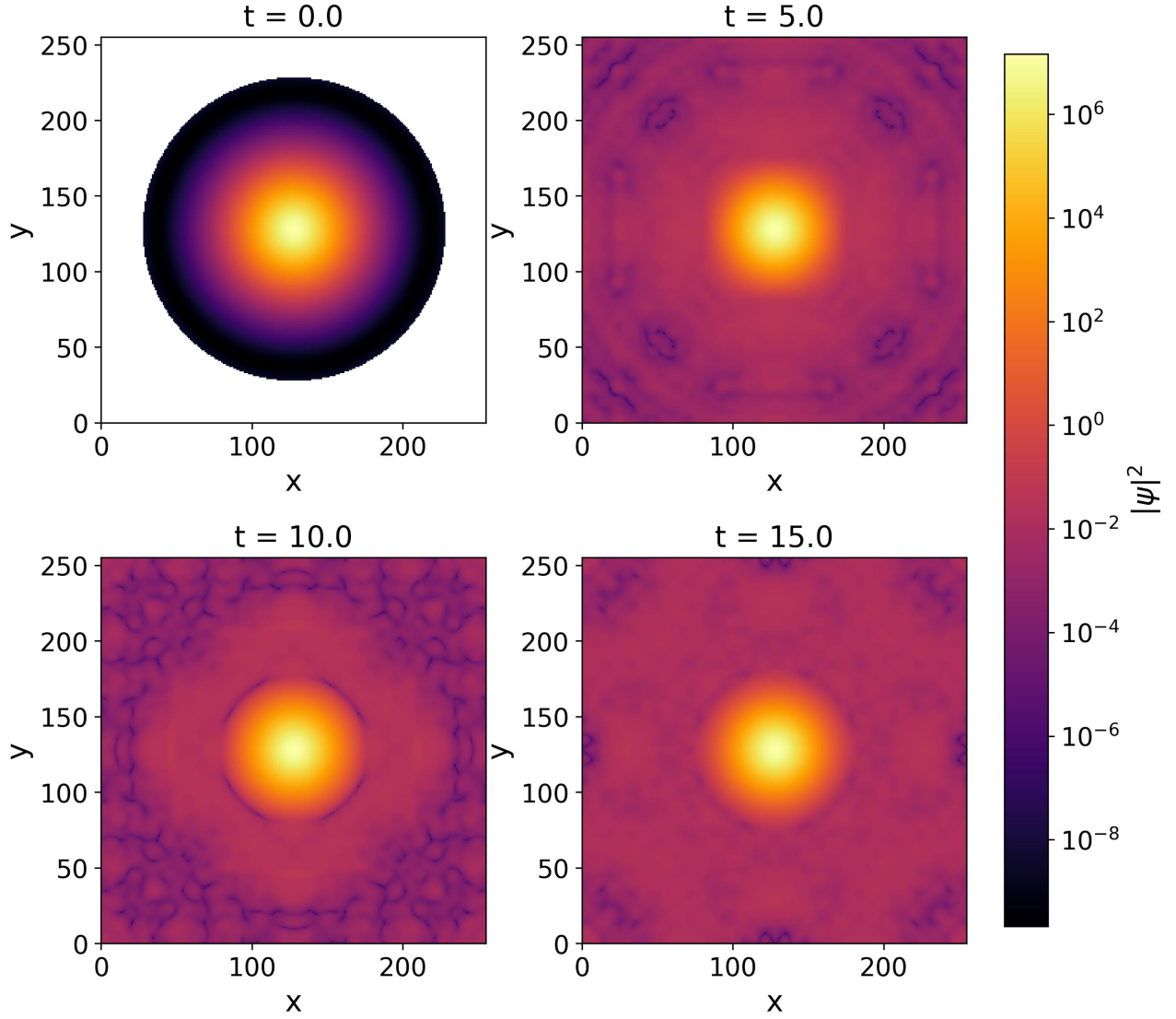


Figure 4.4: Cross-section plots of the wave-function probability density $|\psi(x, y, z)|^2$ in the x-y plane at different times for a simulation with $N = 256$ during the evolution of a static soliton. While the soliton exhibits small oscillations in its density profile, its overall structure is preserved over 15 Gyr of evolution (15000 time steps), indicating numerical stability of the solution.

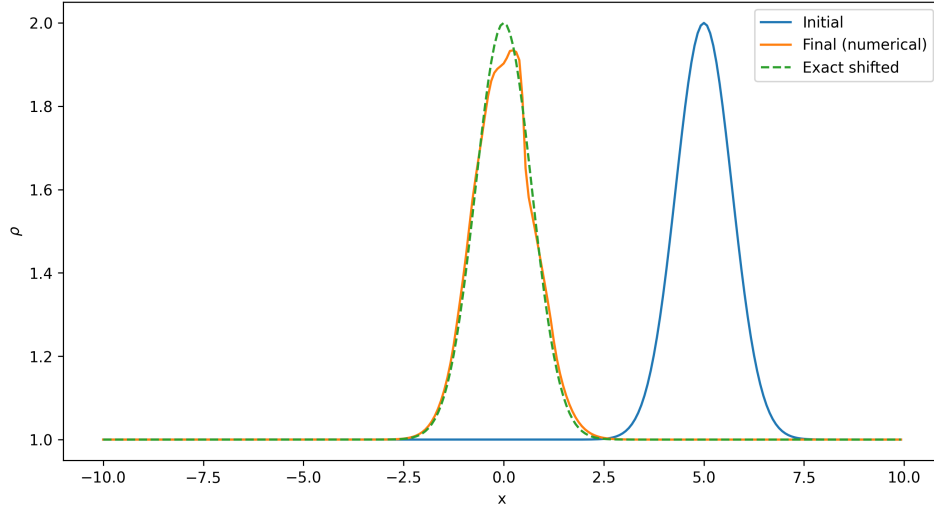


Figure 4.5: Result of the 1D gas translation test, showing a localized density cluster initialized at $x = 5$ with velocity $v_x = -5$, together with its initial profile at $t = 0$ (blue line) and its translated profile at $t = 1$ (orange line).

ULDM Component

The test was carried out by implementing an external static gravitational source placed at the centre of the simulation grid, and placing a soliton at a radial distance r from the source. A perpendicular velocity was then given to the soliton, replicating a circular orbit around the central mass.

We consider a central mass $M = 10^6 M_\odot$, where $M_\odot$ is the solar mass. Taking the gravitational constant in astrophysical units $G = 4.4985 \times 10^{-6} \text{ kpc}^3 M_\odot^{-1} \text{ Gyr}^{-2}$, and placing the soliton at distance $r = 5 \text{ kpc}$, we plug into Equation 4.7:

$$v = \sqrt{\frac{4.4985 \times 10^{-6} \times 10^6}{5}} \approx 0.9485 \text{ kpc/Gyr}.$$

Using this velocity in Equation 4.8, the resulting orbital period is:

$$T \approx \frac{2\pi \cdot 5}{0.9485} \approx 33.1 \text{ Gyr}.$$

Test Results

This velocity was given to the soliton in the simulation and its trajectory was tracked to confirm that it follows a circular orbit. The results of this test can be seen on Figure 4.6.

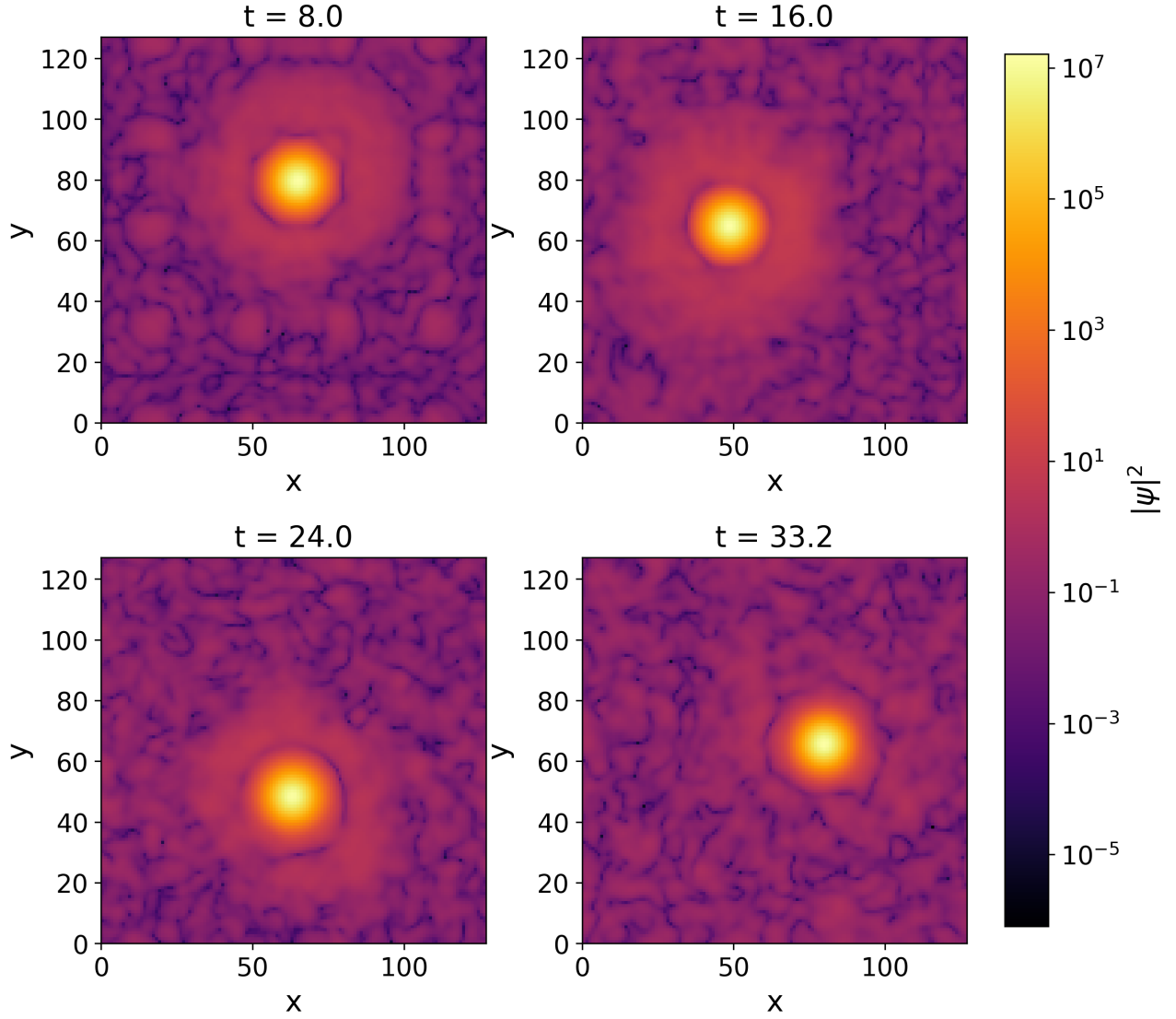


Figure 4.6: Circular motion of a soliton initialized at point $(5, 0, 0)$ kpc around a point like source located at $(0, 0, 0)$ kpc with mass $10^6 M_\odot$. The sequence of snapshots shows the soliton's probability density at various time steps during the orbital motion.

For better visualisation, a graph showing the evolution of the maximum density location is shown in Figure 4.7. As expected, with given velocity $= (0, 0.9485, 0)$ kpc/Gyr the soliton moves in the x - y plane and reaches the same point after approximately 33.1 Gyr¹.

N-Body Component

Similarly to the test in Section 4.3.2, the test particle was created in a static gravitational potential generated by a point-like source with mass $5 \cdot 10^{10} M_\odot$ at the origin of the grid and assigned the corresponding speed computed analogously to Section 4.3.2. The test results shown in Figure 4.8 show the trajectory in the x - y plane that the particle traced over ~ 7 Gyr in Figure 4.8a and the displacement from the origin in Figure 4.8b.

The test showed only a small deviation over 18 000 time steps and therefore the implementation is considered sufficiently consistent to evolve the N-body component.

¹Any discrepancies between the result and the theoretical expectation are due to rounding and finite resolution effects.

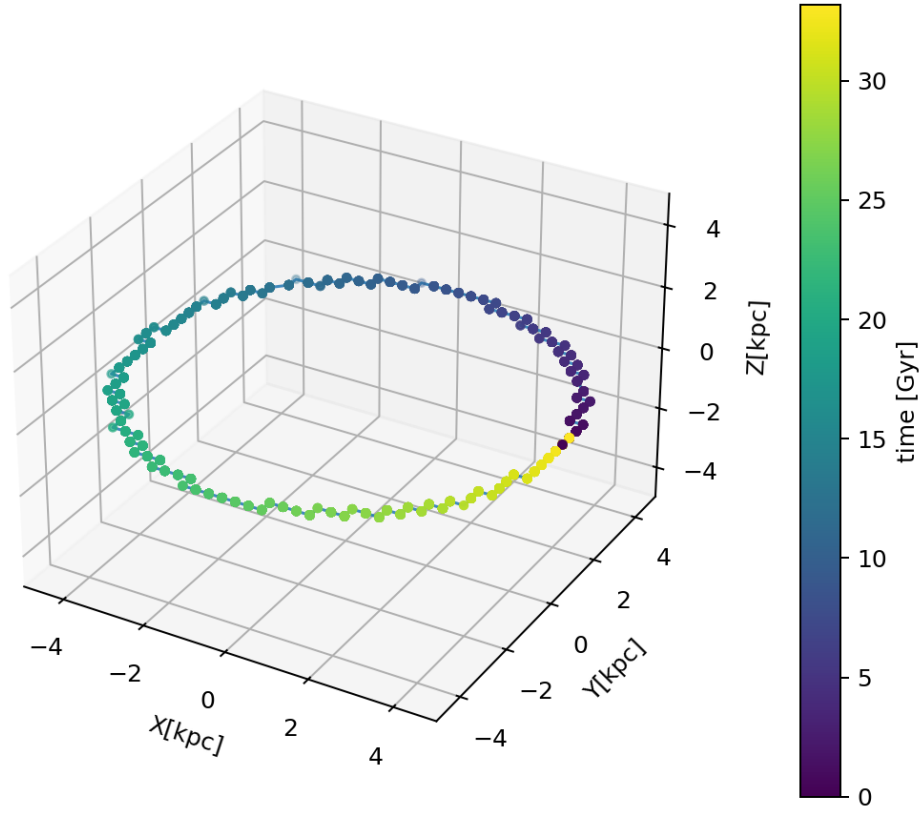
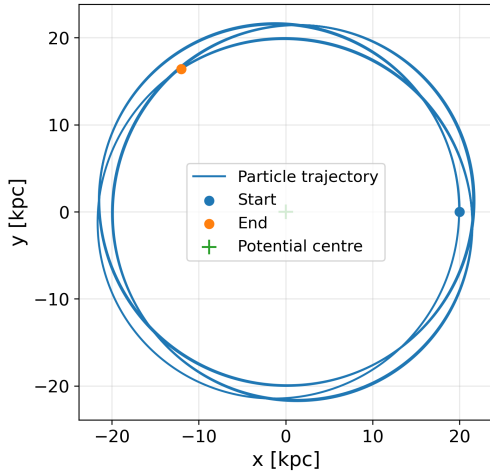
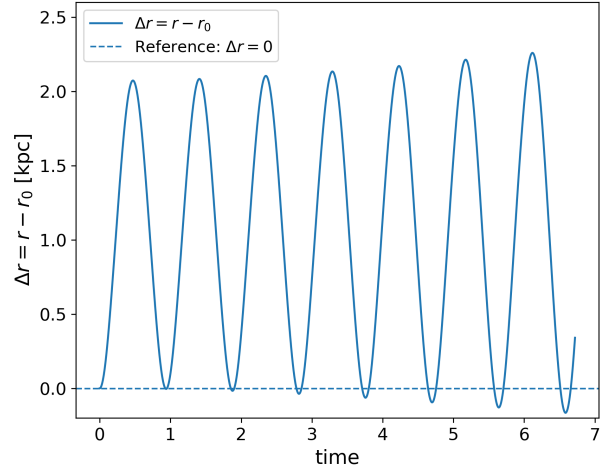


Figure 4.7: Plot showing the trajectory of a soliton initialized with `momenta = [0,0.9485,0]` at point (5,0,0) in time shown in colour [Gyr]. The simulation was set up in a way so that the soliton performs a circular orbit around a massive object located at point (0,0,0).



(a) Trajectory of the test particle in the x - y plane during the circular-orbit test. The particle was initialized at (20.0, 0.0, 0.0) kpc at $t = 0$ Gyr and evolved in the gravitational field of a point-mass source.



(b) Radial deviation $\Delta r = r - r_0$ as a function of time for the same test particle. The result shows a bounded oscillatory deviation from the initial radius, without a strong secular drift.

Figure 4.8: Results of the circular-motion test for the N -body component. A single particle with negligible mass was initialized at (20.0, 0.0, 0.0) kpc and assigned the circular velocity corresponding to a point-like gravitational source of mass $5 \cdot 10^{10} M_\odot$. The left panel shows the orbit in the x - y plane, while the right panel shows the radial deviation from the initial orbital radius over time Δr .

4.4 Sod Shock Tube Test

A more demanding validation of the gas solver was performed using the classical Sod shock tube problem. Unlike the simple translation test discussed previously, this benchmark probes the ability of the numerical scheme to correctly reproduce the characteristic wave structure of the one-dimensional compressible Euler equations, namely a rarefaction wave, a contact discontinuity, and a shock wave. For this reason, it is commonly used as a standard test of hydrodynamic solvers [80, 81].

In one spatial dimension, the gas evolution is governed by the Euler equations written in conservative form as

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} = 0, \quad (4.9)$$

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial}{\partial x} (\rho u^2 + p) = 0, \quad (4.10)$$

$$\frac{\partial E}{\partial t} + \frac{\partial}{\partial x} [u(E + p)] = 0, \quad (4.11)$$

where ρ is the density, u the velocity, p the pressure, and E the total energy density. The system is closed by the ideal-gas equation of state

$$p = (\gamma - 1) \left(E - \frac{1}{2} \rho u^2 \right), \quad (4.12)$$

with adiabatic index γ [81].

The Sod test consists of a Riemann problem with piecewise constant initial data,

$$(\rho, u, p)(x, 0) = \begin{cases} (\rho_L, u_L, p_L), & x < x_0, \\ (\rho_R, u_R, p_R), & x > x_0, \end{cases} \quad (4.13)$$

where in the present case the standard Sod initial conditions were used,

$$(\rho_L, u_L, p_L) = (1.0, 0.0, 1.0), \quad (\rho_R, u_R, p_R) = (0.125, 0.0, 0.1), \quad (4.14)$$

with $\gamma = 1.4$ and the initial discontinuity placed at

$$x_0 = 0.5. \quad (4.15)$$

The formulation of the Riemann problem and the corresponding wave structure follow the standard treatment given in [81].

Test Results

The test was carried out in a one-dimensional domain $x \in [0, 1]$ using resolution of $N = 800$ and no gravitational interaction. The gas was initialised directly from the left and right states given by Equation 4.14, and the numerical solution obtained by the implemented gas solver was compared with the exact solution of the corresponding 1D Riemann problem given at time $t = 0.1$. The results of this test are shown in Figure 4.9.

As can be seen in Figure 4.9, the numerical solution (shown in orange) reproduces the exact wave structure (shown in blue) very well in the central part of the domain, including the rarefaction fan, the contact discontinuity, and the shock front. The remaining discrepancies near the grid boundaries are caused by the use of periodic boundary conditions within the simulation framework. These are essential for solving the SP system (Equation 1.16) using the spectral Fourier method explained in Subsection 2.1.1. Overall, the good agreement between the numerical and exact solutions confirms that the gas solver correctly reproduces the essential features of the gas component.

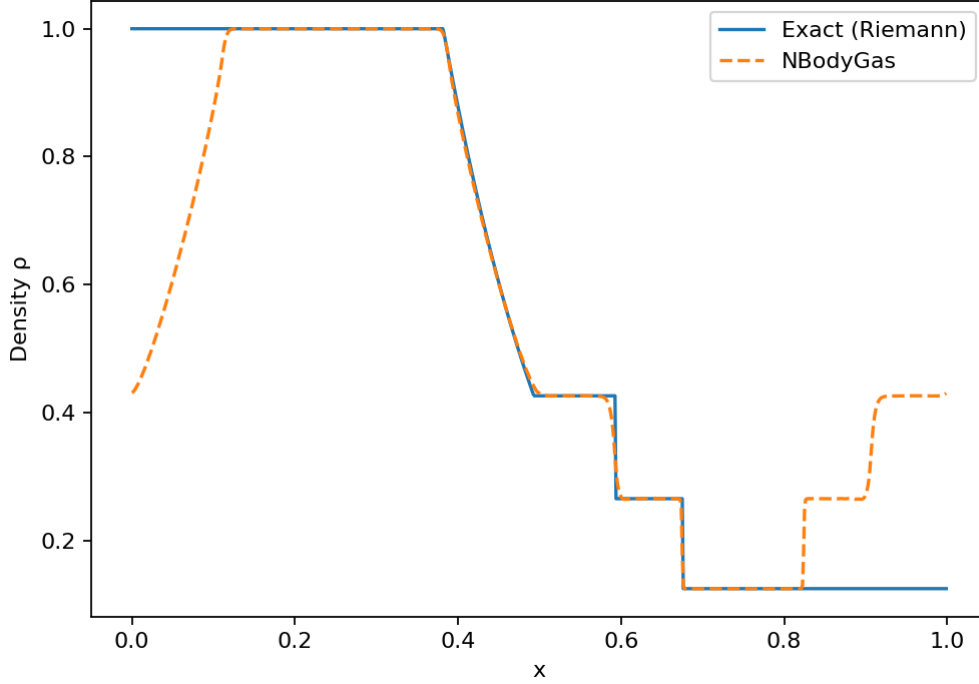


Figure 4.9: Comparison between the numerical gas solution (orange line) and the exact Riemann solution (blue line) for the Sod shock tube test at $t = 0.1$. The test was carried out on a one dimensional grid with box size of one simulation unit length and resolution $N = 800$. Discrepancies near the simulation grid borders are due to periodic boundaries.

4.5 Energy Conservation

The final set of tests focused on verifying energy conservation within the simulation. For this purpose, a simulation was performed with 25 solitons on a grid of size $N = 100$. This number was chosen to be consistent with the setup used in [22], where it was shown that simulations with fewer than 5 initial solitons require simulations of longer times than 30 Gyr, making them computationally expensive. On the other hand, using more than approximately 120 solitons exceeds the spatial resolution and box size that we are using in this work. This sets the allowed range for the number of solitons to $5 \leq N_{\text{sol}} \leq 120$. Our choice of 25 solitons, therefore, represents a practical compromise between having a sufficiently large number of solitons and keeping the computational cost manageable.

For these tests, the total energy of the system was recorded over time using the processes described in Subsection 3.4.7 for its computation. The evolution of the total energy was then analysed by plotting the normalised deviation of the total energy from its initial value. Specifically, the plots show:

$$\frac{\Delta E}{E_0} = \frac{E_t - E_0}{E_0}, \quad (4.16)$$

on their y -axis. Where E_t is the total energy in a time step t and E_0 is the initial total energy in $t = 0$.

Test Results

The primary goal of this analysis was to determine whether the total energy is conserved during the simulation and if so, to what degree. A secondary objective was then to identify which simulation parameters have the greatest impact on the accuracy of energy conservation. Based on the results shown in [22], I expected spatial resolution to be the dominant factor. As

shown in Figure 4.10, increasing the grid resolution from $N = 128$ to $N = 512$ significantly reduces the amplitude of energy fluctuations.

However, resolution did not prove to be the most critical factor. The most crucial parameter turned out to be the time step h , as can be seen in Figure 4.11. Here, simulations with identical setups but different time steps show that smaller values of h result in better energy conservation. Even when both values selected for h shown in Figure 4.11 satisfy the constraint Δt from Equation 3.7, there is a clear change in stability that favours smaller time steps. While yes, reducing the time step by a factor of 10 drastically improves the results, it also increases the total number of steps by the same factor. Therefore, it is almost unusable in practice due to the high computational cost.

Fortunately, this issue can be bypassed by using higher-order split step methods, which were described in Section 3.6. Employing a fourth-order split step method allows for improved energy conservation without reducing the time step, therefore keeping the same number of iterations. This is shown in Figure 4.12, where we can see the energy deviations for simulations with resolutions $N = 128$ and $N = 256$ with different time steps: $h = 0.001$ (blue, using order two method) and $h = 0.01$ (orange, using fourth-order evolution). In practice, using the fourth-order method achieves a comparable reduction in energy deviation as increasing the resolution from $N = 256$ to $N = 512$, or reducing the time step by an order of magnitude while remaining computationally affordable.

However, it is worth noting that the energy deviation in the fourth-order run (orange) exhibits a monotonic upward drift over time. This is unexpected because, while higher-order methods improve local accuracy, they do not guarantee exact global energy conservation, particularly when using non-linear or self-gravitating systems as we are. The split step Fourier method assumes a factorisation of the time-evolution operator:

$$e^{-iH\Delta t} \approx \prod_j e^{-iT_j\Delta t} e^{-iV_j\Delta t},$$

where T_j and V_j represent kinetic and potential contributions to the propagator. For systems governed by the SP system Equation 1.16, the potential depends non-linearly on the evolving wave function, introducing non-commutativity between the kinetic and potential operators. This leads to the operator splitting being only an approximation [82]. Furthermore, the error in simulations is cumulative, meaning that once it begins, it is propagated into the next steps and grows over each iteration. This upward trend can be reduced by making the time step smaller or increasing the grid resolution.

Further plots showing the influence of soliton spin on energy conservation within the simulation framework will be shown in Chapter 5.

4.6 Performance Overview

An additional set of tests that was carried out was a comparison of computational performance for three benchmark simulations carried out on available GPUs. The results are shown in Figure 4.13. All runs used a three-dimensional domain with boundaries $[-25, 25]^3$ kpc, grid resolution $N = 128$, total simulation time $t_{\text{tot}} = 5$ Gyr, global time step $h = 10^{-3}$ Gyr, and `save_every` = 250. The ULDM benchmark used a single ground-state soliton; the N-body benchmark contained 10^6 particles with total particle mass $5 \times 10^7 M_\odot$; and the gas benchmark used a gas-only configuration with density profile $\rho \propto \exp(-R/R_0) \exp(-Z/Z_0)$ and total gas mass $M_{\text{gas}} = 5 \times 10^7 M_\odot$.

For a three-dimensional simulation with grid size N^3 and spin-one ULDM evolved through

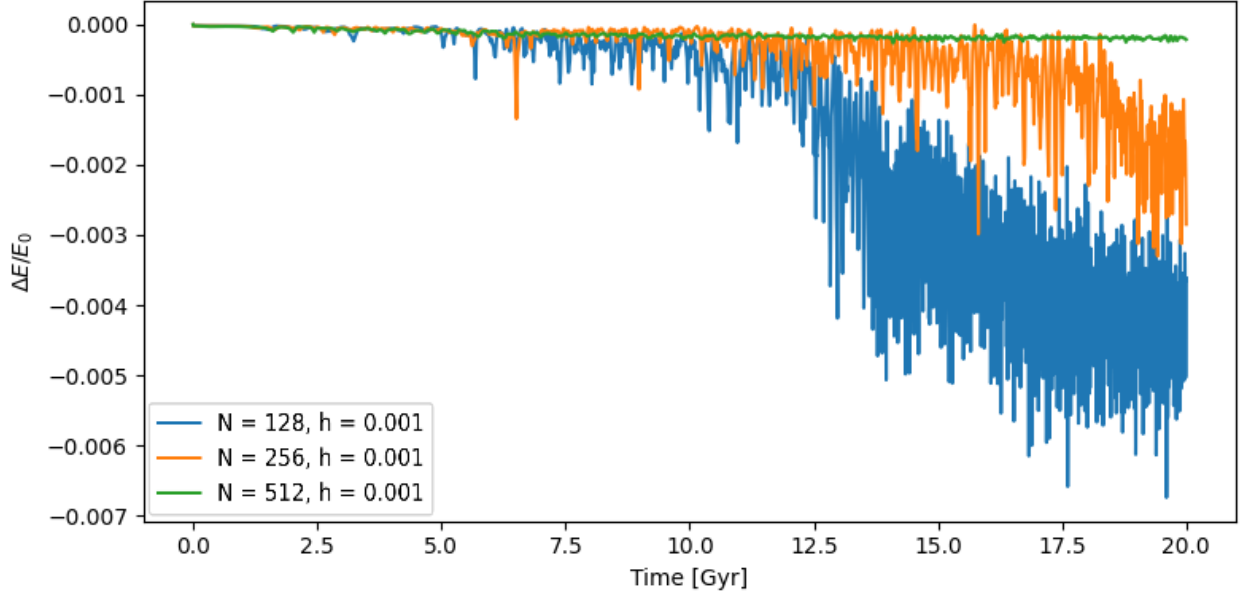


Figure 4.10: Normalized energy deviation $\Delta E/E_0$ for simulations with grid resolutions $N = \{128, 256, 512\}$. The setup for all simulations included a box size of 100, 25 solitons with spin 1, and a time step of $h = 0.001$.

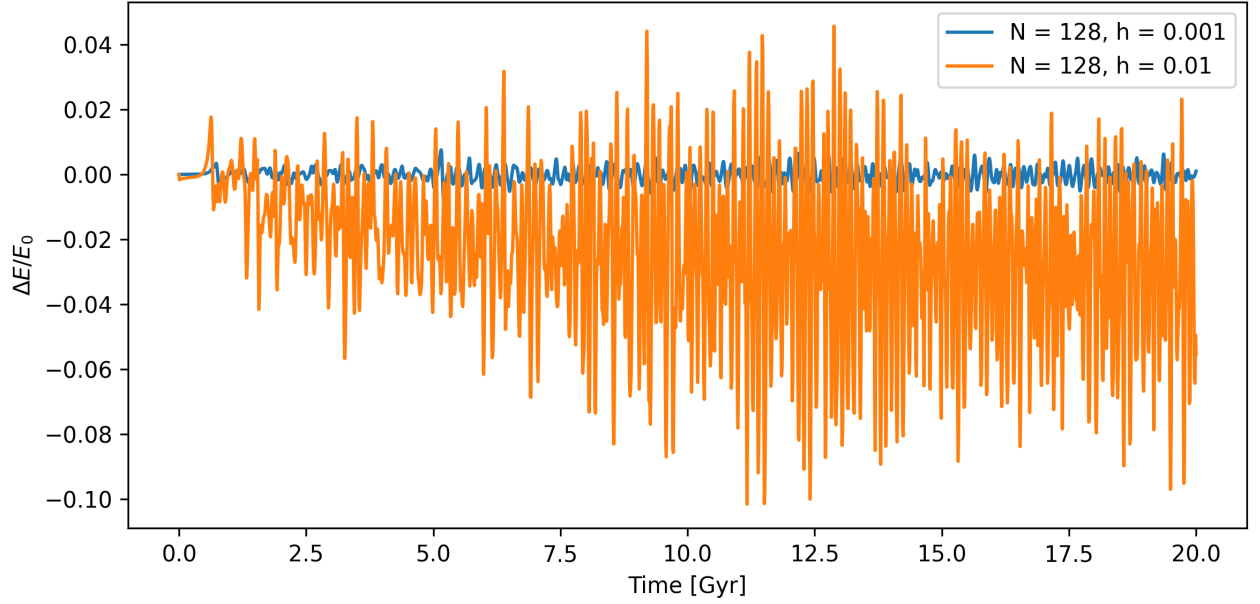


Figure 4.11: Normalized energy deviation $\Delta E/E_0$ for simulations with resolution $N = 128$ and different time steps $h = 0.001$ (blue) and $h = 0.01$ (orange). The setup for both simulations included a box size of 100 and 25 solitons with spin 1.

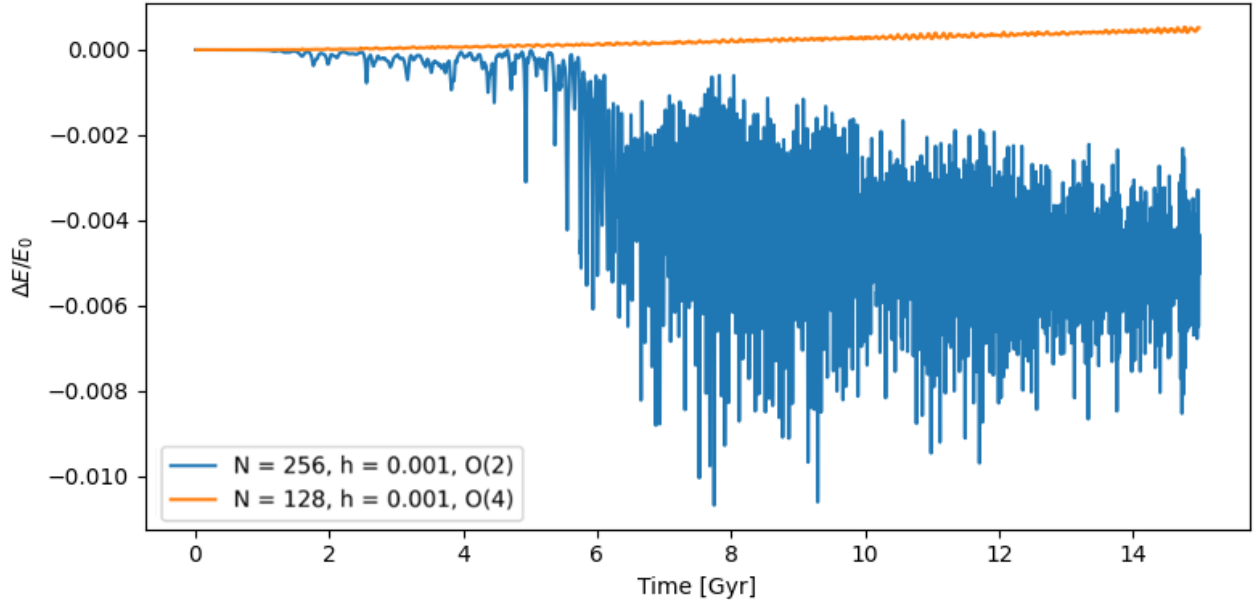


Figure 4.12: Normalized energy deviation $\Delta E/E_0$ for simulations with resolution $N = 256$ and second-order evolution (blue), and resolution $N = 128$ using fourth-order evolution (orange). Time steps were $h = 0.001$ for second-order runs and $h = 0.01$ for the fourth-order run. All simulations used a box size of 100 and 25 solitons with spin 1.

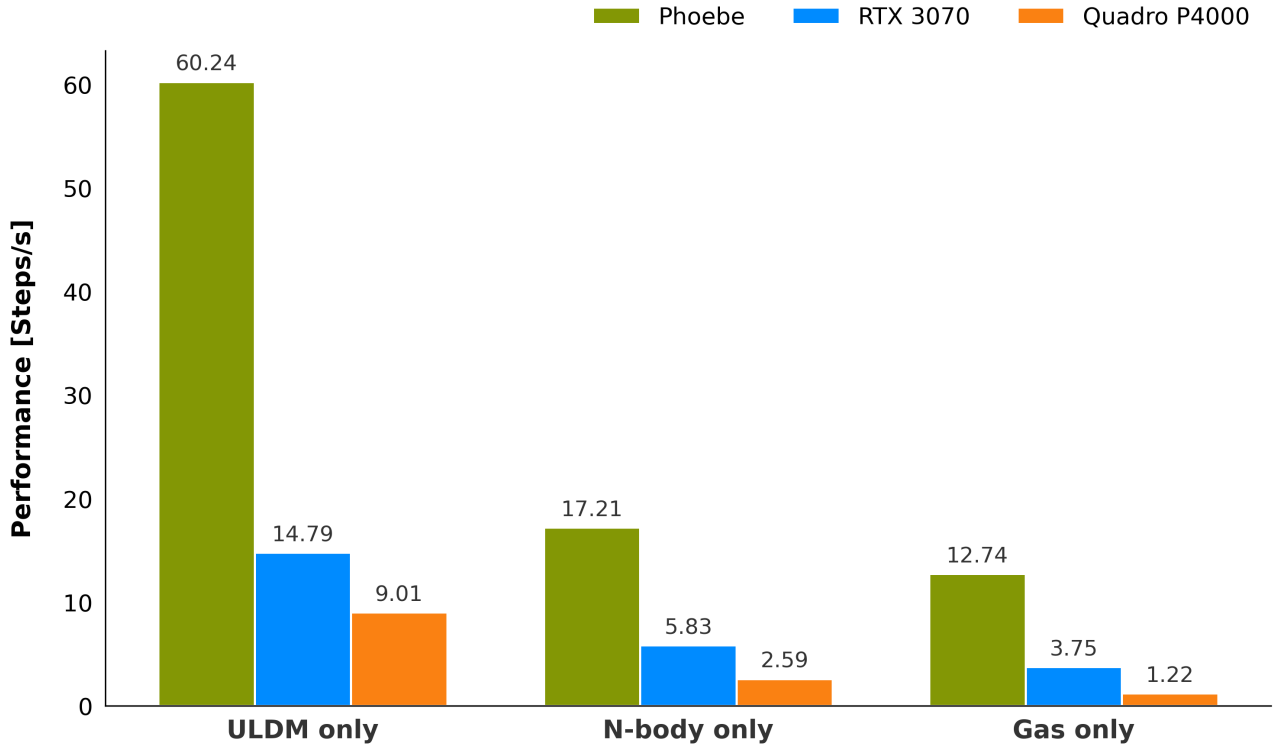


Figure 4.13: Bar-chart visualization of computational performance for three benchmark simulations carried out on different GPUs. All runs used a three-dimensional domain with boundaries $[-25, 25]^3$, grid resolution $N = 128$, total simulation time $t_{\text{tot}} = 5$, global time step $h = 10^{-3}$, and `save_every = 250`. The reported values give the achieved number of simulation steps per second. The ULDM benchmark used a single ground-state soliton, the N-body benchmark contained 10^6 particles, and the gas benchmark used a gas-only exponential disk-like density profile.

three complex field components, the minimal persistent memory requirement scales as

$$M_{\text{bare}}(N) \approx (3 \times 16 + 3 \times 8 + 3 \times 8) N^3 = 96 N^3 \text{ bytes},$$

using `complex128` wave-function arrays and `float64` real-space and Fourier-space grids. Here, the factor 3×16 corresponds to the three complex ULDM field components, each stored as a `complex128` array that requires 16 bytes per grid cell, while the two terms 3×8 account for the three real-space coordinate grids and the three Fourier-space wave-number grids, each stored as `float64` arrays that require 8 bytes per grid cell. In practice, the actual memory footprint is larger because additional arrays are needed for the density, gravitational potential, force components, Fourier-space intermediates, and temporary buffers during FFT-based evolution. The maximum grid size achievable for each GPU is shown in Figure 4.14. It is important to note that a higher resolution grid could be created on Phoebe, $N = 640$ is only the largest $N = 2^x + 2^y$ number tested.

To compare the practically usable problem sizes across different GPUs, a simple throughput-based criterion was introduced. For a fixed spin-one ULDM benchmark in a three-dimensional box with boundaries $[-25, 25]^3$, the grid resolution was gradually increased until the sustained performance fell below 1 simulation step per second. This threshold was adopted as a practical upper bound for routine exploratory runs. In addition to this practical limit, the maximal resolution tested that could still be initialised and evolved without running out of memory was also recorded (shown in blue). The tested values were restricted to a regular sequence of grid sizes chosen for consistency across devices. This restriction was self-imposed in order to avoid irregular grid sizes and to keep the tested resolutions within the structured sequence used throughout the benchmark.

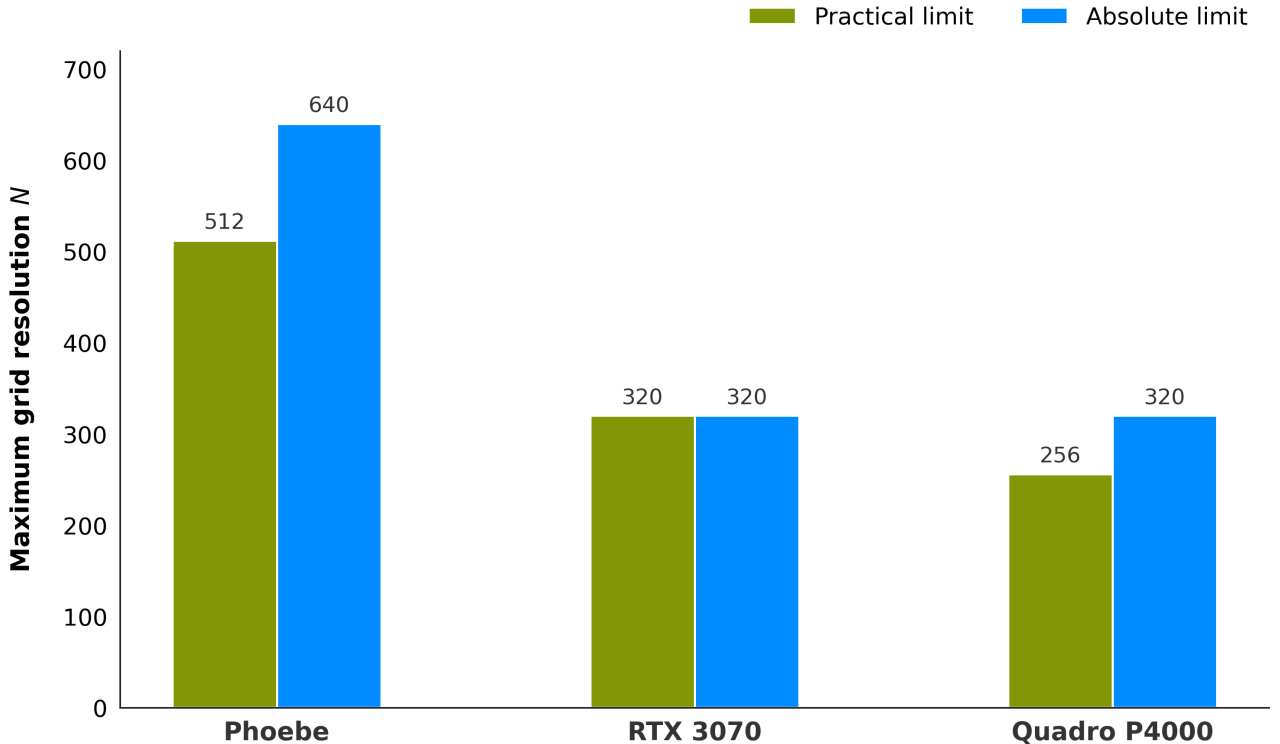


Figure 4.14: Comparison of the maximum feasible grid resolution for ULDM-only simulations on the available GPUs. The practical limit denotes the highest resolution that still achieved approximately one simulation step per second, while the absolute limit denotes the highest resolution that could still be executed within the available memory.

Chapter 5

Results

This chapter presents the main results obtained with the simulation framework developed in this work. Since the primary aim of the thesis was the implementation and validation of the framework itself, the range of physical results is limited to a selected set of representative cases consistent with the scope of a diploma thesis. The chapter is organised according to the individual parts of the study:

1. Dark matter density profiles, Section 5.1
2. Growth of the ULDM core radius, Section 5.2
3. Core mass dependence on the black hole mass, Section 5.3
4. Sink gas accretion rates, Section 5.4
5. composite system mergers, Section 5.5

5.1 Dark Matter Density Profiles

In Figure 5.1, we show the spherically averaged density profiles obtained from simulations of a merger of 25 solitons in a box of size 100 for spins $s = 0, 1, 2, 3$ on a grid with resolution $N = 256$.

These profiles were obtained by fitting the final density field at $t = 30$ Gyr with Equation 2.25. The profiles are normalised to their respective central densities ρ_c and plotted as a function of radius r . A vertical dash-dotted green line marks the transition radius $r_c \simeq 3.5$, where the fit changes from the solitonic profile to the NFW-like outer halo (Equation 1.19). Simulation results reproduce the expected behaviour, we observe a flat-density core extending up to $r \sim r_c$, followed by a power-law decline in the outer halo. The inner parts of all four spin cases align closely with the reference soliton solution, confirming that our split-step evolution correctly preserves the stationary core structure over long integration times.

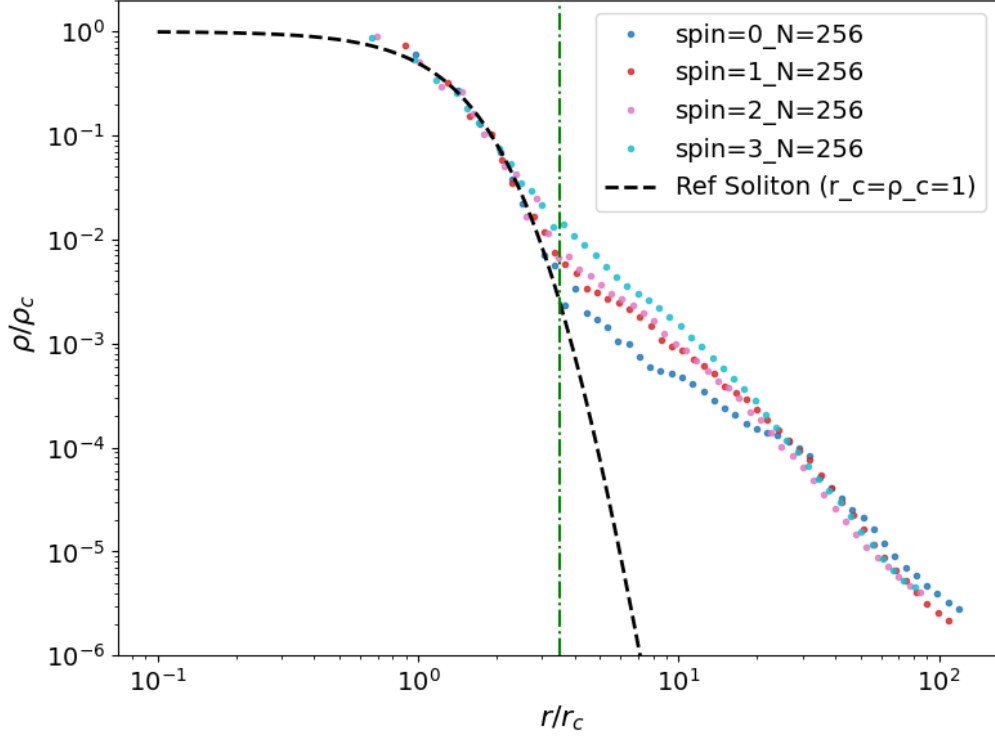


Figure 5.1: Spherically averaged density profiles from simulations of a 25-soliton merger in a box of size 100 for spins $s = 0, 1, 2, 3$ at $t = 30$ Gyr. The profiles are normalized to the central density ρ_c and the radius is scaled by the core radius r_c obtained from fitting Equation 2.25. The dashed black line shows the reference soliton profile with $r_c = \rho_c = 1$, while the vertical dash-dotted green line marks the transition radius $r_e \simeq 3.5$ where the fit switches from the solitonic core to the NFW-like outer halo.

Beyond the transition radius, the simulated profiles fall off more slowly than the pure soliton solution, which is to be expected from the contribution of surrounding halo material. This deviation is most visible at $r/r_c \gtrsim 5$, where interference patterns and residual wave turbulence lead to small-scale oscillations in $\rho(r)$.

The outer slope of the simulated profiles shows variations with spin, and the overall trends are compatible with the results presented in [22]. In our case, higher-spin runs deviate from the reference soliton solution at slightly smaller radii. The $s = 0$ profile follows the reference soliton most closely in both the core and transition region, with the steepest and smoothest decline among all cases, which could be caused by the absence of polarisation mode coupling. In contrast, the $s = 3$ and $s = 2$ profiles show a slightly shallower drop just beyond r_c , which may indicate enhanced mode mixing between polarisation components in the higher-spin cases.

The agreement between our simulated profiles and the reference soliton in the core region is even clearer when looking at the best-fit curves shown in Figure 5.2. Here we plot the fitted profiles for each spin case, with the corresponding values of r_e , r_c and r_s shown in the legend. We do not include fit errors in this work, as the small sample size of our simulations leads to large statistical uncertainties, and showing only the parameter-fit errors could give a misleading impression of accuracy.

Looking at the fitted curves, the overall picture is the same as in Figure 5.1: a very good match to the reference soliton in the core region, and then differences in the outer slope between the spins and the reference soliton. The $s = 0$ profile still shows the steepest and cleanest drop, while the higher-spin cases are slightly shallower just beyond r_c . Using the fits smooths out the small oscillations in the data and makes these differences easier to see.

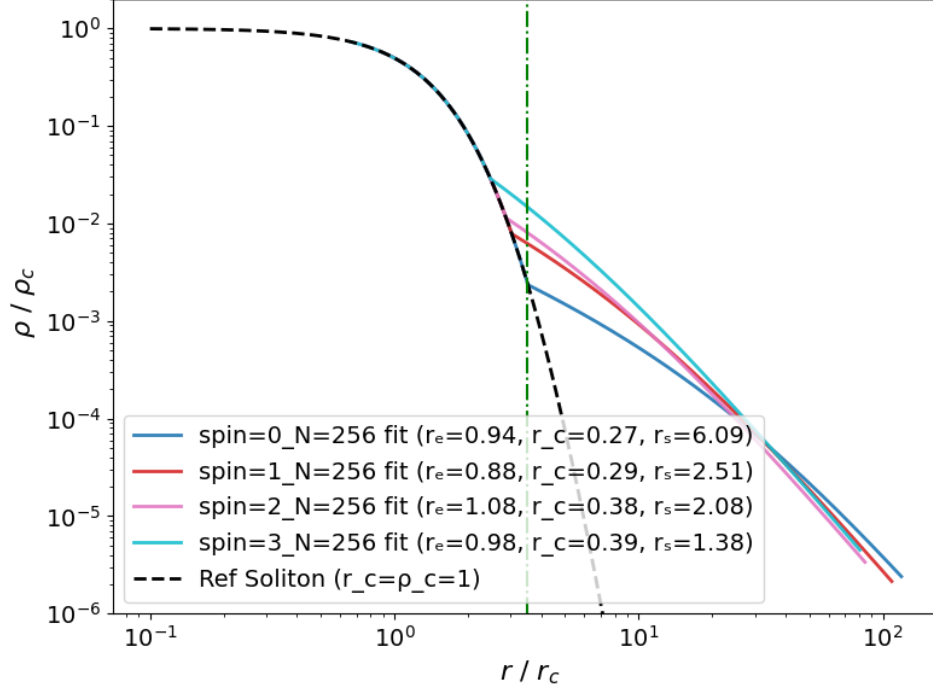


Figure 5.2: Spherically averaged density profiles with the best fits from simulations of a 25 soliton merger in a box of size 100 for spins $s = 0, 1, 2, 3$ at $t = 30$ Gyr. The profiles are normalized to the central density ρ_c and the radius is scaled by the core radius r_c obtained from fitting Equation 2.25. The dashed black line shows the reference soliton profile with $r_c = \rho_c = 1$, while the vertical dash-dotted green line marks the transition radius $r_c \simeq 3.5$ where the fit switches from the solitonic core to the NFW-like outer halo.

5.2 Core Radius Growth

In Figure 5.3 we show the time evolution of the dimensionless core radius parameter $\lambda_\rho(t)$ for spins $s = 0$ (blue), 1 (orange), 2 (green), 3 (red), extracted from our simulations of 25 solitons merger with grid resolution $N = 256$. Here λ_ρ is defined following [22] as the ratio between the initial and central densities at the time step t :

$$\lambda_\rho(t) = \left(\frac{\rho_c^f}{\rho_c^i} \right)^{\frac{1}{4}}, \quad (5.1)$$

and can be interpreted as a measure of core size growth during the merger process. This is because in the soliton profile $\rho_c \propto r_c^{-4}$ (see Equation 2.23), a decrease in the central density ρ_c implies an increase in the core radius r_c . Since $\lambda_\rho(t)$ is defined as in Equation 5.1, values larger than unity correspond to a lower central density and therefore a physically larger solitonic core. The dots represent the raw values obtained at each saved time step, while the solid black lines are exponential fits of the form

$$\lambda_\rho(t) = \lambda_\infty - (\lambda_\infty - 1) e^{-t/\tau} \quad (5.2)$$

which provide a rough estimate of the asymptotic value λ_∞ for each spin case [22]. These λ_∞ values are indicated both in the legend and by the dashed horizontal lines.

In Equation 5.2, λ_∞ denotes the asymptotic value of the core-growth parameter reached at late times, while τ is the characteristic relaxation timescale of the growth process. More precisely, τ corresponds to the time at which the difference from the asymptotic value has

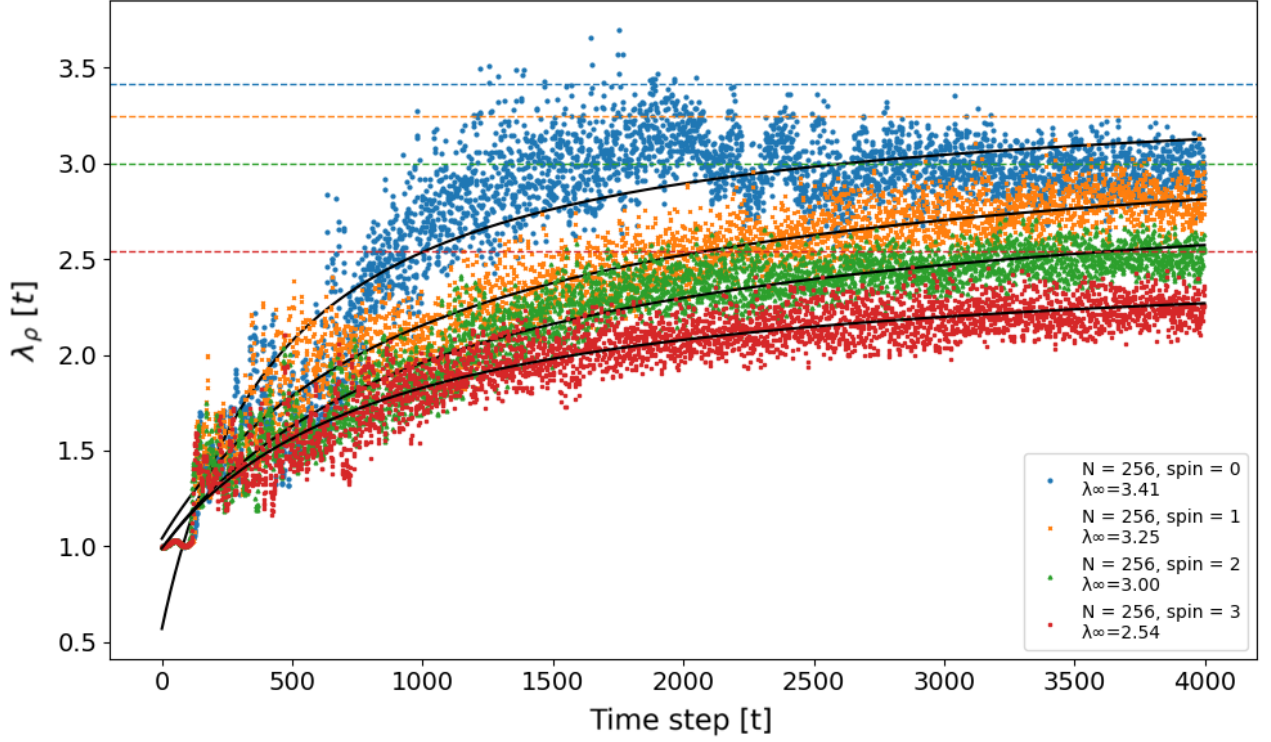


Figure 5.3: Dependence of the dimensionless core radius parameter $\lambda_\rho(t)$ on time from simulations of a 25-soliton merger in a box of size 100 for spins $s = 0$ (blue), 1 (orange), 2 (green), 3 (red) at resolution $N = 256$. Here $\lambda_\rho(t)$, defined in Equation 5.1, is the ratio between the initial and central densities at time step t , so values above unity correspond to a growth of the solitonic core radius during the merger process. Horizontal dashed lines indicate the asymptotic values λ_∞ for each spin.

decreased to $1/e$ of its initial value,

$$\lambda_\infty - \lambda_\rho(\tau) = \frac{\lambda_\infty - 1}{e}.$$

Thus, higher values of τ indicate slower convergence of the core radius towards its final state. For all spins, λ_ρ increases rapidly at the beginning of the simulation and then gradually approaches a plateau, consistent with the core-halo relaxation seen in previous works such as [22].

All of these values are obtained from a single simulation run. Therefore, calculating their errors is of little interest. The $s = 0$ case reaches an asymptotic value of $\lambda_\infty \approx 3.41$, which is slightly larger than the $s = 1$ and $s = 2$ cases, giving $\lambda_\infty \approx 3.25$ and $\lambda_\infty \approx 3.00$, respectively. The case $s = 3$ shows the smallest growth, with $\lambda_\infty \approx 2.54$. These values are in line with the trends reported in [22], where the maximal values of λ_ρ are observed to stabilise at lower levels for higher spins.

In our simulations, the $s = 0$ case appears to continue growing for a longer time before approaching its plateau, while higher-spin cases reach their asymptotic values earlier. Once again, the errors on these parameters are not included here due to the large statistical uncertainties caused by the limited sample size.

The scatter around the fits is mainly due to interference patterns and residual oscillations in the density field after major merger events. Part of these fluctuations can also be attributed to the oscillations seen in Figure 4.3, which arise from discretization effects and numerical noise.

To test whether the trend observed in Figure 5.3 is robust and reproducible and not just a

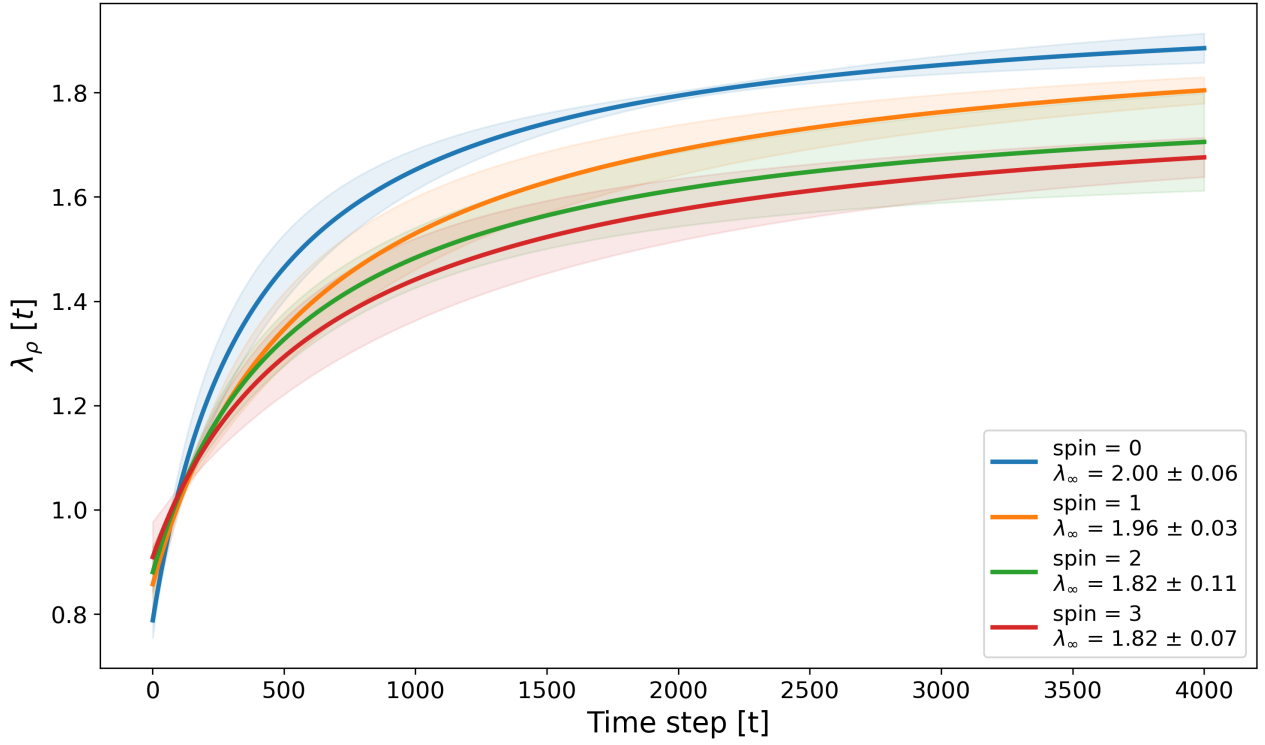


Figure 5.4: Plot showing mean fit curves of the soliton growth parameter $\lambda_\rho(t)$ for spins $s = 0$ (blue), 1 (orange), 2 (green), and 3 (red), averaged across five independent realizations. The lighter shaded regions indicate the 95 % confidence intervals of the mean for each spin.

characteristic phenomenon of a single realisation of a simulation run with a random seed of 1234, the same set of simulations was repeated for five different random seeds ($\text{seed} \in \{1, 2, 3, 4, 5\}$). The simulation setup was fixed for all five of these runs. For each seed, the 25-soliton merger was evolved for spins $s = 0, 1, 2, 3$, and the corresponding $\lambda_\rho(t)$ curves were extracted in the same way as for the single-seed case. Each realisation was then fitted separately with the same saturating function used above, and the fitted curves were subsequently averaged for each spin. The averaged fitted curves with their uncertainties are shown in Figure 5.4, for each spin $s = 0$ (blue), 1 (orange), 2 (green), 3 (red).

The averaged fit parameters are summarized by their mean values and standard deviations. This makes the comparison between different spins more statistically meaningful, as it allows one to distinguish between fluctuations specific to an individual merger and systematic differences that persist across independent realizations.

It should be noted that the averaged values differ from the parameters obtained from the single-seed runs discussed above. This is not unexpected, since individual merger realizations are sensitive to the randomly generated initial positions and phases of the solitonic clumps. A single realization can therefore give a somewhat biased impression of the growth timescale or final saturation level. For this reason, the multi-seed results should be interpreted as a more robust estimate of the spin-dependent trend, while the single-seed runs mainly illustrate the qualitative behaviour of individual mergers. The multi-seed values were recorded into Table 5.1.

The multi-seed analysis confirms that the dependence of core growth on spin remains visible after averaging over different realisations. The lowest-spin cases, $s = 0$ and $s = 1$, reach systematically larger asymptotic values of λ_ρ than the higher-spin cases, $s = 2$ and $s = 3$. Since higher values of λ_ρ correspond to a stronger decrease in the central density and therefore a larger final core radius, this indicates that lower-spin mergers tend to produce more extended

Spin s [-]	n_{runs} [-]	λ_{∞} [-]	τ_{99} [t]
0	5	1.996 ± 0.059	$(2.41 \pm 0.80) \times 10^4$
1	5	1.959 ± 0.032	$(3.64 \pm 1.08) \times 10^4$
2	5	1.825 ± 0.114	$(2.93 \pm 1.08) \times 10^4$
3	5	1.822 ± 0.069	$(3.87 \pm 2.20) \times 10^4$

Table 5.1: Table summarizing the fit parameters obtained from five independent realizations for each spin using the fitting function defined in Equation 5.2. The reported uncertainties correspond to the standard deviations. Here, λ_{∞} is the asymptotic value of the dimensionless core-growth parameter, and τ_{99} is the fitted time required for the curve to reach 99% of this asymptotic value.

relaxed cores. In contrast, the $s = 2$ and $s = 3$ solutions saturate at lower values, suggesting that the final remnant remains more centrally concentrated.

At the same time, the spread of the fitted parameters shows that the precise value of λ_{∞} is still influenced by the stochastic details of the merger process. This is particularly visible for the $s = 2$ case, which exhibits the largest standard deviation in λ_{∞} . Nevertheless, the difference between the low- and high-spin groups is greater than the seed-to-seed scatter within each group, which supports the interpretation that the observed trend is physical rather than purely statistical.

The characteristic times τ_{99} should be interpreted more cautiously. Although they suggest that the saturation time may also depend on spin, the scatter between realisations is relatively large, especially for $s = 3$. Therefore, in the present data set, τ_{99} is best viewed as a secondary indicator of the relaxation process, whereas λ_{∞} provides the more robust measure of the spin dependence of core growth. Overall, the ensemble averaged results support the conclusion that increasing spin suppresses the growth of the solitonic core. This is in qualitative agreement with the single seed simulation and is consistent with the findings shown in [22].

5.3 Core Mass Dependence on Black Hole Mass

5.3.1 The ULDM and BH System

The method for calculating gravitationally stable solitonic cores was introduced in Subsection 2.1.3.

This construction can be modified to include the gravitational potential of a central black hole represented as a point-mass source. The stationary radial problem then takes the form

$$\frac{1}{2} \left(\frac{d^2 \Phi}{dr^2} + \frac{2}{r} \frac{d\Phi}{dr} \right) = [U(r) + U_{\text{BH}}(r) - \omega] \Phi(r), \quad (5.3)$$

$$\frac{d^2 U}{dr^2} + \frac{2}{r} \frac{dU}{dr} = 4\pi \Phi^2(r), \quad (5.4)$$

where $U(r)$ is the self-consistent gravitational potential of the soliton and ω is the eigenvalue of the stationary configuration. The additional contribution from the central black hole is introduced through

$$U_{\text{BH}}(r) = -\frac{GM_{\text{BH}}}{\tilde{\hbar}^2 r}. \quad (5.5)$$

For numerical integration, the system is rewritten as a first-order system by introducing

$$A(r) = \frac{d\Phi}{dr}, \quad B(r) = \frac{dU}{dr}, \quad (5.6)$$

which gives

$$\frac{d\Phi}{dr} = A, \quad \frac{dU}{dr} = B, \quad (5.7)$$

$$\frac{dA}{dr} = 2[U_{\text{tot}}(r) - \omega]\Phi(r) - \frac{2}{r}A, \quad (5.8)$$

$$\frac{dB}{dr} = 4\pi\Phi^2(r) - \frac{2}{r}B. \quad (5.9)$$

The regularity conditions at the origin are imposed as

$$\Phi(0) = 1, \quad U(0) = U_0, \quad \Phi'(0) = 0, \quad U'(0) = 0, \quad (5.10)$$

where the choice $\Phi(0) = 1$ fixes the normalisation of the reference solution, while U_0 and ω are determined numerically so that the resulting profile remains localised and physically acceptable. Because the point-mass potential is singular at $r = 0$, the numerical implementation regularises the radial coordinate near the origin by replacing r with a small cutoff value.

In this way, the original stationary soliton construction is extended to equilibrium configurations in which the solitonic core coexists with a central black hole potential. These solutions can then be used to study how the presence of a black hole modifies the density profile and the allowed equilibrium core mass.

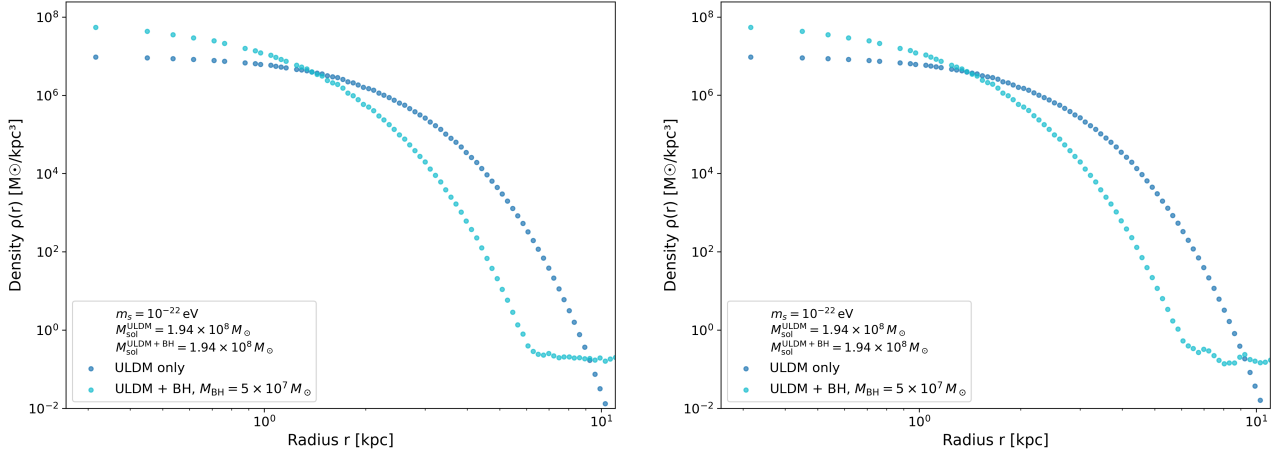
5.3.2 Difference from ULDM Only System

The difference between the two resulting equilibrium configurations is shown in Figure 5.5 through their spherically averaged radial density profiles. ULDM-only results are shown in teal, whereas the ULDM+BH system is shown in blue. Both runs were carried out in a simulation box of size 50 with resolution $N = 256$, for a scalar-field mass $m_s = 10^{-22}$ eV, and evolved to $t = 5$ Gyr. This evolution time was sufficient to verify that both systems relaxed towards stationary equilibrium profiles.

In this comparison, both configurations were initialised with the same soliton mass, $M_{\text{sol}} = 1.94 \times 10^8 M_\odot$ (chosen so that λ from Equation 2.23 is ~ 1). The only difference between the two systems is therefore the presence of a central black hole with mass $M_{\text{BH}} = 5 \times 10^7 M_\odot$ in the ULDM+BH case.

Already at $t = 0$, shown in Figure 5.5a, the ULDM+BH configuration is more centrally concentrated than the isolated ULDM soliton, with a higher central density and a narrower radial profile. This behaviour is consistent with the black hole effectively deepening the central potential well and thereby producing a more compact equilibrium configuration.

For $t = 5$ Gyr, the evolved profiles are shown in Figure 5.5b. The two panels are expected to be nearly identical if the constructed configurations correspond to stable equilibrium solutions, and this is indeed observed. Both the isolated ULDM soliton and the ULDM+BH radial densities are essentially unchanged over the simulated time interval, indicating that the numerical construction yields stable stationary states. A small low-density "foot" appears at large radii, but since it shows up in the outer low-density regions, it is interpreted as a numerical artefact disregarded.



(a) Spherically averaged radial density profiles at $t = 0$. The teal points show the isolated ULDM soliton, while the blue points show the solitonic clump with a central black hole with mass $M_{\text{BH}} = 5 \times 10^7 M_{\odot}$.

(b) Spherically averaged radial density profiles at $t = 5$ Gyr. The teal points show the isolated ULDM soliton, while the blue points show the solitonic clump with a central black hole with mass $M_{\text{BH}} = 5 \times 10^7 M_{\odot}$.

Figure 5.5: Comparison of spherically averaged radial density profiles for two different systems: an isolated ULDM soliton (teal) and a soliton in the presence of a central black hole with mass $M_{\text{BH}} = 5 \times 10^7 M_{\odot}$ (blue). Panel (a) shows the initial profiles at $t = 0$, while panel (b) shows the corresponding profiles after evolution to $t = 5$ Gyr.

5.3.3 Dependence on Mass of the Scalar Field

Another property that can be studied is the dependence of the radial density profile on the scalar-field mass. For a fixed black-hole mass $M_{\text{BH}} = 5 \times 10^7 M_{\odot}$, I considered three representative values of the scalar-field mass,

$$m_s \in \{10^{-21}, 10^{-22}, 10^{-23}\} \text{ eV},$$

and constructed the corresponding equilibrium configurations. The resulting spherically averaged radial density profiles at $t = 5$ Gyr are shown in Figure 5.6. Data for $m_s = 10^{-21}$ eV, 10^{-22} eV, and 10^{-23} eV are shown in blue, brown, and teal, respectively.

A clear dependence on the scalar-field mass is visible. As m_s decreases, the equilibrium configuration becomes both denser and more massive, while the overall qualitative shape of the profile remains similar. In contrast, for the largest scalar-field mass, $m_s = 10^{-21}$ eV, the resulting core is strongly suppressed by the presence of the central black hole, and the profile remains substantially less centrally concentrated. For $m_s = 10^{-22}$ eV and 10^{-23} eV, the profiles retain a pronounced central core and differ primarily in their overall normalisation and radial extent. At larger radii, $r \gtrsim 5$ kpc, both profiles approach a similar low-density tail, although this outermost region should be interpreted with caution because it is more sensitive to numerical noise.

This trend is physically expected. The lower values of m_s correspond to larger characteristic de Broglie wavelengths and therefore to more extended and more massive equilibrium solitons. As a consequence, a fixed central black hole has a weaker relative influence on the inner structure of the configuration when m_s is smaller. Conversely, for larger m_s , the corresponding solitonic core is less massive and can therefore be suppressed more efficiently by the same black-hole potential.

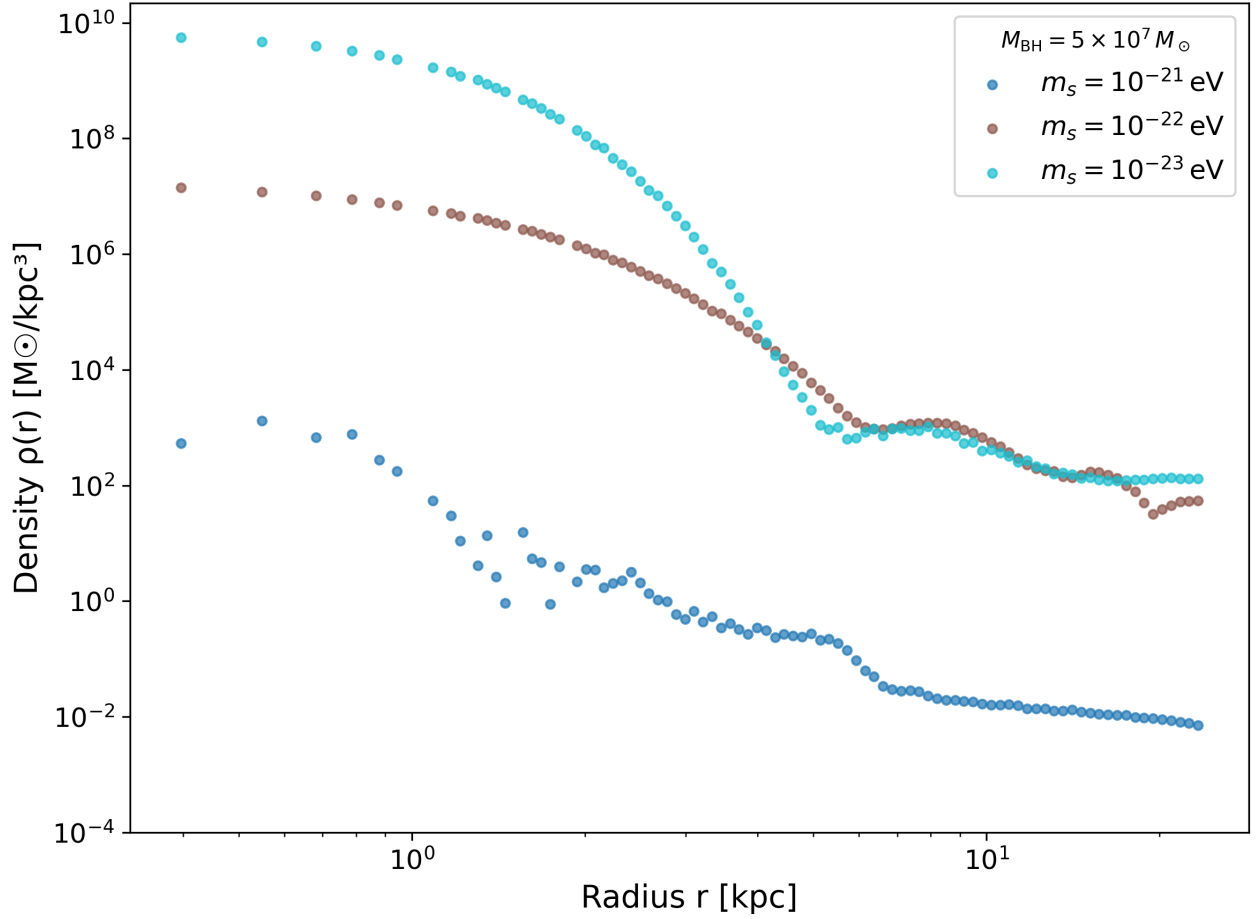


Figure 5.6: Comparison of spherically averaged radial density profiles for three values of the scalar-field mass, $m_s = 10^{-21}$ eV, 10^{-22} eV, and 10^{-23} eV, for a fixed central black-hole mass $M_{\text{BH}} = 5 \times 10^7 M_\odot$.

5.3.4 Scaling Relation

When the relation between the core mass and the black-hole mass is examined in more detail, a tendency is visible for all three cases.

In Figure 5.7 the dependence of the solitonic core mass M_{core} on the black hole mass M_{bh} for the three values of the ULDM scalar field mass is shown. The points represent the core masses obtained from the numerical procedure, described earlier within this section, used to construct equilibrium soliton–black hole systems, while the dashed lines show fits of the form

$$M_{\text{core}}(M_{\text{bh}}) = \frac{M_0}{1 + (M_{\text{bh}}/M_{\text{bh0}})^\alpha}. \quad (5.11)$$

The quantity M_0 is the plateau value of the mass of the core, M_{bh0} is the characteristic black-hole mass at which the suppression of the core begins to become important, and α controls the steepness of the transition.

The shift of the turnover towards larger black-hole masses for smaller values of m_s is also physically expected. Lower ULDM particle masses correspond to larger de Broglie wavelengths and therefore to more extended and typically more massive equilibrium solitons. As a consequence, a more massive black hole is required for its gravitational potential to significantly affect the inner structure of the soliton. This is visible in Figure 5.7, where the $m_s = 10^{-23}$ eV case remains on its plateau up to much larger values of M_{bh} than the $m_s = 10^{-21}$ eV case.

The fitting function given by Equation 5.11 was chosen because it reproduces the two main

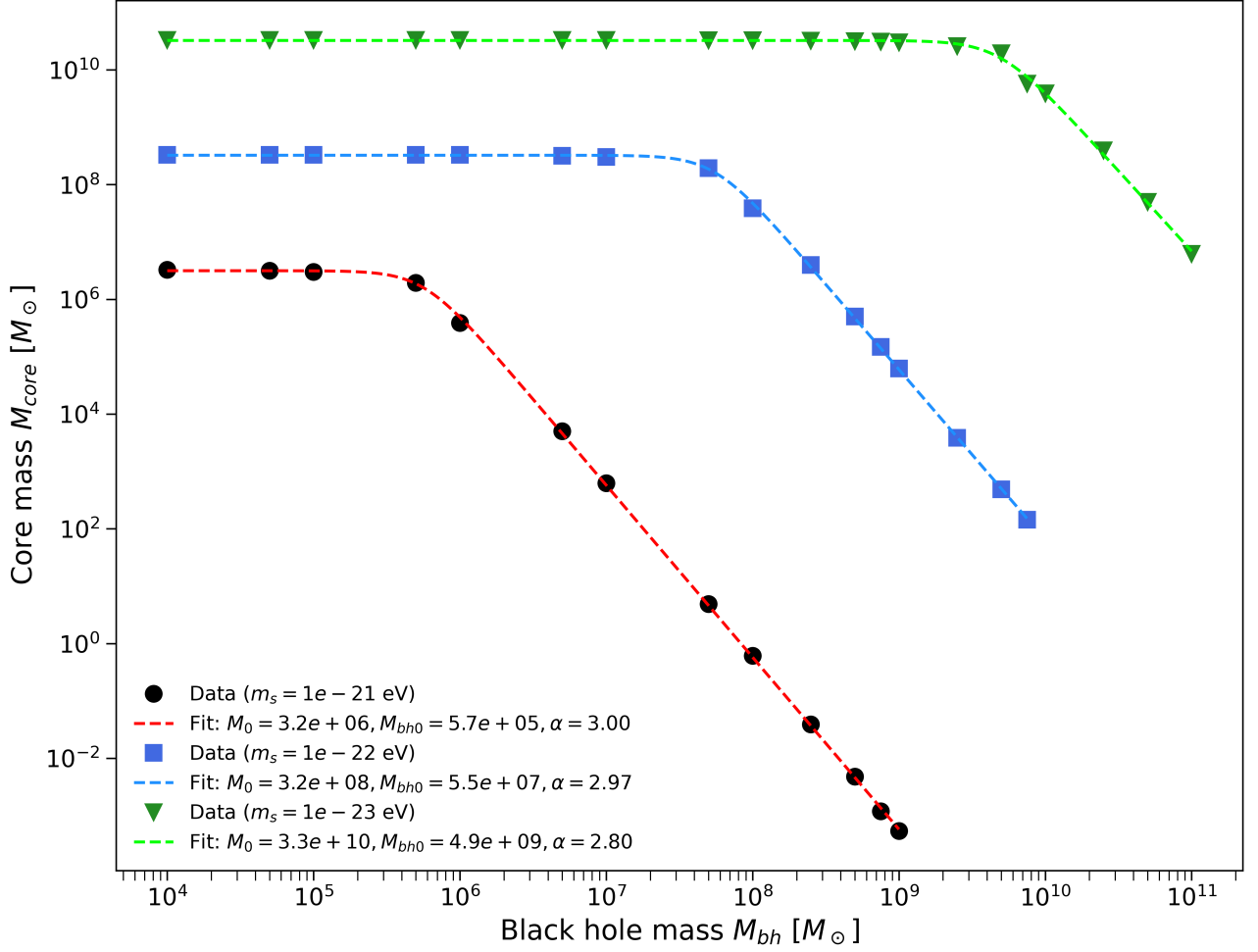


Figure 5.7: Dependence of the solitonic core mass M_{core} on the black hole mass M_{bh} for three values of the ULDM particle mass, $m_s = 10^{-21}$ eV, 10^{-22} eV, and 10^{-23} eV. The points show the numerically obtained equilibrium core masses, while the dashed lines represent fits of the form $M_{\text{core}} = M_0 / [1 + (M_{\text{bh}}/M_{\text{bh0}})^\alpha]$. For each m_s , the core mass remains approximately constant for low black hole masses and then decreases once M_{bh} exceeds a characteristic scale, indicating the increasing influence of the central black hole on the solitonic core structure.

regimes seen in the data while remaining simple enough to interpret. For $M_{\text{bh}} \ll M_{\text{bh0}}$, it approaches $M_{\text{core}} \approx M_0$, capturing the nearly constant core-mass regime. For $M_{\text{bh}} \gg M_{\text{bh0}}$, it reduces to

$$M_{\text{core}} \propto M_{\text{bh}}^{-\alpha}, \quad (5.12)$$

which describes the approximately power-law suppression seen at high black-hole masses shown in Equation 2.27.

Error bars are not shown in Figure 5.7, since the present analysis is based on a limited number of numerically constructed equilibrium configurations and the dominant uncertainty is systematic rather than statistical.

5.3.5 Exploratory scaling estimate

As an additional exploratory exercise, I tested whether the fitted relation from Equation 5.11 can be combined with the black-hole-halo mass relation and the soliton-halo scaling introduced in Section 2.5. For a given black-hole mass M_{BH} , the corresponding host halo mass was first estimated by inverting the empirical black-hole-halo relation. The soliton mass was then esti-

mated from the standard core–halo scaling relation for the selected scalar-field mass m_s . In this way, the two scaling relations provide an approximate prediction for the solitonic core mass associated with a given black-hole mass.

To compare this estimate with the numerical evolution, the soliton–black-hole systems were not evaluated only as static configurations. Instead, each run was initialised with a soliton corresponding to an initial black-hole mass M_{BH}^i . The black-hole mass was then increased during the simulation until it reached a final value M_{BH}^f . The mass growth was implemented by adding a fixed increment $\delta M = (M_{\text{BH}}^f - M_{\text{BH}}^i)/(0.75 \times N_{\text{steps}})$ to the black-hole mass at each time step, so that the final mass was reached after approximately three quarters of the total simulation time, leaving the remaining quarter for the soliton to relax towards a new equilibrium. After the system had evolved, the remaining solitonic core mass within a given radius M_{sol}^f was measured from the final density profile and compared with the value predicted from the scaling estimate for M_{BH}^f .

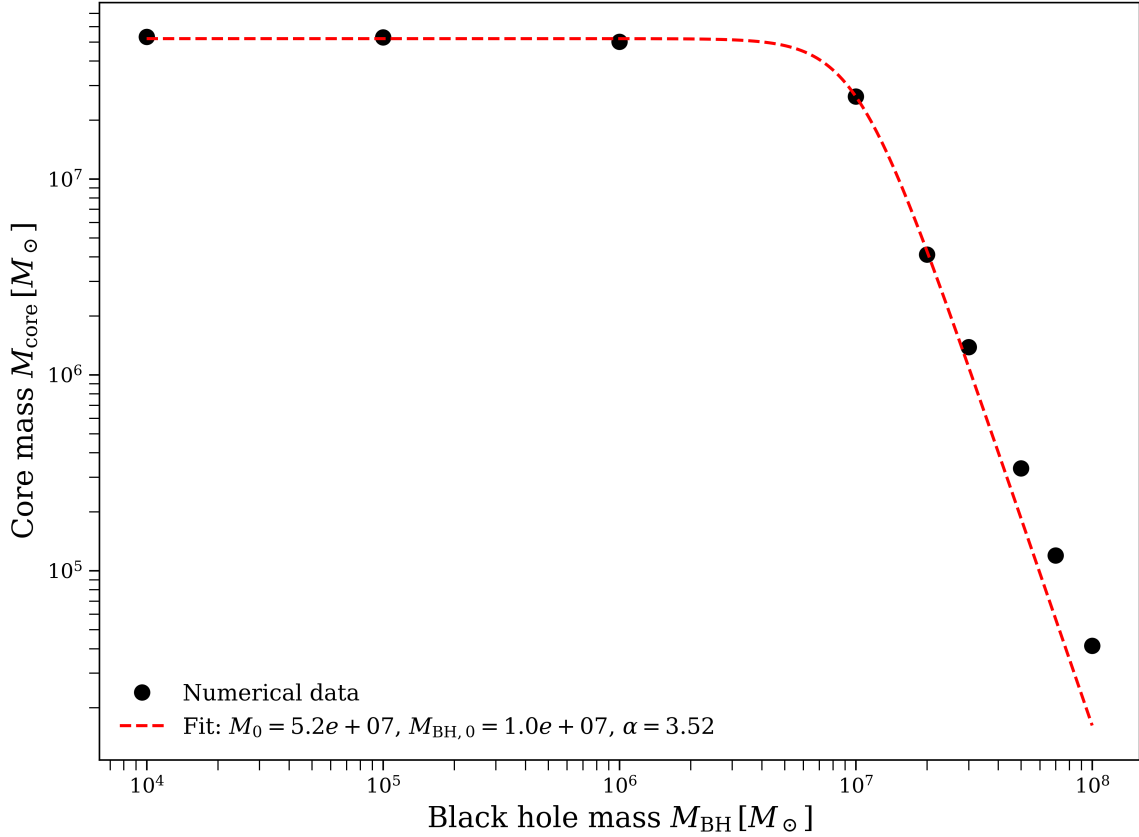


Figure 5.8: Exploratory comparison between the core masses predicted by combining the empirical black-hole–halo relation with the soliton–halo scaling, and the final solitonic core masses measured after dynamical evolution with black-hole mass growth for $m_s = 2.5 \times 10^{-22}$ eV. The plot is intended only as a qualitative proof of concept.

The resulting comparison for $m_s = 2.5 \times 10^{-22}$ eV is shown in Figure 5.8. In this sense, the figure tests whether the combined scaling estimate can approximately predict the final core mass after the black hole has grown inside the solitonic configuration. The agreement is qualitatively reasonable and suggests that the suppression model captures the expected trend over a finite range of black-hole masses. However, this comparison should be interpreted with caution. The procedure combines several approximate ingredients, including the empirical black-hole–halo relation, the core–halo scaling, the numerical construction of the initial solitonic profiles, the imposed black-hole mass growth, and the subsequent measurement of the final core mass. The accumulated uncertainty is therefore substantial, and the present comparison should be

regarded primarily as a proof of concept rather than as a quantitatively robust prediction. Similarly to Figure 5.6, the errors are omitted.

5.4 Gas–Soliton Dynamics and Sink Accretion

To study the dynamics the ULDM may have on a galactic cold gas, I tested the evolution of a coupled self-gravitating gas component and an ULDM soliton, including the formation and subsequent growth of a central unresolved sink.

The purpose of this test is not to model the relativistic formation of a black hole. Instead, the sink particle should be interpreted as a sub-grid representation of an unresolved compact object that can accrete mass. It is created when the gas density exceeds the adopted Truelove-like threshold Equation 3.21, indicating that the local collapse can no longer be resolved on the grid. After the sink has formed, its subsequent growth is followed using the gas accretion prescription implemented in Equation 2.56.

The gas was initialised as a compact Hernquist distribution, using the profile introduced in Equation 2.50, with zero initial velocity. The same gas configuration was evolved in two cases: a gas-only setup and a setup in which the gas evolved in the presence of a central ULDM soliton. In both cases, gas cooling and sink formation were enabled. The cooling time in Equation 2.46 was set to $t_{\text{cool}} = 0.3 \text{ Gyr}$. This value was chosen to make the gas sufficiently dissipative to undergo collapse early in the simulation, while avoiding an immediate loss of pressure support.

A sink particle was created only if the gas density in a given grid cell remained above the critical density threshold for 20 consecutive time steps. This consecutive-step requirement was introduced to suppress sink formation triggered by short-lived density fluctuations and to ensure that the collapse criterion corresponded to a persistent unresolved overly dense region rather than to a transient numerical peak.

To avoid interpreting numerical fragmentation as physical multiple-sink formation, gas-sink creation was restricted to the first central unresolved-collapse event; subsequent evolution was followed only through accretion onto the existing sink.

For each case, five independent realisations were performed using different random perturbation seeds, again selected as $\text{seed} \in \{1, 2, 3, 4, 5\}$. The cumulative accreted gas mass was then averaged over the five runs, and plotted in Figure 5.9. The shaded regions shows the corresponding 95% confidence interval. The initial seed mass of the sink was nearly identical in the two setups, so the comparison isolates the post-formation gas accretion phase.

Figure 5.9 shows that the presence of the soliton, represented by the orange curve, significantly increases the amount of gas accreted by the sink compared to the gas-only runs, shown by the blue curve. The initial small jump in both curves indicates that the sink forms at approximately the same time in both configurations. This is further supported by Figure 5.10, which shows the cumulative gas mass associated with the sink, including both the mass converted into the initial sink seed and the mass subsequently accreted from the gas. The initial jump is almost identical in the two runs, indicating that the mass of the sink at formation is nearly unchanged by the presence of the soliton. The later divergence therefore reflects enhanced post-formation accretion rather than a difference in the initial sink mass.

The scatter between the different seed realisations is very small, as shown by the nearly vanishing confidence intervals. The two curves remain close during the early phase of evolution, but begin to separate approximately 0.3 Gyr, indicating that the soliton mainly affects subsequent post-formation accretion rather than the initial sink-formation event.

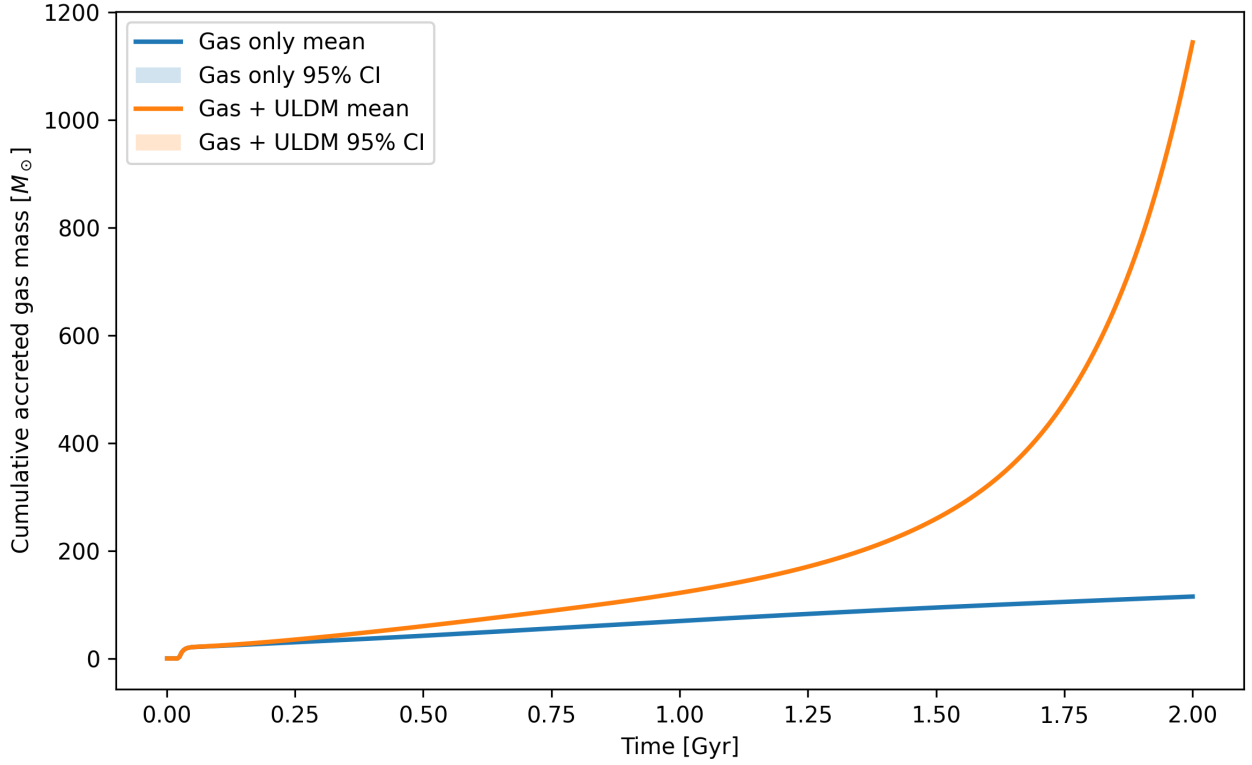


Figure 5.9: Mean cumulative gas mass accreted onto the central sink in gas-only and gas+ULDM simulations. Each curve represents the mean of five independent realizations with different small random perturbations in the initial gas density. The shaded regions show the corresponding 95% confidence intervals.

At $t = 2$ Gyr, the cumulative mass of the accreted gas is:

$$M_{\text{acc}}^{\text{gas-only}}(2 \text{ Gyr}) = 114.95 \pm 0.011 M_{\odot},$$

whereas the gas+ULDM runs give

$$M_{\text{acc}}^{\text{gas+ULDM}}(2 \text{ Gyr}) = 1143.61 \pm 0.12 M_{\odot}.$$

Thus, in this controlled setup, the soliton increases the cumulative accreted mass by a factor of

$$\frac{M_{\text{acc}}^{\text{gas+ULDM}}}{M_{\text{acc}}^{\text{gas-only}}} = 9.949 \pm 0.001,$$

corresponding to a relative increase of $894.9 \pm 0.1 \%$.

To test if the soliton has any impact on the initial mass of the sink created, I tracked the cumulative mass the sink striped from the gas. The results of this test are shown in Figure 5.10, where we can see that the initial sink mass manifested as a jump at the start of the plot is the same for both gas only and gas + ULDM soliton run.

5.5 Composite System Mergers

One of the main goals of this work is to investigate possible ways of including baryonic matter in the ULDM simulation framework and to assess how baryonic components affect the evolution of the system. The preceding sections showed that a central black-hole-like sink can modify the equilibrium solitonic core structure and that the presence of an ULDM soliton can

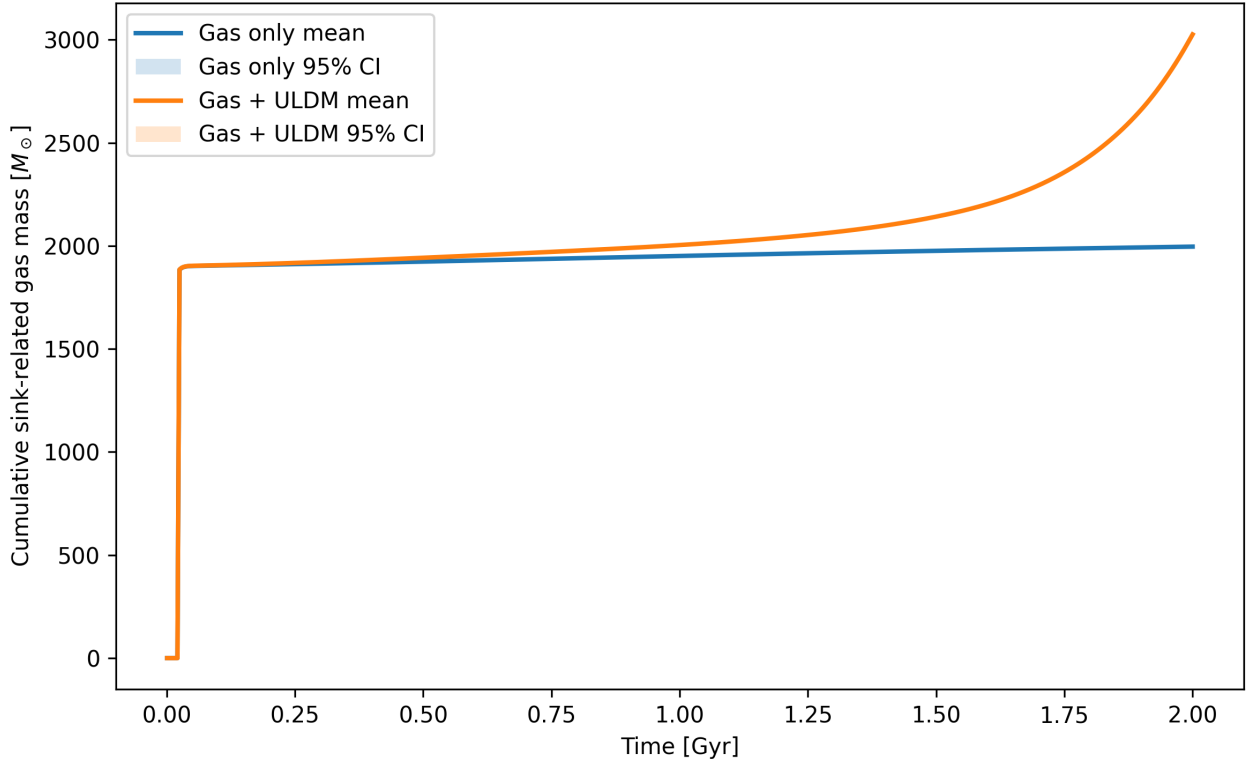


Figure 5.10: Mean cumulative sink-related gas mass for the gas-only (blue) and gas+ULDM (orange) simulations. The plotted quantity includes both the gas mass converted into the initial sink seed at the moment of sink formation and the gas mass subsequently accreted onto the sink. Each curve represents the mean of five independent realizations, and the shaded regions show the corresponding 95% confidence intervals.

enhance post-formation gas accretion onto a central sink. These results motivate the next step: testing whether the implemented components can be combined in dynamically non-trivial multi-component systems. For this reason, I performed a series of exploratory merger simulations of composite ULDM–baryonic systems. These simulations are not intended to represent fully self-consistent galaxy mergers, but rather controlled toy models used to test the coupled dynamics of the implemented components.

The complexity of the system is increased gradually. I first test the stability of isolated composite configurations and then study mergers of systems containing a ULDM soliton and a central sink. Additional collisionless and gaseous baryonic components are then added to examine how they modify the merger dynamics.

5.5.1 Initial Conditions for Composite Systems

The composite merger simulations presented in this section use the galaxy-like setup motivated in Section 2.5, but with reduced component masses chosen to keep the systems numerically stable and well resolved. The aim is not to reproduce a fully self-consistent galaxy merger, but to construct controlled composite systems in which the effect of adding individual baryonic components to an ULDM soliton can be tested.

The central compact object is represented by a sink particle with mass

$$M_{\text{sink}} = 10^6 M_{\odot}. \quad (5.13)$$

This value is smaller than the Milky-Way reference mass discussed in Section 2.5, but remains

in the range of supermassive black-hole proxies and keeps the central potential weak enough that the soliton is not immediately dominated by the sink.

The ULDM component is initialised as a stationary solitonic solution obtained from the static Schrödinger–Poisson eigenvalue problem introduced in Equation 2.15. The soliton mass is set to

$$M_{\text{sol}} = 3.3 \times 10^8 M_{\odot}. \quad (5.14)$$

This value was chosen because it corresponds to a stable, well-resolved soliton for the adopted grid spacing and does not require an additional strong rescaling of the equilibrium profile after solving the eigenvalue problem. This is important because excessive rescaling can make the soliton core too compact compared to the grid resolution, causing artificial relaxation before the actual merger dynamics begins.

The baryonic mass is chosen as a local fraction of the resolved ULDM mass $M_{\text{bar}} = 0.15 M_{\text{sol}}$. This gives a total baryonic mass comparable, in relative terms, to the cosmic baryon fraction discussed in Section 2.5, but applied only to the resolved central system rather than to the full host halo. The baryonic mass is split into a collisionless Hernquist component and a gaseous disk as

$$M_{\text{H}} = 0.6 M_{\text{bar}} = 3.0 \times 10^7 M_{\odot}, \quad (5.15)$$

and

$$M_{\text{gas}} = 0.4 M_{\text{bar}} = 2.0 \times 10^7 M_{\odot}. \quad (5.16)$$

This choice keeps the baryonic components dynamically relevant while preventing them from dominating the gravitational potential of the ULDM soliton. In particular, using a substantially larger gas mass led to stronger hydrodynamic sub-cycling through the CFL condition and to less stable initial configurations.

The collision-less stellar component is initialised with a Hernquist profile. The scale radius and truncation radius are set to

$$a_{\text{H}} = 10 \text{ kpc}, \quad r_{\text{trunc,H}} = 10 \text{ kpc}. \quad (5.17)$$

These values produce an extended and relatively diffuse collisionless component. This is useful for the present simulations, because a more compact Hernquist distribution would create a stronger central baryonic potential and would lead to stronger initial relaxation of the composite system. The Hernquist component was initialised using $N_{\text{particles}} = 1 \times 10^6$ particles. This value was chosen as a compromise between sufficiently smooth sampling of the collisionless density profile and the computational cost of particle deposition and force interpolation.

The gas component was created as an exponential disk with density shape

$$\rho_{\text{gas}}(R, z) \propto \exp\left(-\frac{R}{R_0}\right) \exp\left(-\frac{|z|}{Z_0}\right), \quad (5.18)$$

where

$$R_0 = 4 \text{ kpc}, \quad Z_0 = 0.8 \text{ kpc}. \quad (5.19)$$

The disk is smoothly tapered at large radii using a cut-off radius

$$R_{\text{cut}} = 15 \text{ kpc}, \quad (5.20)$$

with transition width

$$dR_{\text{cut}} = 1.5 \text{ kpc}. \quad (5.21)$$

The smooth taper avoids a sharp density discontinuity at the edge of the gas disk, which would otherwise generate numerical artifacts during the hydrodynamic evolution.

Before performing merger simulations, each composite configuration is evolved in isolation. This allows us to check whether the chosen combination of soliton, sink, collision-less particles, and gas remains approximately stable on its own. Only after this stability test are two copies of the system placed into the simulation box and allowed to merge.

The values used in the simulations presented in this section are summarised in Table A.1. The same setup was used for both the three- and four-component runs, with the only difference being that the gaseous component was not present in the simpler system.

5.5.2 Stability Testing

To assess the stability of the initialised composite systems, I monitored their virial ratio. For a gravitationally bound system in virial equilibrium, twice the kinetic energy should approximately balance the absolute value of the gravitational potential energy. Since the code already tracks the total energy and its individual components, as outlined in Subsection 3.4.7, this can be tested by plotting

$$2K + W \approx 0 \quad \Longleftrightarrow \quad \frac{2K}{|W|} \approx 1, \quad (5.22)$$

and checking whether the system remains bounded and oscillates around the value of one. Additional quantities, such as the total tracked energy and the density distribution, were monitored during the runs, but are not shown separately, since the purpose of this test is only to verify that the isolated composite systems do not undergo rapid disruption before being used as initial conditions for the merger simulations.

The resulting graphs tracking the virial ratio for four different systems of increasing complexity are shown in Figure 5.11. The first tested system contains only the ULDM soliton, Figure 5.11a. The second system contains the ULDM soliton together with the central sink particle, Figure 5.11b. The third system contains the ULDM soliton, the sink particle, and a collisionless N-body component initialised with the Hernquist profile given in Equation 2.50. Its virial ratio is shown in Figure 5.11c. Finally, the fourth system includes all components, i.e. the ULDM soliton, the sink particle, the Hernquist N-body component, and the gaseous disk. All simulation runs were carried out in a simulation box with $L = 100$ and resolution $N = 256$. The 3 more stabilised run were simulated until the total time of 2 Gyrs, while the fourth system required more time to get to a more stabilised configuration and was therefore simulated until the total time of 10 Gyrs. The resulting virial ratio for the 4 component run is shown in Figure 5.11d. All component masses and initial parameters are the same as those introduced in the previous section.

From the virial-ratio plots for the ULDM-only and ULDM + sink particle runs, Figure 5.11a and Figure 5.11b, we can see that both systems remain very close to the virial value of one. The mean values oscillate only slightly around

$$\mu_{\text{ULDM}} = 1.003 \pm 0.001$$

and

$$\mu_{\text{ULDM+BH}} = 1.013 \pm 0.001.$$

This indicates that the isolated soliton is well virialized and that the addition of the central sink particle does not significantly disturb its equilibrium.

For the two more complex systems, shown in Figure 5.11c and Figure 5.11d, the behaviour is less ideal but still remains bounded. In the three-component case, consisting of the ULDM soliton, sink particle, and Hernquist N-body component, the virial ratio first increases and then settles into oscillations around

$$\mu_{\text{ULDM+BH+H}} = 1.102 \pm 0.018.$$

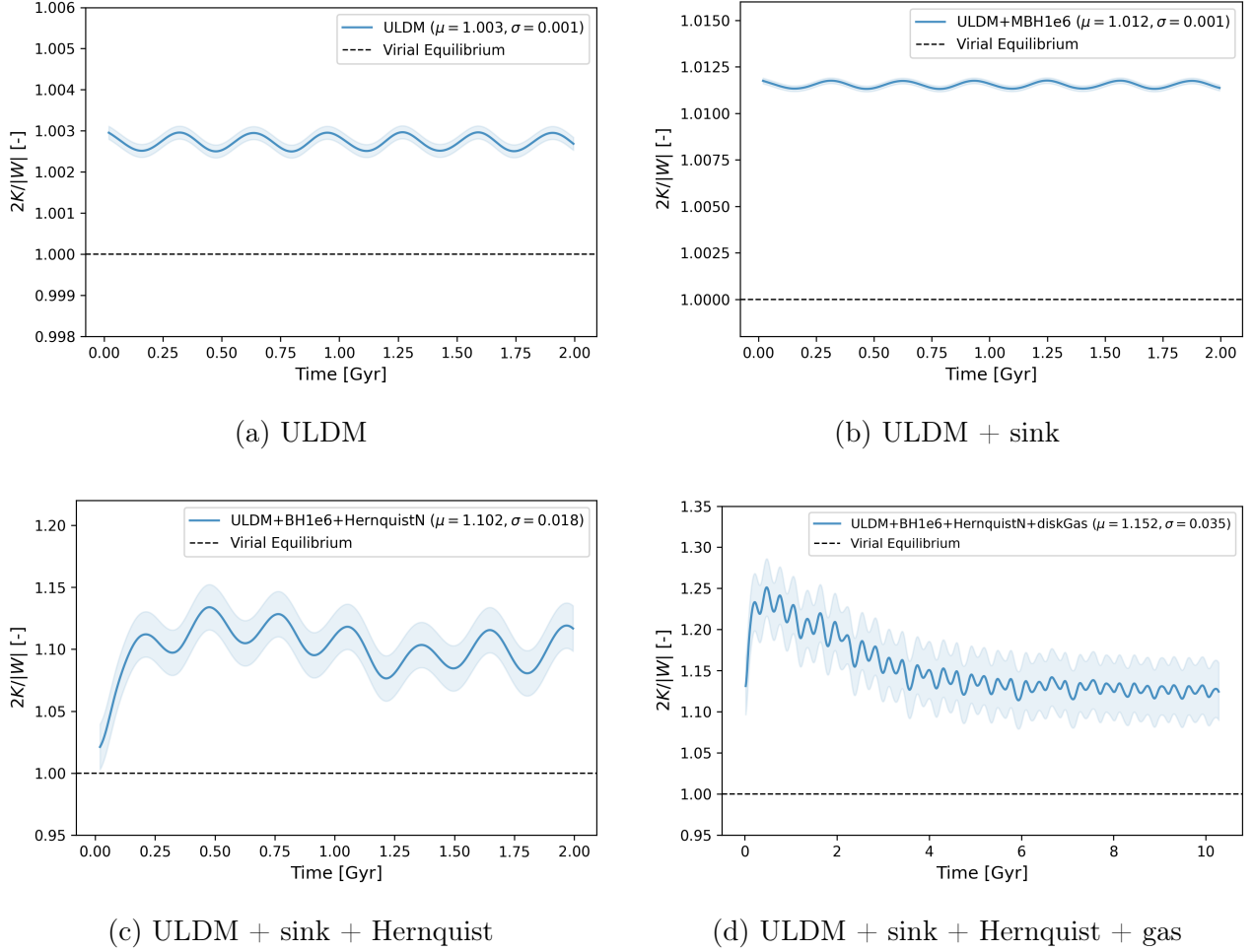


Figure 5.11: Virial ratio $2K/|W|$ for isolated systems of increasing complexity: ULDM only, ULDM + sink, ULDM + sink + Hernquist N body component, and ULDM + sink + Hernquist N body component + gas disk. The dotted line marks the virialized value $2K/|W| = 1$. All configurations use the same initial soliton mass $M_{\text{sol}} = 3.3 \times 10^8 M_{\odot}$. The added component masses are $M_{\text{sink}} = 1.0 \times 10^6 M_{\odot}$, $M_{\text{Hernquist}} = 3.0 \times 10^7 M_{\odot}$, and $M_{\text{gas}} = 2.0 \times 10^7 M_{\odot}$. The values of μ and σ in the panel titles denote the mean and standard deviation of the virial ratio over the analysed time interval.

This suggests that the system is not in perfect virial equilibrium at the start of the sim, but relaxes into a quasi-stable state, which is sufficient for our simulations.

The four-component system, which additionally includes the gaseous disc, shows the largest departure from unity. During the first giga-year of the evolution, the virial ratio rises to values of around 1.25 and then decreases gradually, indicating that the initial configuration is still relaxing. After approximately 6 Gyrs, the system reaches bounded oscillations around

$$\mu_{\text{all}} = 1.152 \pm 0.035.$$

The larger offset and scatter are expected, since the gas component introduces additional hydrodynamic evolution and is not generated as an exact equilibrium solution of the full composite gravitational potential.

Another reason why composite systems do not exactly reach the virial value is that the initial soliton is not an exact equilibrium solution of the full gravitational potential of all components. As discussed in Section 5.3, the stationary solitonic profile is obtained by solving the radial Schrödinger–Poisson eigenvalue problem (Equation 2.15). This construction can be extended by including additional spherically symmetric contributions to the gravitational potential. For

example, the contribution of a Hernquist component can be included through the analytic potential

$$\Phi_{\text{H}}(r) = -\frac{GM_{\text{H}}}{r + a_{\text{H}}}, \quad (5.23)$$

[59], so that the effective potential entering the radial eigenvalue problem becomes

$$\Phi_{\text{eff}}(r) = \Phi_{\text{sol}}(r) + \Phi_{\text{sink}}(r) + \Phi_{\text{H}}(r). \quad (5.24)$$

This is still compatible with the radial solver, because the Hernquist profile is spherically symmetric.

The situation is less straightforward for the gaseous disc. Since it is flattened, its gravitational potential is not spherically symmetric, whereas the soliton solver assumes spherical symmetry. Therefore, the disc cannot be included exactly in the radial eigenvalue problem. At most, it can be represented by a spherically averaged approximation, which captures only the monopole part of its gravitational field. As a result, the resulting soliton is only approximately adapted to the full four-component system. This explains why the configuration containing even the gas components shows a longer relaxation phase and a larger offset of the virial ratio from unity.

Overall, the stability tests show that all isolated configurations remain gravitationally bound and do not undergo rapid disruption. The more complex systems are not exact equilibrium solutions, but they relax into quasi-stable configurations. Therefore, the final state of the three- and four-component run can be used as a blueprint for the toy galaxy merger simulations.

5.5.3 System Mergers

The stable systems obtained from the simulations in the previous subsection were duplicated in a box of size $L = 100$ with a resolution of $N = 256$. This corresponds to a spatial resolution of $dx = 100/256 \approx 0.39$ kpc. The two systems were created at positions $\vec{p}_1 = (10, 10, 0)$ and $\vec{p}_2 = (-10, -10, 0)$ and assigned initial velocities $\vec{v}_1 = (-2, 0, 0)$ and $\vec{v}_2 = (2, 0, 0)$, respectively. This setup was chosen to mimic a merger in the $x - y$ plane with a non-zero impact parameter and non-trivial relative motion.

As a reference case, a system containing only two sink particles, each with a mass of $M_{\text{BH}} = 10^6 M_{\odot}$, was simulated. The resulting trajectories are shown in Figure 5.12. As expected, in the absence of an extended surrounding medium, the two sinks fall towards each other and merge after one or two close approaches. Therefore, this case is used only as a reference, while the following analysis focuses on the more complex composite systems.

5.5.4 ULDM + Sink + N-body Mergers

A more interesting scenario is obtained for a three-component system, with masses distributed according to the setup described in Subsection 5.5.1. The two systems were again initialised at positions $\vec{p}_1 = (10, 10, 0)$ and $\vec{p}_2 = (-10, -10, 0)$ and assigned initial velocities $\vec{v}_1 = (-2, 0, 0)$ and $\vec{v}_2 = (2, 0, 0)$.

The trajectories of the two central sink particles are immediately more interesting. In Figure 5.13, the two systems approach each other and merge into a single bound remnant. Following the two central sink particles provides a simple way of tracking how the two systems rearrange and settle into the common potential. After the first close encounter, the sinks form a long-lived bound pair and continue to orbit around their common centre. The result qualitatively resembles a stalled bound pair on the resolved scales of the simulation. The large-scale merger forms a bound central pair orbiting one another, but the subsequent reduction of the sink separation becomes inefficient on the resolved scales of the simulation.

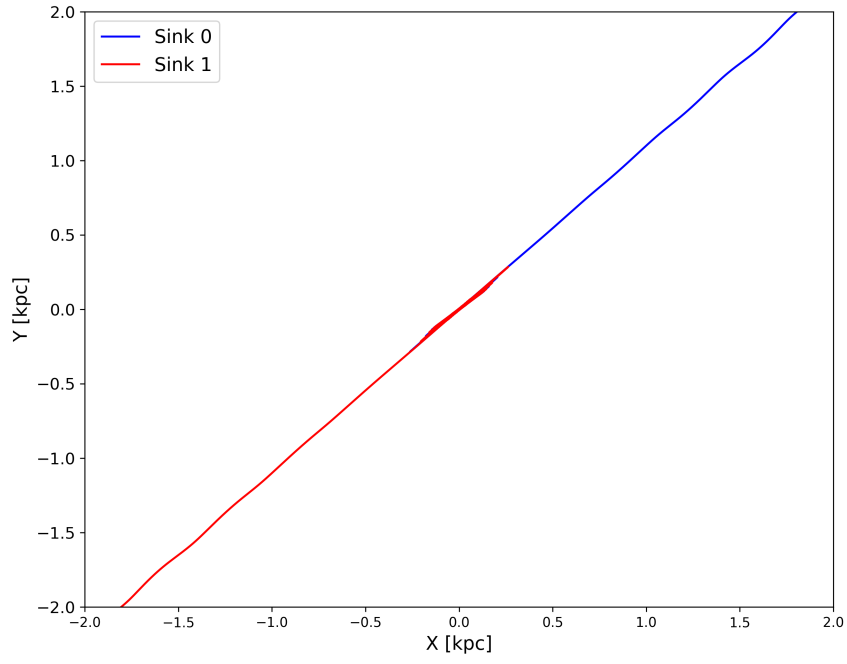


Figure 5.12: Close up to trajectories of two sink particles initialized at $\vec{p}_1 = (10, 10, 0)$ and $\vec{p}_2 = (-10, -10, 0)$ with initial velocities $\vec{v}_1 = (-2, 0, 0)$ and $\vec{v}_2 = (2, 0, 0)$, respectively.

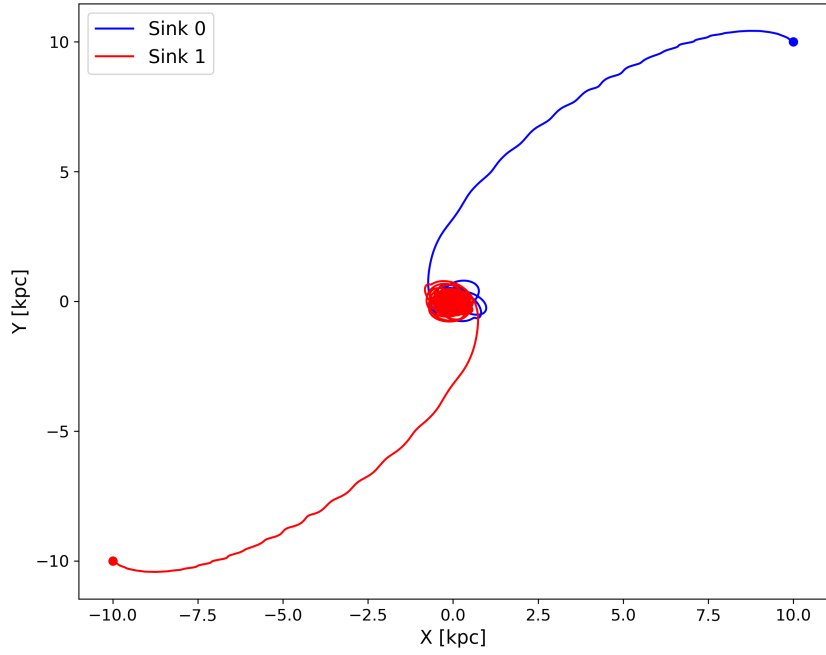


Figure 5.13: Trajectories of the two central sink particles in the ULDM + sink + N-body merger simulation. The two systems have been initialized at positions $\vec{p}_1 = (10, 10, 0)$ and $\vec{p}_2 = (-10, -10, 0)$ with velocities $\vec{v}_1 = (-2, 0, 0)$ and $\vec{v}_2 = (2, 0, 0)$.

To quantify the post-merger evolution, the separation between the two sinks was analysed after the first close approach, which corresponds to the main merger event. Since the separation oscillates as the two sinks orbit around each other, local maxima of the separation were extracted and fitted with an exponential envelope,

$$d_{\text{env}}(t) = d_{\text{floor}} + A \exp[-\gamma(t - t_0)], \quad (5.25)$$

where A is the initial amplitude of the envelope, γ is the decay rate, d_{floor} is the asymptotic separation floor, and t_0 is the time of the first maximum used in the fit.

For this metric to be comparable with the latter case, where the system also contains a gas component, the characteristic dynamical time was defined as

$$t_{\text{dyn}} = \sqrt{\frac{R_0^3}{GM_{\text{tot}}}}, \quad (5.26)$$

where R_0 is the initial separation of the two systems and M_{tot} is their total mass. The fitted decay rate can then be expressed in dimensionless form as

$$\gamma_{\text{scaled}} = \gamma t_{\text{dyn}}. \quad (5.27)$$

For the ULDM + sink + N-body merger, this procedure gives

$$\gamma t_{\text{dyn}} = 0.98 \pm 0.29, \quad (5.28)$$

corresponding to a physical decay rate of

$$\gamma = 1.01 \pm 0.30 \text{ Gyr}^{-1}. \quad (5.29)$$

The associated decay timescale is approximately

$$\tau = \frac{1}{\gamma} = 0.99 \pm 0.29 \text{ Gyr}. \quad (5.30)$$

The resulting fitted data are shown in Figure 5.14, where the sink separation is fitted using the exponential envelope defined in Equation 5.25. Due to the finite spatial resolution, a significant fraction of the small-amplitude oscillations occur close to or below the resolution limit. Nevertheless, the plot shows that after the first close encounter, the two sinks remain in a bound configuration and continue to orbit with a gradually decreasing separation amplitude.

This behaviour is different from the sink-only reference run, where the two sinks merge almost immediately. In the composite system, the surrounding ULDM and N-body components modify the local gravitational environment and the relative motion of the sinks. As a result, the sink pair remains separated on the resolved scales of the simulation, even though the large-scale systems have already merged.

5.5.5 Composite Mergers with Gas

The final frontier of this work is a simulation of a merger of two toy-galaxies which include a component for ULDM, BH proxy, collisionless N-body representing stars, and galactic gas. As a blueprint for this system, a snapshot of all the simulated components have been taken from the end of the stabilisational run shown in Figure 5.11d. Once again, two copies of this model have been created at positions $\vec{p}_1 = (10, 10, 0)$ and $\vec{p}_2 = (-10, -10, 0)$ with velocities $\vec{v}_1 = (-2, 0, 0)$ and $\vec{v}_2 = (2, 0, 0)$.

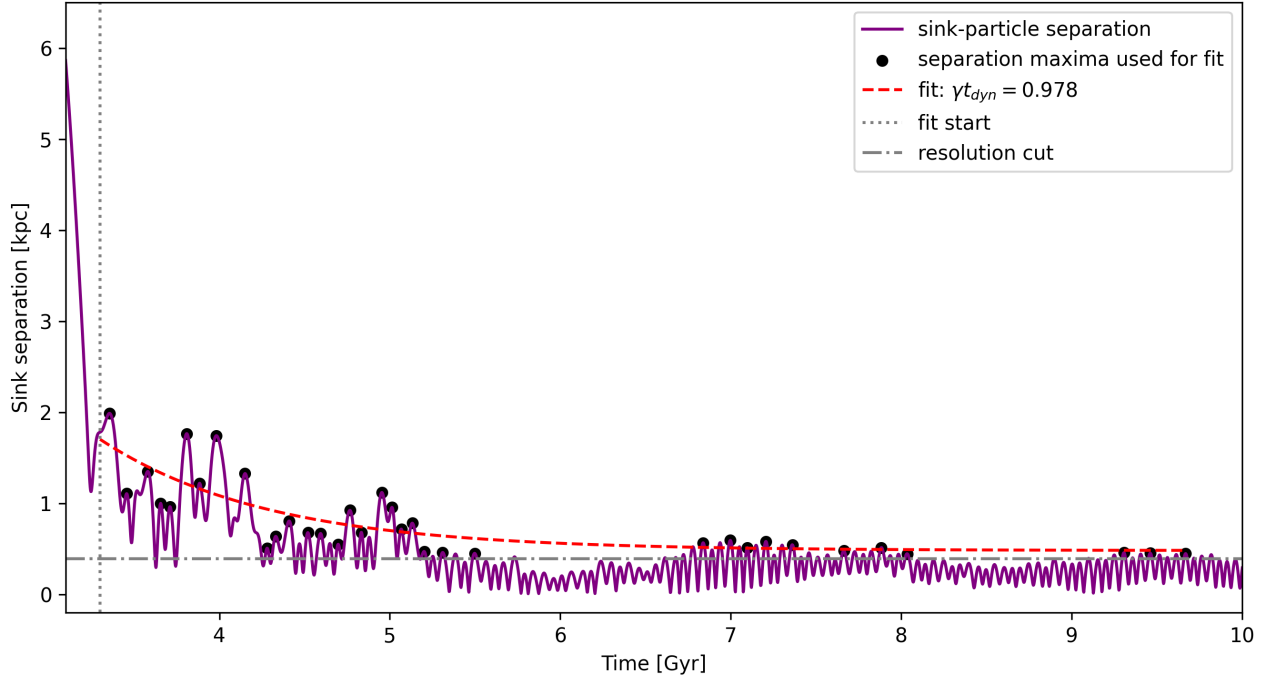


Figure 5.14: Time evolution of the separation between the two central sink particles in the ULDM + sink + N-body merger simulation. Local maxima of the separation after the first close encounter were fitted with an exponential envelope to estimate the decay rate of the bound sink pair. The horizontal line marks the spatial resolution limit of the simulation and the vertical line marks the first close approach corresponding to the merger event.

To further characterise the dynamics of the merger, Figure 5.15 shows the time evolution of the individual energy components. The first close encounter at $t \approx 3.0$ Gyr is clearly visible as a sharp transient in all components, after which the system enters a new oscillatory post-merger state. The increased amplitude of the oscillation after the encounter reflects the conversion of the relative orbital kinetic energy into the internal motions of the merged remnant.

The quantum-pressure contribution U_{quantum} , defined in Equation 1.14, shows particularly strong post-merger oscillations. This is consistent with enhanced wave interference and density redistribution in the ULDM component following the violent relaxation event.

The total energy remains negative throughout the simulation, indicating that the post-merger remnant stays gravitationally bound. The absolute value of E_{tot} is not special by itself, since it depends on the adopted mass, length, and time units. More important is that E_{tot} remains bounded and does not show a systematic long-term drift after the merger, which supports the numerical stability of the run.

Again, the distance between the two central sinks was studied and the decay rate of the separation envelope was calculated. The relative distance between the two central sinks is shown in Figure 5.16. For this run, the fitted decay rate is noticeably lower than in the ULDM + sink + N-body case,

$$\gamma t_{\text{dyn}} = 0.33 \pm 0.13, \quad (5.31)$$

corresponding to

$$\gamma = 0.34 \pm 0.14 \text{ Gyr}^{-1}. \quad (5.32)$$

The associated decay timescale is

$$\tau = 2.94 \pm 1.19 \text{ Gyr}. \quad (5.33)$$

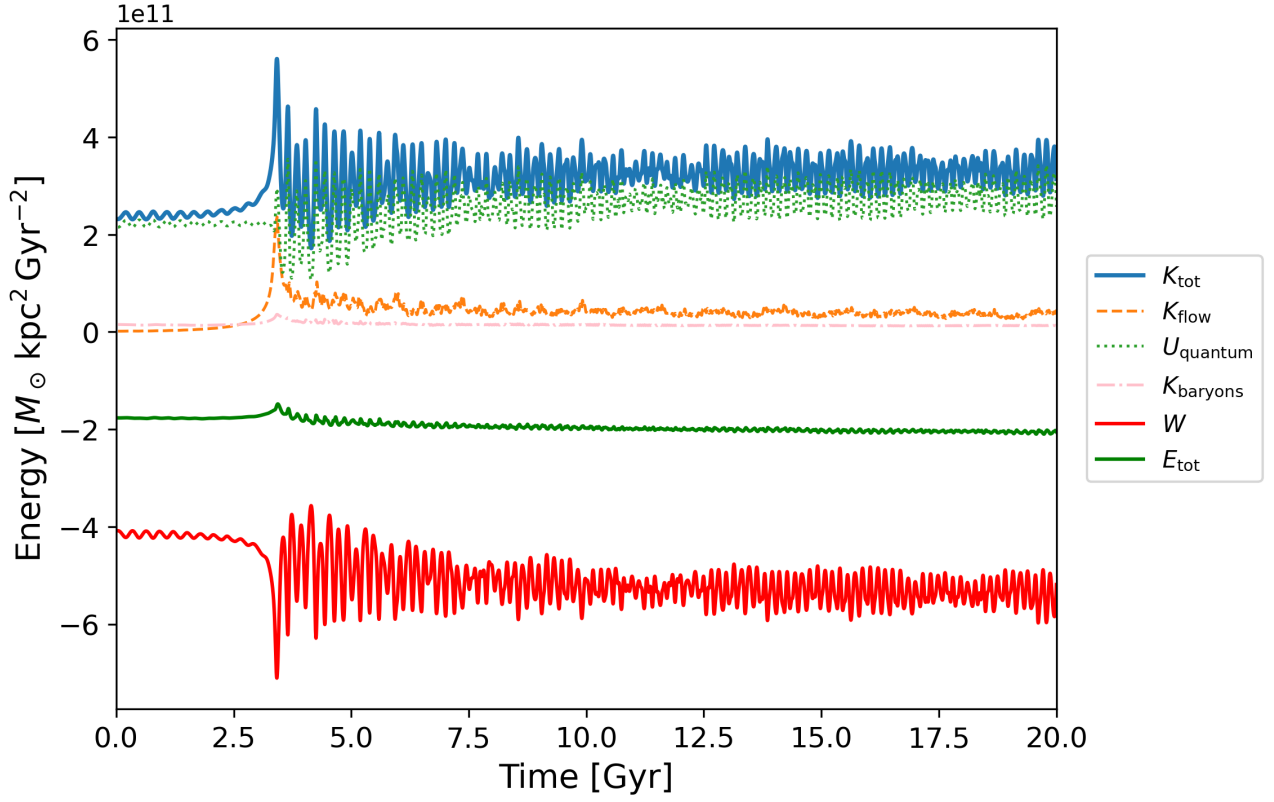


Figure 5.15: Time evolution of the kinetic energy K_{tot} , gravitational potential energy W , total energy E_{tot} , and selected kinetic-energy components K_{flow} and U_{quantum} during the full ULDM + sink + N-body + gas merger simulation.

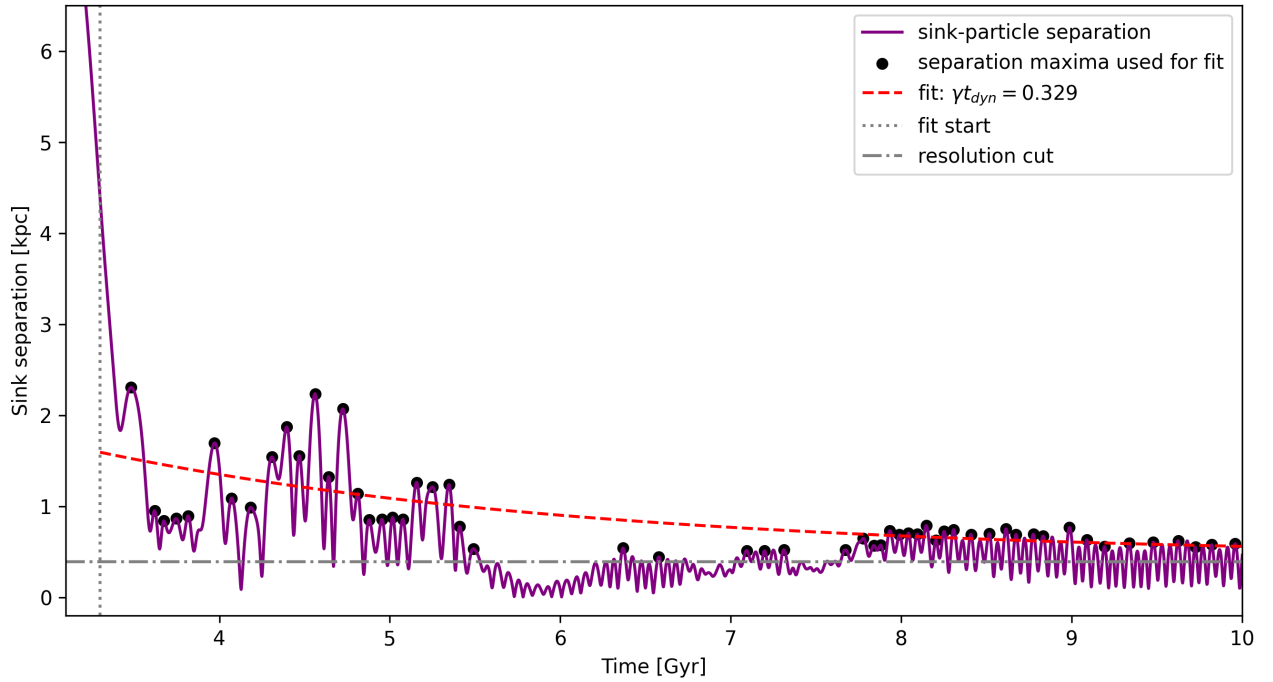
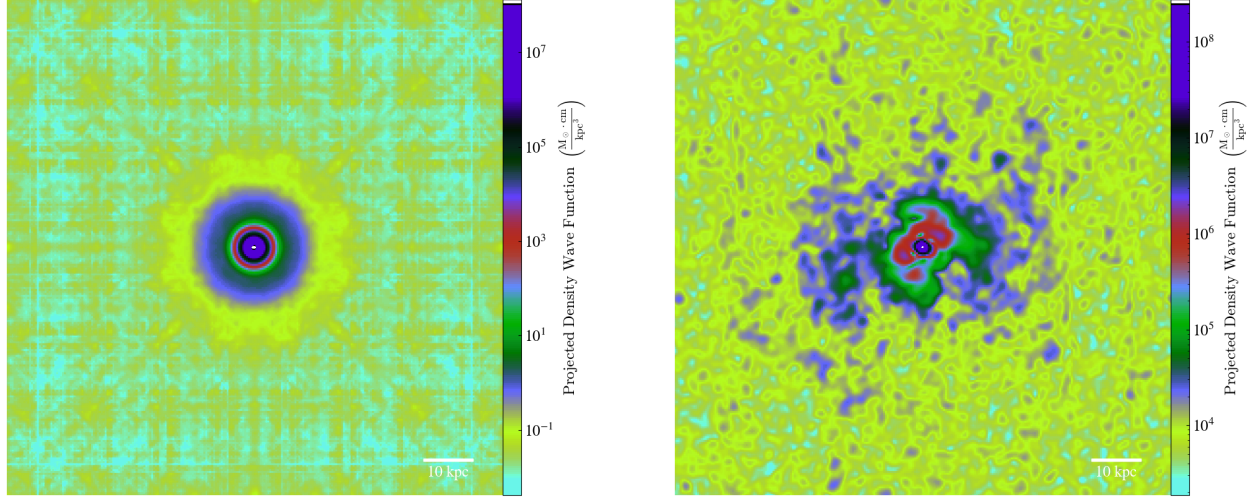


Figure 5.16: Time evolution of the separation between the two central sink particles in the full ULDM + sink + N-body + gas merger simulation. Local maxima after the first close encounter were fitted with an exponential envelope to obtain the empirical decay rate, considering only points above the grid resolution, shown by the gray lines.



(a) $t = 10$ Gyr, stabilization

(b) $t = 20$ Gyr, after merger

Figure 5.17: Projected ULDM density at $t = 10$ Gyr (Figure 5.17a) for the initial composite merger setup and at time $t = 20$ Gyr after the merger (Figure 5.17b). The soliton starts out spherically symmetric and is later disturbed into a centrally concentrated remnant.

Further analysis of the gas component can give us an idea about the “galacticity” of our toy galaxy. In other words, we can ask whether the initialised gas behaves at least qualitatively like a rotating galactic gas component, or whether it is completely destroyed by the merger.

Before studying the gas component it is useful to examine what happens to the ULDM component during the merger. This is shown in the projection plots in Figure 5.17. Figure 5.17a shows the ULDM soliton projection obtained after a stabilisation run of 10 Gyr, which served as the blueprint for the solitonic cores within the composite merger systems. After the merger, the solitonic core is significantly disturbed, as can be seen in Figure 5.17b, but remains centrally concentrated rather than being dispersed across the simulation volume.

The initial gas distribution, together with its velocity map, is shown in Figure 5.19a. The gas was originally set up as a rotating disc with initial circular velocity scale $v_0 = 35.0 \text{ km s}^{-1}$ and mass $M_{\text{gas}} = 2 \times 10^7 M_{\odot}$. This structure was perturbed and deformed during the merger, but the final gas distribution still forms a centrally concentrated, roughly circular remnant. The resulting central gas structure is shown in Figure 5.19b. It is important to note that by the end of the merger simulation the total gas mass is no longer exactly equal to $2M_{\text{gas}}$, although two gas discs were initialised. Instead, the remaining gas mass is approximately $M_{\text{gas, end}} \sim 3.8 \times 10^7 M_{\odot}$, as part of the gas was accreted by the sink particles throughout the initialisation run + the merger run.

However, the sink particles do not accrete only gas, and it turns out that accretion from the N-body component is much more important in this run. During the stabilisation period, the N-body component lost

$$\Delta M_{\star \rightarrow \text{sink}}^{\text{stab}} = 2.09 \times 10^6 M_{\odot}. \quad (5.34)$$

Therefore, the two systems were used for the merger, had a total N-body mass of

$$M_{\star, 0}^{\text{tot}} = 5.58 \times 10^7 M_{\odot}. \quad (5.35)$$

At the end of the merger simulation, the measured N-body masses of the two remnants were

$$M_{\star, 1} = 2.72 \times 10^7 M_{\odot}, \quad M_{\star, 2} = 2.74 \times 10^7 M_{\odot}. \quad (5.36)$$

This gives the N-body mass transferred to the sinks during the merger as

$$\Delta M_{\star \rightarrow \text{sink}}^{\text{merge}} = M_{\star, 0}^{\text{tot}} - (M_{\star, 1} + M_{\star, 2}) = 1.19 \times 10^6 M_{\odot}. \quad (5.37)$$

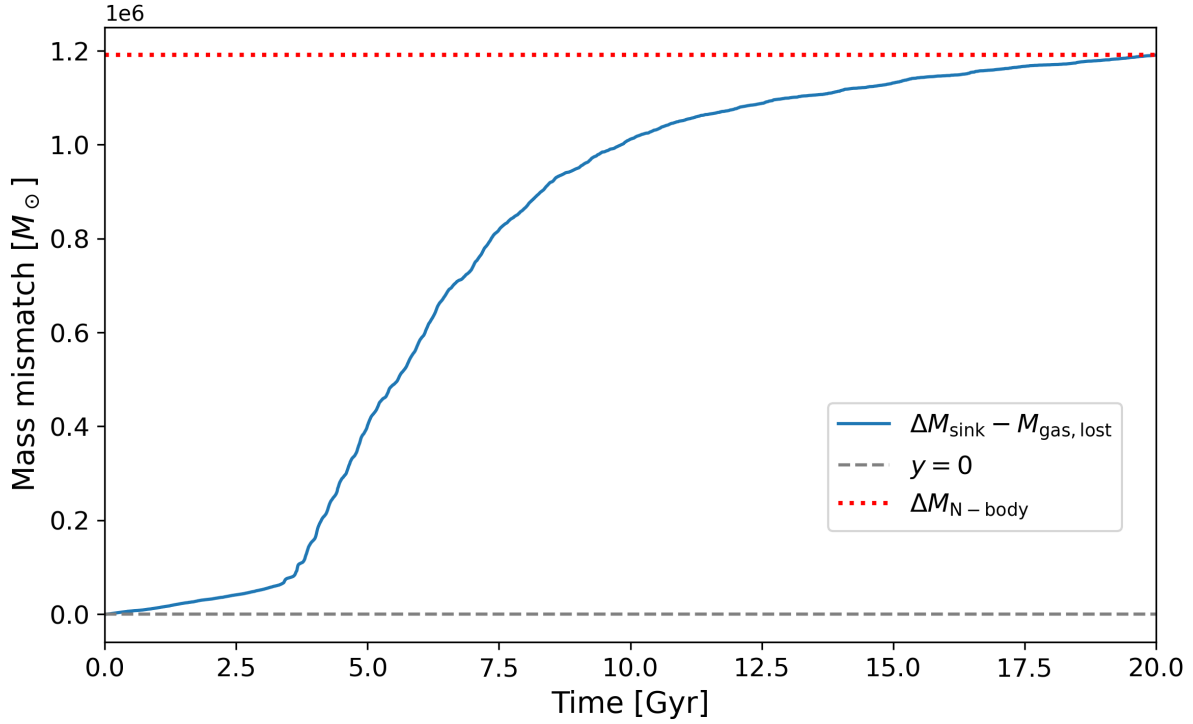


Figure 5.18: Mass mismatch between the total sink mass growth ΔM_{sink} and the gas mass removed from the grid during the merger $M_{\text{gas,lost}}$. The horizontal line shows the independent N-body mass transferred to the sinks during the merger, $\Delta M_{\star \rightarrow \text{sink}}^{\text{merge}} = 1.19 \times 10^6 M_{\odot}$ measured at $t = 20$ Gyr.

This value can be directly compared to Figure 5.18, where the plotted quantity is

$$M_{\text{missing}} = \Delta M_{\text{sink}} - M_{\text{gas,lost}} \quad (5.38)$$

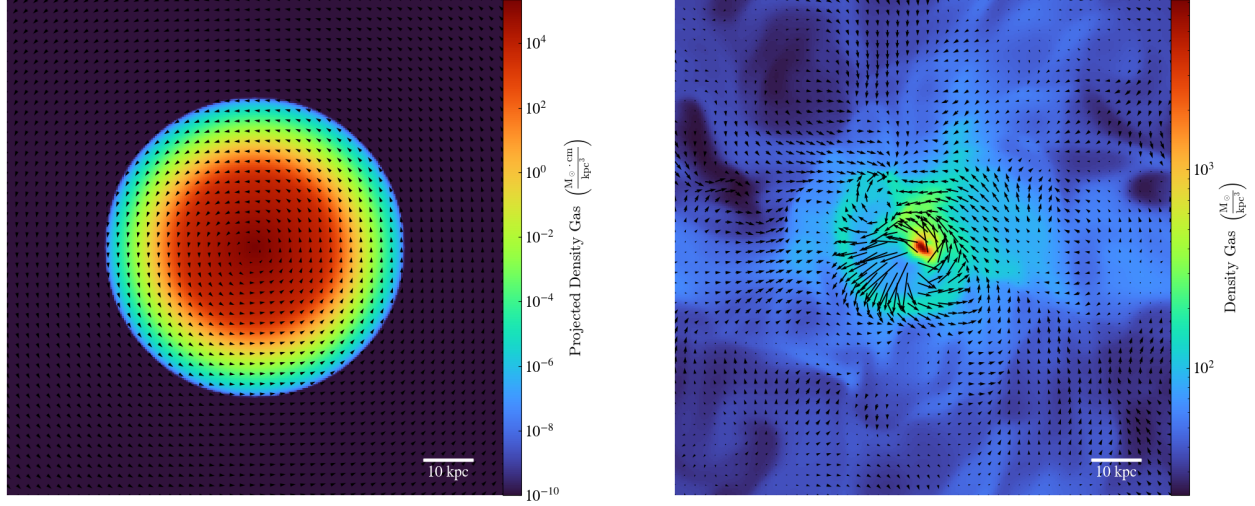
shows the part of the sink mass growth that cannot be explained by gas removal from the grid. The horizontal line marks the independent estimate of the mass loss of the N-body, $\Delta M_{\star \rightarrow \text{sink}}^{\text{merge}}$. The agreement between these two quantities shows that the apparent mass mismatch is not a mass conservation problem but is explained by accretion from the collisionless N-body component.

The plots in Figure 5.19 give us the density and velocity maps of the gas at the initialisation and the end of the simulation. Figure 5.19a shows the gas at the start of the simulation, where it was set up as a rotating disc to prevent strong immediate gravitational collapse. Figure 5.19b then shows the gas velocity field after the merger. While the final velocity field is not perfectly circular and its uniformity leaves a lot to be desired, the gas still retains a clear rotational component. Therefore, for the purposes of this toy model, it is not perfect, but good enough to be interpreted as a disturbed rotating gas disk.

Having a rotating gas disc within the galaxy merger, we can also ask whether its rotational profile agrees with the expected circular velocity curves. For this purpose, two simple predictions were constructed: one using only the baryonic components and one also including the ULDM solitonic core. The predicted circular velocity was calculated as

$$v_{\text{circ}}(R) = \sqrt{v_{\text{BH}}^2(R) + v_{\text{gas}}^2(R) + v_{\text{bulge}}^2(R) + v_{\text{sol}}^2(R)}, \quad (5.39)$$

where individual terms can be included or omitted depending on the model.



(a) $t = 0$ Gyr, initialization

(b) $t = 20$ Gyr, after merger

Figure 5.19: Gas velocity maps at the moment of initialization (Figure 5.19a) and after the merger of two composite systems (Figure 5.19b). The gas is initialized as a rotating disk and later evolves into a disturbed but still rotating gas structure.

The contribution from the central sink particles was approximated as a point-mass term,

$$v_{\text{BH}}(R) = \sqrt{\frac{GM_{\text{BH}}}{R}}, \quad (5.40)$$

where $M_{\text{BH}} = 2 \times 10^6 M_{\odot}$ is the initial total mass of the two central sink particles. The collisionless stellar component was approximated by a Hernquist profile,

$$v_{\text{bulge}}(R) = \sqrt{\frac{GM_{\text{bulge}}R}{(R + a_{\text{bulge}})^2}}, \quad (5.41)$$

with $M_{\text{bulge}} = 2 \times 3 \times 10^7 M_{\odot}$ and $a_{\text{bulge}} = 10$ kpc.

For the gas disc, the thin exponential disc approximation was used,

$$v_{\text{gas}}^2(R) = 4\pi G \Sigma_0 R_d y^2 [I_0(y)K_0(y) - I_1(y)K_1(y)], \quad y = \frac{R}{2R_d}, \quad (5.42)$$

where $R_d = 4$ kpc, $\Sigma_0 = M_{\text{gas}}/(2\pi R_d^2)$, and I_n and K_n are modified Bessel functions. This procedure was inspired by [83]. The value of M_{gas} was taken directly from the gas mass measured in the simulation snapshot.

The exact baryonic mass distribution in the simulation may differ slightly from the idealised values used in the prediction, mainly due to sink accretion during the stabilisation and merger runs. However, as shown in Figure 5.18, the total baryonic mass budget is accounted for by the combination of gas removal and N-body accretion onto the sinks. Since the ULDM component remains the dominant mass contribution in this setup, these small baryonic redistribution effects are not expected to change the qualitative comparison between the predicted curves.

Finally, the ULDM solitonic component was approximated by a Plummer-like profile,

$$v_{\text{sol}}(R) = \sqrt{\frac{GM_{\text{sol}}R^2}{(R^2 + a_{\text{sol}}^2)^{3/2}}}, \quad (5.43)$$

where $M_{\text{sol}} = 2 \times 3.3 \times 10^8 M_{\odot}$ and $a_{\text{sol}} = R_{1/2}/1.305$ [84]. The half-mass radius of the solitonic component was estimated from the final wave-function snapshot as the radius of a sphere,

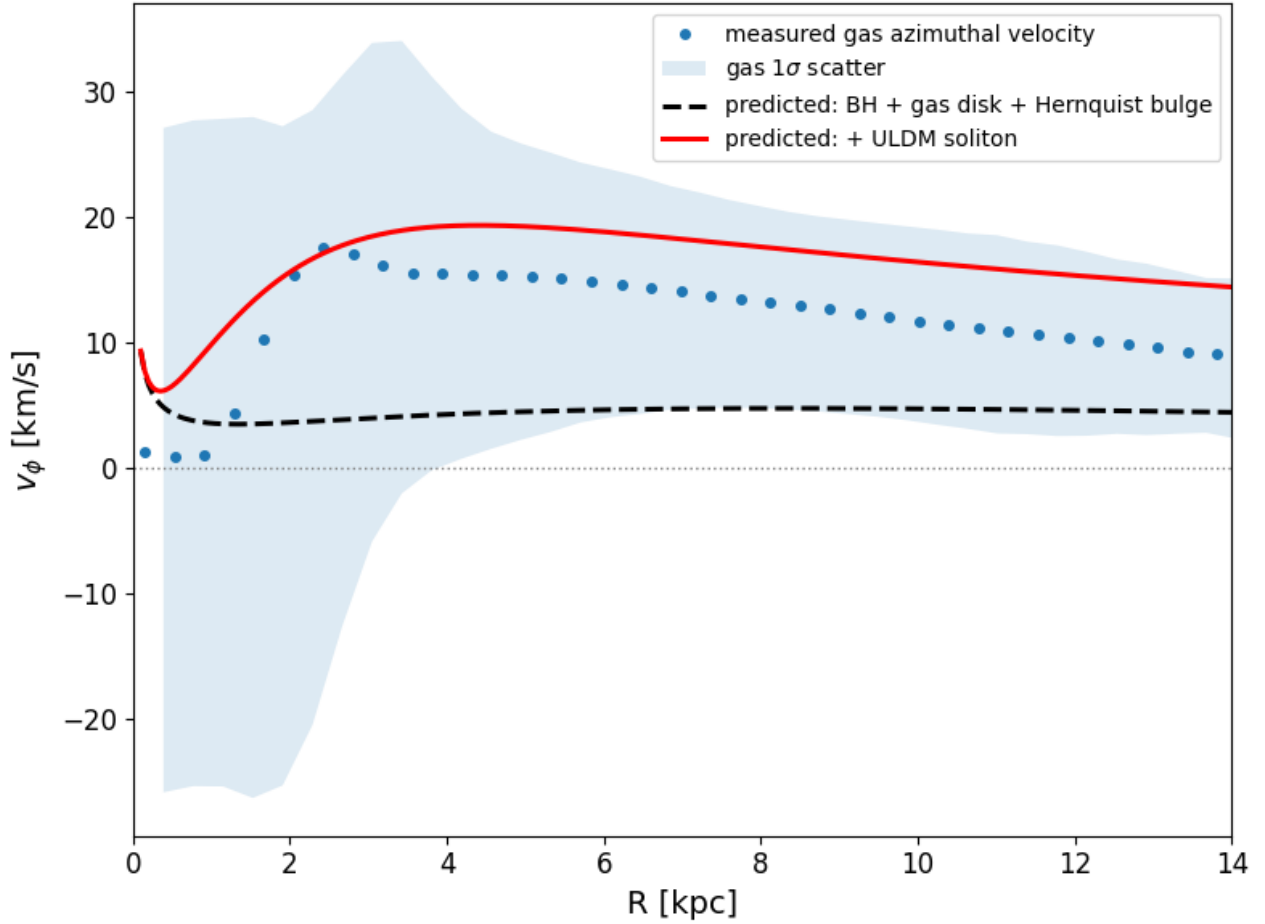


Figure 5.20: Density-weighted azimuthal gas velocity v_ϕ as a function of cylindrical radius R at $t = 20$ Gyr, corresponding to the end of the full composite merger simulation. The data points show the measured gas profile, with shaded uncertainty corresponding to the 1σ scatter in each radial bin. The black dashed curve shows the predicted circular velocity from the baryonic components only, while the red curve also includes the ULDM solitonic component.

centred on the density maximum, enclosing 50% of the total ULDM mass in the simulation. This gave $R_{1/2} \approx 3.9$ kpc, and therefore $R_{1/2} = 4.0$ kpc was used in the prediction.

The measured gas profile was obtained directly from the simulation data. In cylindrical coordinates around the z -axis, the azimuthal velocity was calculated as

$$v_\phi = \frac{-yv_x + xv_y}{R}, \quad R = \sqrt{x^2 + y^2}. \quad (5.44)$$

The gas cells were then binned by cylindrical radius R , and the density-weighted mean value of v_ϕ was calculated in each radial bin. The error bars shown in Figure 5.20 correspond to the 1σ scatter of v_ϕ within each bin.

The resulting profile is shown in Figure 5.20. The black dashed curve corresponds to the prediction using only the baryonic components, namely the sink particles, gas disc, and collisionless Hernquist component. The red curve also includes the ULDM solitonic component. The difference between these two predictions is large, which is expected, since the ULDM component is the dominant mass contribution in this galaxy system. The measured gas points broadly follow the order of magnitude of the prediction including ULDM, although the large scatter shows that the gas is not on perfectly circular orbits after the merger.

Chapter 6

Discussion

Before interpreting the physical results, it is important to note that the validation tests in Chapter 4 show that the individual components of the framework behave as expected in controlled settings and therefore the obtained results should be physically valid. The ULDM solver preserves stable solitonic profiles (Section 4.2), the N-body component reproduces orbital motion in a point-mass potential (Subsection 4.3.2), and the gas solver captures the main structure of the Sod shock tube test (Section 4.4). These tests support the use of the framework for the exploratory simulations discussed below, although they do not remove the limitations associated with resolution, timestep choice, simplified initial conditions, and the absence of cosmological boundary conditions.

6.1 Spin Dependence of Density Profiles

In Section 5.1, I simulated mergers of 25 solitons in a box of size 100 for different values of the scalar-field spin. The higher-spin solitonic profiles were constructed using the method outlined in Subsection 2.1.5. The density-profile results in Section 5.1 show that the formation of a central solitonic core is robust for all tested spin values. The spin dependence appears mainly outside the core, in the transition region between the soliton-like centre and the outer halo-like component. These results reproduce and extend the qualitative trends observed in [22].

The spin dependent behaviour in the outer profile can be linked to the internal structure of the higher-spin field. In the spin-0 case, the scalar field has a single polarisation component, and the merger relaxation proceeds through a relatively simple redistribution of mass and energy between the solitonic core and the surrounding halo. For higher-spin fields, the additional polarisation components introduce internal degrees of freedom that are absent in the spin-0 case. During the merger, energy can be exchanged between these components, which may reduce the efficiency of mass transfer into the central core and a correspondingly shallower outer profile. This interpretation is consistent with the observation in Figure 5.1 that the deviation from the reference soliton profile begins at slightly smaller radii for higher spins, suggesting that the transition between the core-dominated and halo-dominated regimes shifts inward as the number of polarisation components increases.

However, it should be noted that this interpretation remains qualitative. A more rigorous treatment would require tracking the energy content of individual polarisation components separately throughout the merger, which is beyond the scope of this work. nevertheless, the results suggest that the spin of the scalar field may leave a distinguishable imprint on the outer halo structure, even when the central solitonic core remains largely unaffected.

6.2 Spin Dependence of Solitonic Core Growth

The core-growth analysis in Section 5.2 shows that the remnant of the merger undergoes a clear increase in the characteristic soliton size for all spin values tested. This is consistent with the expected relaxation of a multi-soliton system into a single central core surrounded by a more extended halo-like component. The growth parameter $\lambda_\rho(t)$ therefore provides a compact way to quantify how strongly the central density decreases, and equivalently how the effective radius of the core increases, during merger.

The single-seed simulations show the same qualitative ordering as averaged runs, with lower-spin systems reaching larger asymptotic values of λ_∞ than higher-spin systems. However, the absolute values of λ_∞ differ between the single-seed and multi-seed analysis. This is expected since one merger realisation is sensitive to the particular initial positions, phases, and interference pattern of the solitons. A single fit can therefore be influenced by oscillations specific to the given run and by the exact timing of the relaxation process.

For this reason, the averaged multi-seed results provide a more reliable estimate of the spin trend. They show that the $s = 0$ and $s = 1$ cases reach systematically higher values of λ_∞ , while the $s = 2$ and $s = 3$ cases saturate at lower values. This suggests that increasing spin of the scalar field may reduce the growth of the relaxed solitonic core. Physically, this may be connected to the larger number of internal components in the higher-spin fields, which can redistribute energy and mass differently during the merger. In the spin-0 case, all of the kinetic energy liberated during the merger must be redistributed between the translational and internal degrees of freedom of a single scalar component. In contrast, higher-spin fields possess multiple polarisation components that can exchange energy among themselves. This additional way of redistributing energy may act as a hindrance and be absorbing a fraction of the energy that would otherwise be deposited into the growth of the central core. As a result, the resulting clump in higher-spin cases retains a more compact and centrally concentrated configuration, consistent with the lower values of λ_∞ reported in Table 5.1.

Relaxation timescales should be interpreted with more caution than λ_∞ . Their uncertainties are relatively large, especially for the higher-spin cases, and the fitted values depend more strongly on the time evolution of each realisation. Therefore, the main conclusion of this section is not a precise spin-dependent relaxation time, but rather the qualitative trend that higher spin leads to a smaller final core-growth parameter.

This spin dependence of the core growth may have a potentially observable consequence. Since the solitonic core mass scales with the core radius, a suppressed core growth implies a more compact central soliton in the relaxed remnant. For models with higher spin, this would shift the predicted core properties towards smaller radii and higher central densities compared to the spin-0 case, which could in theory be tested against kinematic observations of dwarf spheroidal galaxies, where the solitonic core is expected to leave an imprint on the stellar velocity dispersion profile [21].

6.3 Study of ULDM Black Hole Systems

As shown in Figure 5.5, the modified eigenvalue construction introduced in Section 5.3 is capable of producing gravitationally stable ULDM+BH configurations. These systems can then be used to study how the resulting equilibrium profile depends on the scalar-field mass, as shown in Figure 5.6.

6.3.1 Scalar Field Mass Dependence

The profiles show that changing m_s mainly affects the central solitonic core. Lower scalar-field masses lead to denser and more massive equilibrium configurations. This is consistent with the larger de Broglie wavelength associated with smaller m_s , which increases the characteristic soliton scale and changes the balance between gravity and quantum pressure. In contrast, the outer parts of the profiles for $m_s = 10^{-23}$ eV and $m_s = 10^{-22}$ eV approach similar density levels. This may indicate that, once the central configuration is sufficiently massive, the outer low-density region is less sensitive to the precise scalar-field mass than the inner core.

The case $m_s = 10^{-21}$ eV behaves differently, since the corresponding solitonic configuration is much less massive and is more strongly affected by the fixed central black-hole potential. Its lower density profile should therefore be interpreted as a consequence of the stronger relative importance of the black hole for larger scalar-field masses. This already suggests that the effect of the black hole cannot be described only by its absolute mass, but by its mass relative to the solitonic configuration supported for a given m_s .

To more systematically quantify this transition, the equilibrium core mass was then studied as a function of the black-hole mass in Figure 5.7. The fit in Equation 5.11 should be understood primarily as a phenomenological interpolation between the weak-BH and strong-BH regimes, rather than as an analytical derivation. Its main purpose is to provide a compact way to quantify where the transition occurs and how sharply the core mass decreases once the black hole starts to dominate the central potential.

The convergence of the outer density profiles for $m_s = 10^{-22}$ eV and $m_s = 10^{-23}$ eV in radii $r \gtrsim 5$ kpc deserves a comment. One possible interpretation is that the outer low-density envelope is primarily shaped by the gravitational potential of the black hole, which mass is non-negligible compared to that mass of the soliton, rather than by the wave properties of the scalar field, making it less sensitive to the precise value of m_s once the central configuration is sufficiently massive. Alternatively, this convergence may reflect a numerical effect, since the outermost regions of the profile are more susceptible to grid noise and boundary artefacts. Distinguishing between these two possibilities would require higher-resolution runs or a systematic study of the profile behaviour as a function of box size, which is left for future work.

6.3.2 Mass Scaling Relations

The behaviour shown in Figure 5.7 directly reflects the competition between the self-gravity and quantum support of the solitonic core on one hand, and the external gravitational potential of the central black hole on the other. For sufficiently small black-hole masses, the system supports a solitonic core similar to the isolated value. Once the black-hole mass exceeds a characteristic value, the equilibrium core mass must be significantly reduced. This shows that a central black hole can significantly modify, or partially suppress, one of the characteristic small-scale signatures of ULDM in galactic centres.

The value $M_{\text{BH}} = 5 \times 10^7 M_\odot$ used in the profile comparison was chosen as a representative case in which the black-hole contribution is already dynamically visible, while remains numer-

ically well behaved and retains a clearly identifiable core-like equilibrium structure within the numerical construction described in Section 5.3. At sufficiently large black-hole-to-core mass ratios, the interpretation of the constructed profiles should be treated with caution. In this regime, the central point-mass potential may dominate the inner structure so strongly that the configuration is no longer best understood as a weakly perturbed solitonic core.

This result may have observational implications. Solitonic cores are among the most distinctive small-scale predictions of ULDM models, and observational searches for them typically rely on their expected mass and radius as a function of the host halo mass. However, the results shown in Figure 5.7 demonstrate that the presence of a central black hole can significantly reduce the equilibrium core mass, shifting it below the value predicted by the standard core–halo scaling relation. This means that in galaxies hosting a sufficiently massive central black hole, the solitonic core may appear smaller or less massive than naive scaling arguments would suggest, potentially leading to an underestimate of the ULDM particle mass when fitting observational data. This effect is expected to be most relevant for systems where the black hole mass approaches or exceeds the characteristic suppression scale $M_{\text{bh},0}$. Dwarf galaxies with relatively massive central compact objects, as well as more massive systems with well-constrained black hole masses, would therefore be the most informative targets for testing this prediction observationally.

A final caveat is that the plotted configurations are numerically obtained equilibrium solutions. Although the smooth behaviour of the curves suggests a meaningful trend, a full analysis of dynamical stability would require additional time-evolution tests. The results should therefore be interpreted primarily as a scaling study of equilibrium core masses in the presence of a central black hole.

6.3.3 Scaling Model Estimate

As an exploratory exercise, I tested whether the scaling relations introduced in Section 2.5 can be used to predict the soliton mass from a given black-hole mass. For each final value of M_{BH} , the corresponding halo mass was first estimated from the empirical black-hole–halo relation, and the expected soliton mass was then calculated using the soliton–halo scaling relation. The resulting prediction was compared with the soliton mass measured in simulations, where the central sink particle was initialised with a lower mass and then increased to the final value used in the prediction. The comparison is shown in Figure 5.8.

This procedure combines several approximate ingredients: the empirical black-hole–halo relation, the soliton–halo scaling relation, the numerical construction of the initial soliton profile, the imposed growth of the central sink mass, and the later measurement of the remaining solitonic core. The result should therefore be interpreted only as a proof-of-concept test rather than as a quantitatively reliable prediction. Nevertheless, the comparison shows the same qualitative behaviour, which is reasonably captured by the phenomenological suppression model of Equation 5.11. This suggests that combining the scaling estimate with the suppression fit may provide a useful approximate way to predict how much of the original solitonic core can survive in the presence of a growing central black hole.

The deviations visible in the latter part of Figure 5.8 suggest that the quasi-equilibrium description breaks down when the black hole mass grows on a timescale comparable to or shorter than the relaxation time of the solitonic configuration. In this regime, the soliton does not have sufficient time to readjust its internal structure to the changing central potential and is no longer able to relax towards the evolving equilibrium configuration. The present comparison therefore not only tests the scaling model, but also provides a rough indication of the timescale on which the black hole mass must grow in order for the core to remain stable.

6.4 Gas–Soliton Dynamics and Sink Accretion

The results presented in Section 5.4 suggest that the solitonic core primarily affects the post-formation accretion phase rather than the initial sink-formation event. The nearly identical sink seed masses in both runs suggest that the soliton does not significantly alter the conditions that trigger collapse, but instead modifies the subsequent gas infall environment.

This enhancement can be understood from the form of the sink accretion prescription, which is Bondi-like, as introduced in Equation 2.56. In such a model, the accretion rate scales with the local gas density around the sink ρ_∞ and inversely with the relative velocity between the gas and the sink v_{rel} . The presence of the ULDM soliton deepens the central gravitational potential and keeps the gas more concentrated around the sink after its formation. This increases the local density entering the accretion prescription and can also reduce the relative gas–sink motion once the gas becomes bound to the central potential. Both effects act in the same direction, increasing $\dot{M}_{\text{acc}}$, so even a moderate deepening of the central potential can lead to substantially larger cumulative accreted mass over time. This is consistent with the factor of approximately ten enhancements reported in Section 5.4.

The small width of the confidence intervals across the five independent realisations indicates that this result is robust with respect to the specific perturbation seed and is therefore a genuine dynamical effect of the soliton rather than an artefact of the initial conditions. Nevertheless, the result should be interpreted as a controlled numerical experiment rather than a quantitative prediction for black-hole growth in a realistic galaxy. A more realistic treatment would require a broader parameter study covering different soliton masses, gas fractions, and cooling prescriptions, as well as additional physical processes such as feedback from the accreting sink and a more detailed model of the unresolved accretion region.

The results from Figure 5.10 confirm that the sink formation threshold is governed by the local gas density exceeding the Truelove criterion rather than by the depth of the gravitational potential well, and that the influence of the soliton manifests itself only in the subsequent accretion phase.

6.5 Composite System Mergers

Section 5.5 focused on showing the simulation framework’s capacity to simulate multi-component dynamics in a non-trivial scenario and produce physically interpretable results.

Galaxy mergers are among the most energetic and dynamically complex events in the Universe and are expected to play a significant role in the assembly of present-day galaxies. Within the ULDM framework, mergers of composite systems represent a particularly interesting scenario, since the wave nature of the dark matter component introduces dynamics that are not captured in standard collisionless N-body mergers.

6.5.1 Stable Systems

In order to study mergers of stable systems, it is first necessary to construct configurations that are close to equilibrium. In Subsection 5.5.2 it was shown that following the methods outlined in Section 2.5, Section 5.3 and Subsection 5.5.2 itself, it is possible to create systems that are either in a virialized state or close to it, as can be seen in Figure 5.11. This is the main way to determine whether the system is gravitationally bound and quasi-stable or not. Further diagnostics such as checking the energy conservation constraints and overall total mass budget within the simulation run were carried out, but not included for the brevity of this work.

The mean values of virial ratios were $\mu_{\text{ULDM}} = 1.003 \pm 0.001$, $\mu_{\text{ULDM+BH}} = 1.013 \pm 0.001$, $\mu_{\text{ULDM+BH+H}} = 1.102 \pm 0.018$, and $\mu_{\text{all}} = 1.152 \pm 0.035$ for systems of increasing complexity, showing that the ULDM and ULDM+BH systems are the most stable and properly virialized, while the more complex systems are not in a perfectly virialized state. This offset could be reduced by running a simulation where solitons, sinks, gas, and particle clumps are randomly distributed across the grid and left to collapse and virialize naturally, but this would require significant computational resources and was not necessary, as quasi-stable systems are sufficient for the purposes of this work.

6.5.2 System Mergers

Understanding how the solitonic core, the central black hole proxy, and the baryonic components respond to and interact during a merger is therefore a natural extension of the framework developed in this work. Since soliton-only mergers were already studied in Section 5.1 and Section 5.2, the focus here is on the two more complex composite systems. Two stabilised systems were created at positions $\vec{p}_1 = (10, 10, 0)$ and $\vec{p}_2 = (-10, -10, 0)$ with initial velocities $\vec{v}_1 = (-2, 0, 0)$ and $\vec{v}_2 = (2, 0, 0)$, chosen to produce a merger with a non-zero impact parameter. As a reference, an equivalent run containing only sink particles was carried out.

A key property of the mergers investigated in this study is the distance between the two central sink particles. In the reference run, shown in Figure 5.12, the sinks simply fell toward each other. However, for more complex systems, a more interesting trajectory was observed. This trajectory is shown for the ULDM + BH + Hernquist system in Figure 5.13, although it was very similar for the 4-component run. Here, we can see that introducing the baryonic and ULDM components results in sink trajectories that resemble long-lived binary systems, rather than a direct collision. This effect could be caused by the surrounding mass gravitationally tugging on the sinks from varying distances, causing slight orbital shifts. Consequently, whereas the two naked sinks would have simply collided, they are now displaced from a direct impact trajectory.

To quantify the orbital evolution of the binary system of bound sinks, the local maxima of the sink separation were fitted with the exponential envelope defined in Equation 5.25, yielding the resulting plots shown in Figure 5.14 and Figure 5.16. The rescaled decay rates obtained from this are $\gamma t_{\text{dyn}} = 0.98 \pm 0.29$ for the three-component system and $\gamma t_{\text{dyn}} = 0.33 \pm 0.13$ for the four-component system. Counter-intuitively, the decay rate is lower in the four-component case despite the additional mass contributed by the gas. This suggests that the gas does not act simply as an additional source of effective orbital damping, but instead redistributes mass and angular momentum during the merger in a way that reduces the net orbital damping experienced by the sink pair.

The time cut-offs used to initiate the fitting were chosen based on the energy composition plots shown in Figure 5.15 for the four-component case. Prior to the merger at approximately $t \approx 3$ Gyr, all energy components evolve smoothly with only small opposing oscillations between the total kinetic energy K_{tot} and the potential energy W , consistent with a quasi-stable pre-merger configuration. The sharp transient visible at $t \approx 3$ Gyr marks the first close encounter and was therefore used as the starting point for the envelope fitting.

During and after the merger, both K_{flow} and U_{quantum} rise sharply at the expense of the potential energy W . Within the Madelung decomposition (Equation 1.14) of the ULDM kinetic energy, K_{flow} corresponds to the bulk flow energy associated with phase gradients of the wave field, while U_{quantum} is the quantum pressure contribution arising from spatial variations in the density amplitude $\sqrt{\rho}$. The rise in K_{flow} therefore reflects an increase in the bulk velocity of the ULDM field during the encounter, while the simultaneous rise in U_{quantum} indicates enhanced density gradients within the wave field, consistent with increased wave interference

and soliton deformation during the merger. Both effects are driven by the conversion of gravitational potential energy into internal kinetic energy of the ULDM component as the two systems interact.

Following the turbulent merger, the system slowly relaxes into a more stable configuration, as indicated by the decreasing amplitude of oscillations across all energy components. The small peak in the total energy visible at the moment of the merger could be attributed to the gravitational potential changing rapidly on a timescale comparable to the simulation time step, temporarily violating the slow-variation assumption underlying the operator splitting. After the merger, the physical oscillations of the relaxing remnant similarly introduce small time-stepping errors that accumulate over time. Both effects are considered negligible for the purposes of this work.

The evolution of the density distribution of the ULDM component during the merger is shown in Figure 5.17. Prior to the merger, the solitonic core was spherically symmetric and centrally concentrated, with a smooth density profile extending outward as expected for a virialized configuration. After the merger, the core remains centrally focused but is visibly disturbed. The post-merger ULDM density field shows significant large-scale fluctuations and interference patterns across the grid, which are characteristic of the wave nature of the scalar field responding to the violent dynamical event. Some fluctuations are also visible for the profile in Figure 5.17a, but of negligible magnitude when compared to those in Figure 5.17b. These fluctuations reflect the excitation of higher-order modes in the ULDM field during the merger, and the fact that they persist at $t = 20$ Gyr suggests that the ULDM component has not yet fully relaxed into a new equilibrium configuration. The central density peak has also increased by roughly an order of magnitude compared to the pre-merger soliton, consistent with the two solitonic cores having merged into a single more massive remnant.

As an additional diagnostic of the stability within the simulation and to verify that the sink accretion from both the gas and star is mass conserving, I plotted the mass mismatch between the total sink mass growth ΔM_{sink} , which includes both the sink and the slowly draining reservoir together and the gas mass removed from the grid $M_{\text{gas,lost}}$. The accretion reservoir acts as an intermediary, releasing the accreted mass to the sink incrementally over successive time steps rather than instantaneously, which improves energy conservation during the accretion process. The resulting plot in Figure 5.18 also includes a dotted red line indicating a mass that has been stripped away from the N body component at the final point of the simulation. The fact that the line meets the tracked mismatch in blue at $t = 20$ Gyr shows that the overall mass budget within the simulation has been conserved.

Knowing that no mass and energy leaked out during the simulation, we can discuss the results shown in Figure 5.19. Initially, the gas is arranged in a regular rotating exponential disc. After the merger, the disc has been significantly disrupted, but not destroyed. The gas has been redistributed into a centrally concentrated remnant that still retains an approximately circular shape, suggesting that a significant fraction of the initial angular momentum has been preserved throughout the merger event.

The plot also shows the corresponding gas velocity maps. Before the merger, the velocity field shows a clean rotational pattern consistent with the initialised disc. After the merger, the velocity field is no longer perfectly ordered, but a rotational component remains visible. While the gas is clearly not in a relaxed equilibrium state after such a violent event, the fact that coherent rotation survives at all is encouraging and suggests that the original angular momentum is at least partially conserved. For the purposes of this toy model, the post-merger gas can therefore be interpreted as a disturbed but still rotating quasi-disc, which motivates the comparison of the rotation curves shown in Figure 5.20.

A comparison of the measured azimuthal velocity profile with the two analytical predictions

is shown in Figure 5.20. The red curve includes all components, including the ULDM soliton, while the black dashed curve accounts for the baryonic components only. The measured profile broadly follows the shape of the red curve more closely than the black dashed one, suggesting that the ULDM soliton contributes meaningfully to the central gravitational potential and that its inclusion is necessary to reproduce even the order-of-magnitude behaviour of the rotation curve. This is consistent with the ULDM component being the dominant mass contribution in this system.

At larger radii, the measured velocities fall off more steeply than both predictions, which is expected given that the post-merger gas disc is more centrally concentrated than the idealised exponential profile assumed in the analytical model and therefore lacks mass at large radii. In the innermost region, the measured velocities approach zero, which is physically plausible for a centrally concentrated remnant, although the large errors in this region make a quantitative interpretation questionable.

Overall, while the agreement is not quantitative, the qualitative consistency between the measured profile and the ULDM-inclusive prediction demonstrates that the framework is capable of producing physically interpretable results even for the most complex multi-component merger simulations carried out in this work.

Conclusion

This work presents the development of a modular and extensible simulation framework for studying ultra-light dark matter in the presence of baryonic matter. The primary goal was to implement and validate a numerical solver based on the split-step Fourier method for evolving the Schrödinger–Poisson system, which describes the non-relativistic dynamics of bosonic dark matter on astrophysical scales. To move beyond isolated ULDM configurations, suitable models for the individual baryonic components were studied and implemented, enabling simulations of composite multi-component systems consisting of a ULDM wave field, a collisionless stellar component, a grid-based gas, and sink particles acting as a black hole proxy.

To achieve this, the theoretical background of dark matter and its models was studied (Chapter 1), a physical model in which all components of the framework coexist and interact was designed (Chapter 2), and the resulting framework was implemented as described in Chapter 3.

A series of validation tests (Chapter 4) were carried out that included:

1. The Quantum Harmonic Oscillator Test, to verify the correct functioning of the Fourier solver,
2. Stability Checks, to confirm that increasing spatial resolution reduces numerically induced density oscillations and that solitonic solutions remain stable over time,
3. Kinematic Tests, to reproduce the predicted circular motion around a point-like mass source,
4. Energy Conservation Analysis to demonstrate the numerical stability of the simulation and to quantify the impact of spatial resolution, time-step size, and integrator order on energy conservation.

With the validated framework, a series of simulations were performed (Chapter 5). For multi-soliton mergers, density profiles and soliton core growth were studied across different scalar field spins, reproducing the expected solitonic core-halo structure and revealing spin-dependent differences in both the outer profile slopes and the asymptotic core growth parameter. The influence of a central black hole on the equilibrium solitonic core mass was studied, showing that a sufficiently massive black hole can significantly suppress the core mass, with the characteristic suppression scale depending on the ULDM particle mass. The interaction between the sink particle component and the ULDM component was also explored, demonstrating that the solitonic core enhances post-formation gas accretion onto the central sink. Finally, mergers of composite multi-component systems were performed, revealing long-lived bound sink pairs and a non-trivial dependence of the orbital decay rate on the presence of the gas component. The interpretation of these results was given in Chapter 6. Overall, the framework developed in this work provides a flexible basis for future studies of coupled ULDM–baryon dynamics in increasingly realistic astrophysical scenarios.

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Appendix A

Table of Initial Conditions

Table A.1: Initial conditions used for the composite-system stability tests. The subsequent merger runs were initialized from the final states of the corresponding stabilized systems. The three-component merger corresponds to the same setup without the gas component.

Component / parameter	Value	Comment
Simulation box size	$L = 100$ kpc	cubic domain
Grid resolution	$N = 256$	same in all spatial directions
ULDM scalar-field mass	$m_s = 10^{-22}$ eV	fiducial value
ULDM soliton mass	$M_{\text{sol}} = 3.3 \times 10^8 M_{\odot}$	stationary SP solution
Sink mass	$M_{\text{sink}} = 10^6 M_{\odot}$	central SMBH proxy
Total baryonic mass	$M_{\text{bar}} = 0.15 M_{\text{sol}}$	local baryonic loading
Hernquist mass	$M_{\text{H}} = 3.0 \times 10^7 M_{\odot}$	collisionless component
Hernquist scale radius	$a_{\text{H}} = 10$ kpc	diffuse stellar component
Hernquist truncation radius	$r_{\text{trunc,H}} = 10$ kpc	imposed truncation
Number of particles	$N_{\text{particles}} = 1 \times 10^6$	Hernquist sampling
Gas mass	$M_{\text{gas}} = 2.0 \times 10^7 M_{\odot}$	omitted in three-component runs
Gas radial scale length	$R_0 = 4$ kpc	exponential disk
Gas vertical scale height	$Z_0 = 0.8$ kpc	exponential disk
Gas cut-off radius	$R_{\text{cut}} = 15$ kpc	smooth taper
Gas taper width	$dR_{\text{cut}} = 1.5$ kpc	smooth transition

Appendix B

A Note on Self-Interaction

As mentioned in Chapter 1, ultra-light dark matter models do not have to be restricted to the purely gravitational interactions. In more general scalar-field models, additional self-interaction terms can be included in the effective Schrödinger equation. The optional self-interaction implemented in the present code follows the prescription used in [33].

In the implementation, the self-interaction is controlled by the boolean parameter `use_self_int` and by the coupling parameter `a_s`. The additional potential contribution is computed as

$$V_{\text{SI}} = \frac{\lambda \tilde{\hbar}^2}{4c} (\psi^2 + 2\rho\psi),$$

where

$$\lambda = \frac{32\pi a_s c}{\hbar}.$$

Here, ρ is the density entering the evolution, ψ is the evolved wave function, and a_s sets the overall strength of the self-interaction. This contribution is then added to the gravitational potential before the potential-step propagator is constructed.

The simulations presented in the main part of this thesis were performed with this option disabled, i.e. in the purely gravitational Schrödinger–Poisson limit. For this reason, the self-interaction term is not discussed in the main results and is included here only as an additional implemented feature of the framework.